

Regeneration of a Pharmaceutical Waste Solvent by Batch Extractive Distillation

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Final Project

Regeneration of a Pharmaceutical Waste Solvent by Batch
Extractive Distillation

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Abstract

This Final Project investigates the regeneration of a pharmaceutical waste solvent by batch extractive distillation, focusing on the recovery of high-purity isopropyl alcohol (IPA) from an IPA–water mixture using dimethyl sulfoxide (DMSO) as entrainer, due to the fact that IPA and water form a minimum-boiling azeotrope, which prevents effective recovery by conventional batch distillation. ChemCAD simulations were used to study the influence of operating parameters, charge composition, entrainer addition, reflux ratio, feed conditions, and number of stages on product purity, recovery, and process duration.

The selected operating conditions allowed the process to exceed the required IPA purity specification of 99.75 mol%, achieving a maximum IPA distillate purity of 99.856 mol% during the main distillation step. The energy demand of the base process was evaluated, and heat-recovery alternatives were studied to reduce operating cost. Two main heat recovery strategies were compared: a pre-heater using column top vapour and a heat pump configuration based on vapour compression. Although the pre-heater required lower additional capital investment, the heat pump method provided significantly greater long-term energy savings and a shorter estimated payback period.

Material recovery strategies were also investigated by recycling off-cut streams and recovering DMSO-rich residues. These methods were assessed for their ability to reduce solvent loss, decrease waste generation, and maintain the required IPA purity. In addition, preliminary equipment sizing was carried out for the packed distillation column, storage tanks, condenser, reboiler, vacuum pump, and heat-recovery equipment. A cost estimation was then prepared for the base distillation system and the proposed heat-recovery alternatives.

Overall, the results show that batch extractive distillation with DMSO is a technically feasible method for regenerating high-purity IPA from pharmaceutical waste solvent streams. The proposed process combines solvent recovery, material recycling, and heat integration, making it a promising approach for reducing waste and improving the economic and environmental performance of solvent regeneration.

Nomenclature

Latin Symbols

Symbol	Meaning	Unit
A	Area	m^2
A_{calc}	Calculated heat-transfer area	m^2
A_{cond}	Condenser heat-transfer area	m^2
A_{tube}	Tube-side flow area	m^2
a	Bundle-diameter correlation constant	-
b	Bundle-diameter correlation exponent	-
B	Batch number	-
B_{ij}	Binary interaction parameter from component i to component j	cal/mol
B_{ji}	Binary interaction parameter from component j to component i	cal/mol
C	Cost	\$
C_1	Compressor 1	MJ/h
C_2	Compressor 2	MJ/h
C_A	Corrosion allowance	mm or m
C_{shell}	Shell clearance	m
C_i	Installed cost	\$
C_p	Purchase cost	\$
C_{total}	Total installed cost	\$
C_{add}	Additional installed cost	\$
$C_{pre,saving}$	Pre-heater utility saving per batch	$\text{\$/batch}$
$C_{HP,steam\ saved}$	Value of steam saved by the heat-pump method per batch	$\text{\$/batch}$
$C_{HP,electricity}$	Electricity cost of the heat-pump method per batch	$\text{\$/batch}$
$C_{HP,saving}$	Net saving of the heat-pump method per batch	$\text{\$/batch}$
c_p	Specific heat capacity	$\text{J/(kg}\cdot\text{K)}$
D	Diameter	m
D_{bundle}	Tube bundle diameter	m
E	Weld joint efficiency / Young's modulus	- / Pa
$E_{recovery}$	Energy required for the recovery distillation step	GJ
E_{pre}	Heat recovered by the pre-heater per batch	GJ/batch
$E_{HP,steam}$	Steam energy replaced by the heat-pump method per batch	GJ/batch

Symbol	Meaning	Unit
$E_{HP,electricity}$	Electricity energy required by the heat-pump compressors per batch	GJ/batch
F	F-factor / LMTD correction factor	$\sqrt{Pa}$ / -
F_c	Cost correction factor	-
F_t	Compressor correction factor	-
H	Height	m
$HETP$	Height equivalent to one theoretical plate	m
HP	Horsepower	hp
HP_1	Horsepower of compressor 1	hp
HP_2	Horsepower of compressor 2	hp
I	First component in binary interaction table	-
J	Second component in binary interaction table	-
k	Overall heat-transfer coefficient	W/(m ² ·K)
K_{vap}	Heat-transfer coefficient for vapour	W/(m ² ·K)
K_{dist}	Heat-transfer coefficient for distillate	W/(m ² ·K)
K_{calc}	Calculated heat-transfer coefficient	W/(m ² ·K)
L	Length	m
L_{shell}	Shell length	m
m	Mass	kg
m_{Total}	Total mass	kg
$\dot{m}$	Mass flow rate	kg/s or kg/h
MS	Marshall and Swift index	-
N	Number of stages	-
N_t	Number of tubes	-
N_{cycles}	Number of operating cycles	-
n	Amount of substance	kmol
n_{IPA}	Amount of IPA	kmol
n_{H_2O}	Amount of water	kmol
Nu	Nusselt number	-
Nu_w	Water-side Nusselt number	-
P	Pressure	Pa, bar, or mbar
P_1	Initial pressure	bar abs
P_2	Final pressure	bar abs
p	Pitch	m
p_t	Tube pitch	m
Pr	Prandtl number	-
Pr_w	Water-side Prandtl number	-
Q	Heat duty	W or MJ/h
Q_{max}	Maximum heat duty	W or MJ/h
Q_{reb}	Reboiler heat duty	W or MJ/h
Q_{feed}	Feed-heating energy	W or MJ/h
Q_{total}	Total energy demand	GJ
R	Reflux ratio	-
R_f	Fouling resistance	m ² ·K/W
Re	Reynolds number	-

Symbol	Meaning	Unit
r	Latent heat of vaporization	J/kg
S	Allowable stress / required pumping speed	Pa / m ³ /s
T	Temperature	K or °C
$\bar{T}$	Average temperature	K or °C
$T_{1,1}$	Hot-stream inlet temperature	K or °C
$T_{1,2}$	Hot-stream outlet temperature	K or °C
$T_{2,1}$	Cold-stream inlet temperature	K or °C
$T_{2,2}$	Cold-stream outlet temperature	K or °C
$T_{comp.}$	Compressor outlet temperature	°C
T_{HX1}	Temperature after heat exchanger 1	°C
t	Time	h or s
V	Volume	m ³
$\dot{V}$	Vapour volumetric flow rate	m ³ /s
V_{max}	Maximum required volume	m ³
V_{tank}	Selected tank volume	m ³
$V_{packing}$	Packing volume	m ³
VF	Vapour fraction	-
$VF_{comp.}$	Vapour fraction after compression	-
VF_{HX1}	Vapour fraction after heat exchanger 1	-
v	Velocity	m/s
x	Liquid mole fraction / wall thickness	- / m or mm
x_{IPA}	IPA mole fraction	-
x_{H_2O}	Water mole fraction	-
x_{DMSO}	DMSO mole fraction	-
x_{tube}	Tube wall thickness	m
x_{shell}	Shell wall thickness	m
y	Vapour mole fraction	-

Greek Symbols

Symbol	Meaning	Unit
α	Binary interaction parameter in NRTL table	Cal/mol
α_{cond}	Condensation-side heat-transfer coefficient	W/(m ² ·K)
α_w	Water-side heat-transfer coefficient	W/(m ² ·K)
$\alpha_{boiling}$	Boiling-side heat-transfer coefficient	W/(m ² ·K)
α_{shell}	Shell-side heat-transfer coefficient	W/(m ² ·K)
α_{st}	Steam condensation coefficient	W/(m ² ·K)
ΔP	Pressure jump or pressure drop	Pa or mbar
ΔP_{total}	Total pressure jump across packed section	mbar
ΔT	Temperature difference	K or °C
ΔT_1	First terminal temperature difference	K
ΔT_2	Second terminal temperature difference	K
ΔT_{lm}	Logarithmic mean temperature difference	K
ΔT_m	Mean temperature difference	K

Symbol	Meaning	Unit
ΔT_{log}	Logarithmic mean temperature difference	K
ΔC	Difference in capital cost	\$
λ	Thermal conductivity	W/(m·K)
μ	Dynamic viscosity	Pa·s
ν	Poisson's ratio	-
π	Mathematical constant pi	-
ρ	Density	kg/m ³
σ	Surface tension	N/m
∞	Infinite reflux condition	-

Subscripts, indices and Conventions

Symbol / convention	Meaning
i	First component index
j	Second component index
1	First point, side, or terminal value
2	Second point, side, or terminal value
1, 1	Hot-stream inlet
1, 2	Hot-stream outlet
2, 1	Cold-stream inlet
2, 2	Cold-stream outlet
in	Inlet or inside basis
out	Outlet or outside basis
avg	Average value
real	Real or estimated value
calc	Calculated value
clean	Clean condition without fouling
dirty	Dirty condition including fouling
cond	Condensation or condenser side
boiling	Boiling side
shell	Shell side or shell geometry
tube	Tube side or tube geometry
wall	Heat-transfer wall
film	Film condition
w	Water or cooling water
st	Steam
l	Liquid phase
v	Vapour phase
IPA	Isopropyl alcohol or isopropanol
H ₂ O	Water
DMSO	Dimethyl sulfoxide
RO	Run-off stream
Chg.	Charge
Dist.	Distillate

Symbol / convention	Meaning
L	Liquid stream
V	Vapour stream
top	Column top
bottom	Column bottom
external	External pressure condition
cr	Critical value
max	Maximum value
final	Final selected value
lm	Logarithmic mean
log	Logarithmic mean
m	Mean or corrected mean
reb	Reboiler
main	Main IPA distillation step
feed	Relating to the feed
process	Complete process
total	Total value
recovery	Recovery step
purchase	Purchase cost
random	Random packing
unit	Unit cost
packing	Packing
base	Base distillation system
pre	Pre-heater
HP	Heat-pump method
add	Additional cost
HX	Heat exchanger
HPS	High-pressure steam
comp	Compressor
int	Inter-cooler
tank	Tank
col	Column
bundle	Tube bundle
$\dot{(\)}$	Flow rate or rate with respect to time
$\overline{(\)}$	Average value
[-]	Dimensionless value
$\leq$	Less than or equal to
$\geq$	Greater than or equal to
AEL	TEMA exchanger type code
BKU	TEMA exchanger type code

Abbreviations

Abbreviation	Meaning
IPA	Isopropyl alcohol / isopropanol

Abbreviation	Meaning
DMSO	Dimethyl sulfoxide
H ₂ O	Water
NRTL	Non-random two-liquid model
BIP	Binary interaction parameter(s)
VLE	Vapour-liquid equilibrium
HETP	Height equivalent to one theoretical plate
LMTD	Logarithmic mean temperature difference
RO	Run-off
Chg.	Charge
Dist.	Distillate
HX	Heat exchanger
HX1	Heat exchanger 1
HX2	Heat exchanger 2
IC	Inter-cooler
VF	Vapour fraction
HP	Horsepower
HPS	High-pressure steam
MS	Marshall and Swift index
TEMA	Tubular Exchanger Manufacturers Association
MJ	Megajoule
GJ	Gigajoule
kW	Kilowatt
W	Watt
Pa	Pascal
mbar	Millibar
bar abs	Absolute bar pressure
mol%	Mole percent
mass%	Mass percent
kmol	Kilomole
kg/h	Kilogram per hour
m ³ /h	Cubic meter per hour
W/m ² K	Watt per square meter kelvin
m ² K/W	Square meter kelvin per watt
Cal/mol	Calorie per mole
\$/GJ	Dollars per gigajoule

Chapter 1

Introduction



Figure 1.1: Distillation columns [1].

The main goal of this research project is to investigate a batch extractive distillation process using dimethyl sulfoxide (DMSO) as a solvent to recover a distillate containing at least 99.75 mol% isopropyl alcohol (IPA). This separation is especially relevant to the pharmaceutical industry, where IPA is commonly used and is often required in very high concentrations. The mixture to be separated contains water, IPA, and a negligible amount of toluene. Dimethyl sulfoxide will be used as feed. Batch distillation cannot be used effectively for this regeneration process, since water and IPA form a homogeneous minimum-boiling azeotrope at approximately 87–88 mass% IPA and 80.2–80.4 °C [2].

Furthermore, the project aims to analyze how operational and geometrical parameters affect the performance of the distillation process and the column dimensions required for efficient operation. In addition, the study also examines different strategies for improving solvent and material recovery, reducing waste, and increasing the energetic efficiency of the process.

Extractive distillation involves the continuous addition of a usually low-volatility extra component to a mixture when separation by conventional distillation is not possible. This additional component increases the relative volatility difference between the components to be separated, thereby allowing the azeotrope to be bypassed. It differs from azeotropic distillation, in which

the third component, often referred to as an entrainer or extractive agent, forms a new low-boiling azeotrope with one of the original components and is recovered from the distillate rather than from the bottom of the column [3].

DMSO was selected as the solvent for this process because of its very high boiling point, which is significantly greater than that of both water and IPA. It is a stable, cost-effective, and non-hazardous solvent with a strong affinity for water, while also being easily recyclable; most importantly, it has a favorable effect on the relative volatility. These properties make it suitable for extractive distillation processes where high-purity IPA recovery is required [2].

For this project, ChemCAD modeling software was used to simulate the batch extractive distillation process, predict phase compositions and mixture behavior, obtain numerical data, and optimize the operating conditions. Modeling is essential because it allows the separation process to be evaluated and improved without the need to perform numerous time-consuming laboratory experiments. Several ChemCAD simulations were carried out to study not only the original separation process, but also different recycling and energy-saving configurations.

Further simulations were conducted to investigate the possibility of recycling off-cut material. In one approach, off-cut material from a previous batch was added directly to the new charge, while the remaining DMSO recovered from the previous simulation was reused as the entrainer. In another approach, the IPA-water off-cut material was mixed with the DMSO-water off-cut product from the previous distillation, and the resulting mixture was added to the new charge. These simulations were performed to evaluate whether valuable material could be recovered and reused while maintaining the required IPA purity in the distillate.

Energy-saving alternatives were also investigated. One simulation considered preheating the feed using vapor from the top of the distillation column, with the aim of reducing the energy required to heat the feed before separation. A further dynamic simulation was performed in which the top vapor was compressed and its heat was used in the reboiler to heat the charge. This configuration was studied as a potential heat-integration strategy to decrease the overall energy demand of the batch distillation operation.

In addition to the process simulations, the two main heat exchangers of the system, namely the condenser and the reboiler, were designed. Their main dimensions, number of tubes, wall thicknesses, and suitable exchanger types were determined. Finally, a cost estimation was carried out for the complete distillation column, including the main equipment required for operation. This provides a broader technical and economic evaluation of the proposed separation process.

Among the benefits of this distillation strategy are its versatility, reduced waste generation, improved solvent recovery, scalability, and potentially lower cost compared with alternative separation methods. Since DMSO can be applied in a wide range of pharmaceutical processes and enables the recovery of high-purity IPA from azeotropic mixtures, the results of this study may provide a sustainable and industrially relevant solution for handling IPA-water mixtures.

Chapter 2

Theoretical Background

2.1 Batch Extractive Distillation

In batch extractive distillation, a mixture made up of a more volatile and a less volatile component is heated up to achieve the component separation, by causing the more volatile component to evaporate in a higher concentration and then condense for its recovery. This process is done in batch, where only a portion of the mixture is treated in a single cycle; a fixed amount of it is to be separated at a time [4].

The distillate of the mixture is separated from the less volatile substances that are left in the residue or at the bottom of the tanks, by being brought back to its liquid phase by a condenser and then collected. However, part of this condensate will return to the column as reflux to improve the mass transfer between phases and therefore the efficiency of the column, at a ratio called reflux ratio (R). It is the quotient of the liquid used as reflux and the total liquid withdrawn (distillate). Under constant reflux ratio, the composition of the distillate will steadily change (the more volatile component's mole fraction will decrease in the distillate) as the procedure progresses.

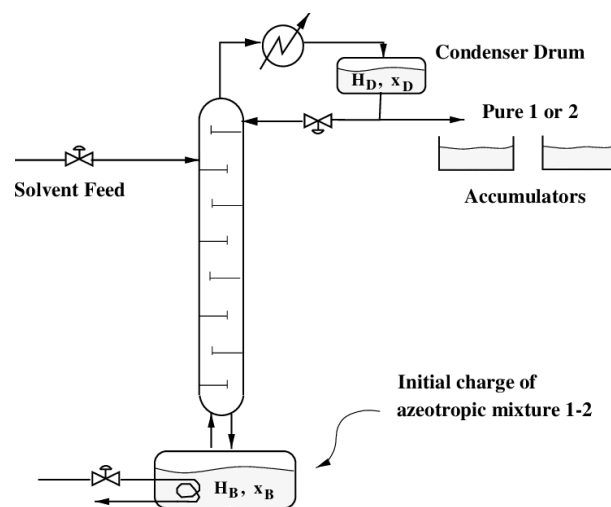


Figure 2.1: A diagram illustrating batch extractive distillation [5].

2.2 Azeotropes

An azeotrope is a specific mixture composition where the liquid and vapour phases in vapour–liquid equilibrium have the same composition. In ordinary distillation, the vapour phase is usually richer in the more volatile component than the liquid phase, which allows separation to occur. At the azeotropic composition, however, the vapour produced during boiling has the same composition as the liquid, meaning that no further enrichment of one component is achieved by simple vaporization and condensation. As a result, conventional distillation cannot separate the mixture beyond the azeotropic composition without changing the operating conditions or introducing an additional separating agent.

Both phases meet on the main diagonal line of the equilibrium curve diagram; this would mean that $x = y$ (Figure 2.2). This condition prevents any more separation from happening, since the equilibrium reached eliminates the gradient needed for further change in the concentrations [4]. In the case, the azeotrope presents at a mass fraction of 87-88% and a temperature of 80.2 – 80.4 °C. This is due to the interactions between the molecules (see Table 3.1) of each component in the solution.

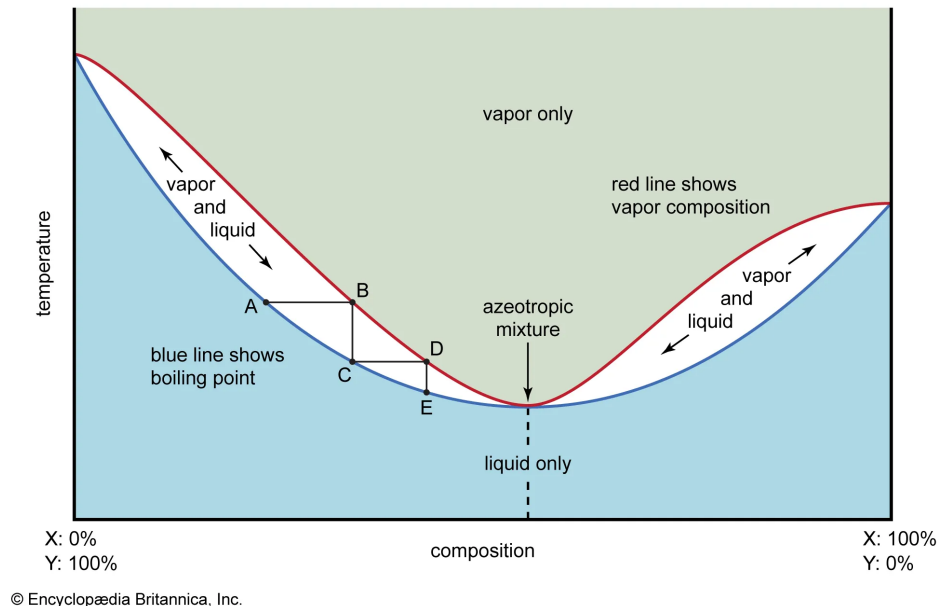


Figure 2.2: Visual representation of an azeotropic mixture [6].

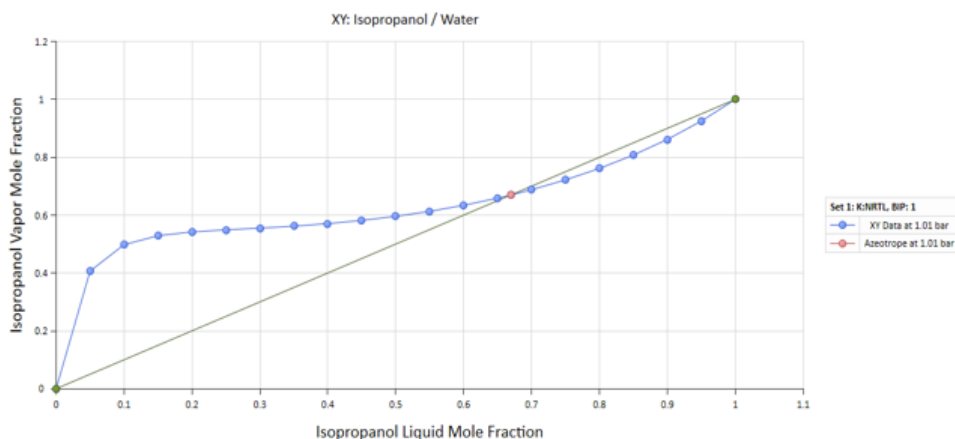


Figure 2.3: Azeotrope present in the water/IPA mixture; obtained from ChemCAD.

2.3 Material Recovery in Extractive Distillation

Material recovery is an important part of extractive distillation, since the process requires the addition of a third component. The purpose of this component is to change the relative volatility of the original components and make the separation possible. However, because the solvent is not the desired final product, it should be recovered and reused whenever possible. This reduces the amount of fresh solvent required, lowers operating costs, and decreases waste formation. In extractive distillation, the solvent is usually chosen to have a much higher boiling point than the components being separated, so it is mostly recovered from the bottom product rather than from the distillate [7].

In batch extractive distillation, the recovery of materials is especially relevant because the composition of the liquid in the still pot changes continuously with time. As the operation progresses, different fractions may be obtained: some fractions may meet the desired purity specification, while others may contain valuable amounts of IPA, water, or DMSO but not be pure enough to be considered final products. These intermediate or run-off streams can often be recycled to a later batch or treated in a separate recovery step.

For this project, material recovery is studied through the recycling of run-off material and the recovery of DMSO from previous distillation residues. Since DMSO acts as the extractive solvent, its recovery is especially necessary for improving the economic and environmental performance of the process.

2.4 Heat Exchangers

Heat exchangers are devices used to transfer heat between two fluids at different temperatures. In most industrial heat exchangers, the fluids do not mix directly, but are separated by a solid wall through which heat is transferred. The heat transfer process usually involves convection from the hot fluid to the wall, conduction through the wall, and convection from the wall to the cold fluid. Heat exchangers are widely used in chemical and mechanical engineering for heating, cooling, condensation, evaporation, and heat recovery [8]. The most common configuration in the case of condensers and boilers is the shell and tube, single-pass heat exchanger; see Figure 2.4 for an illustration.

In distillation systems, the two most important heat exchangers are the condenser and the reboiler. The condenser removes heat from the vapor leaving the top of the column and converts it into liquid. Part of this liquid may be withdrawn as distillate, while the remaining part is returned to the column as reflux. The reboiler, on the other hand, supplies heat to the liquid at the bottom of the column, producing vapor that rises through the column and allows mass transfer between vapor and liquid phases.

The design of a heat exchanger depends mainly on the required heat duty, the temperature difference between the fluids, the heat transfer coefficient, and the available heat transfer area. A common design equation is:

$$Q = kAF\Delta T_{lm} \quad (2.1)$$

where Q is the heat duty, k is the overall heat transfer coefficient, A is the heat transfer area, ΔT_{lm} is the logarithmic mean temperature difference, and F is a correction factor. In this project, this theory is applied to the design of the distillation column's condenser and reboiler.

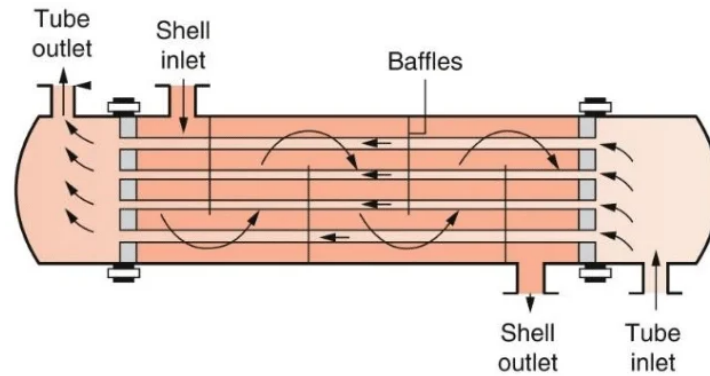


Figure 2.4: Diagram of a single-pass shell and tube heat exchanger [9].

2.5 Heat Recycling in the Industry

Heat recycling, also called heat recovery, is the reuse of heat that would otherwise be rejected from a process. In many industrial systems, some streams must be cooled while others must be heated. Instead of using only external utilities, such as cooling water and steam, heat can be transferred from hot process streams to cold process streams. This reduces the energy demand of the plant and improves overall process efficiency. [10].

In distillation, heat recycling is especially important because the process usually requires significant energy input in the reboiler and significant heat removal in the condenser. One simple form of heat recycling is feed preheating, where heat from a hot stream, such as overhead vapor or bottom product, is used to increase the temperature of the feed before it enters the column. This decreases the amount of heat that must later be supplied by the reboiler.

A more advanced method is vapor recompression. In this method, the vapor leaving the top of the column is compressed, which increases its temperature and pressure. The compressed vapor can then be used as a heating medium in the reboiler. The goal is for the net output thermal energy of the recycling system to be higher than the required input energy.

Chapter 3

Mixture Composition

3.1 Initial Conditions

Our mixture to be separated has a volume of 3500 liters, and a composition of 66.6 mol% isopropanol (IPA), and 33.4 mol% water. The time available to complete the distillation process is 22 hours. Inside the batch distillation column, the temperature will gradually increase until it reaches the boiling point of the mixture, 80.2 °C. Our column was assigned a number of theoretical stages of $n = 12$ (not including the total condenser and the still pot). The feed stage initially is the 5th, counted from the top. As for the entrainer, dimethyl sulfoxide (DMSO) will be fed to the column at 100% purity, room temperature (25 °C) and 1.013 bar in pressure. The chosen starting feed flow rate is 5 kmol/h. These parameters are to be changed depending on the results.

Table 3.1: Binary interaction parameters.

I	J	B_{ij} [Cal/mol]	B_{ji} [cal/mol]	α [cal/mol]
Isopropanol	Water	20.0554	832.981	0.3255
Isopropanol	Dimethyl Sulfoxide	115.279	-25.0123	0.4
Water	Dimethyl Sulfoxide	1203.77	-524.822	0.6615

Table 3.1 shows the binary interaction parameters (BIPs) for the mixture, which are the coefficients to be used in the NRTL thermodynamic model to describe the interactions between pairs of molecules in the mixture. The BIPs describe the excess Gibbs free energy of mixing [11].

The process will start with total reflux ($R = \infty$) and after the steady-state is reached, it will be working with an initial reflux ratio of $R = 6$, subject to change as the analysis progresses. In the same manner, It will start with an initial feed temperature of 20°C and 12 theoretical stages for the column.

3.2 Residue Curves

Residue curves (Figure 3.1) depict how the liquid composition in the distillation still-pot evolves over time during simple batch distillation at 1.01 bar. Each vertex of the triangle repre-

sents a pure component, and for the IPA-water-DMSO system the curves start at the azeotrope, showing the need for DMSO to carry out the separation. They, too, illustrate the separation trajectory: the concentration of DMSO increases as time progresses for every case, revealing that the distillation of this mixture will generally lead to DMSO accumulation in the pot, while IPA and water are removed. Similarly, the diagram tells us that DMSO forms no azeotropes with the other two components.

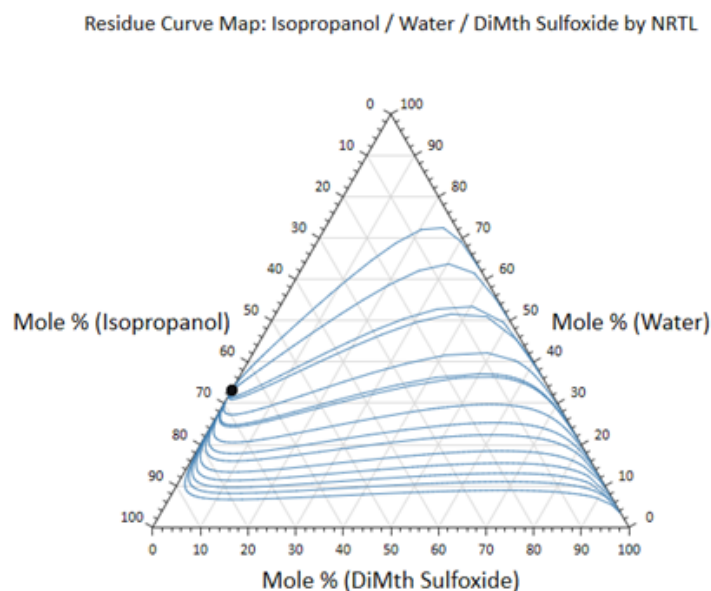


Figure 3.1: Residue curves of the mixture, obtained from ChemCAD.

3.3 Effect of DMSO on the Azeotrope

The altered activity coefficients increase the relative volatility of IPA compared to water. This shifts the VLE curve, eliminating the original azeotropic point where $x = y$. The high boiling point of DMSO (189°C) ensures it mostly remains in the liquid phase, accumulating in the column's extractive section. This prevents IPA-water azeotrope reformation and allows IPA to distill overhead. As it can be seen in Figure 3.2, The IPA-water-DMSO system no longer exhibits an azeotrope, enabling distillation to achieve >99.75% IPA purity.

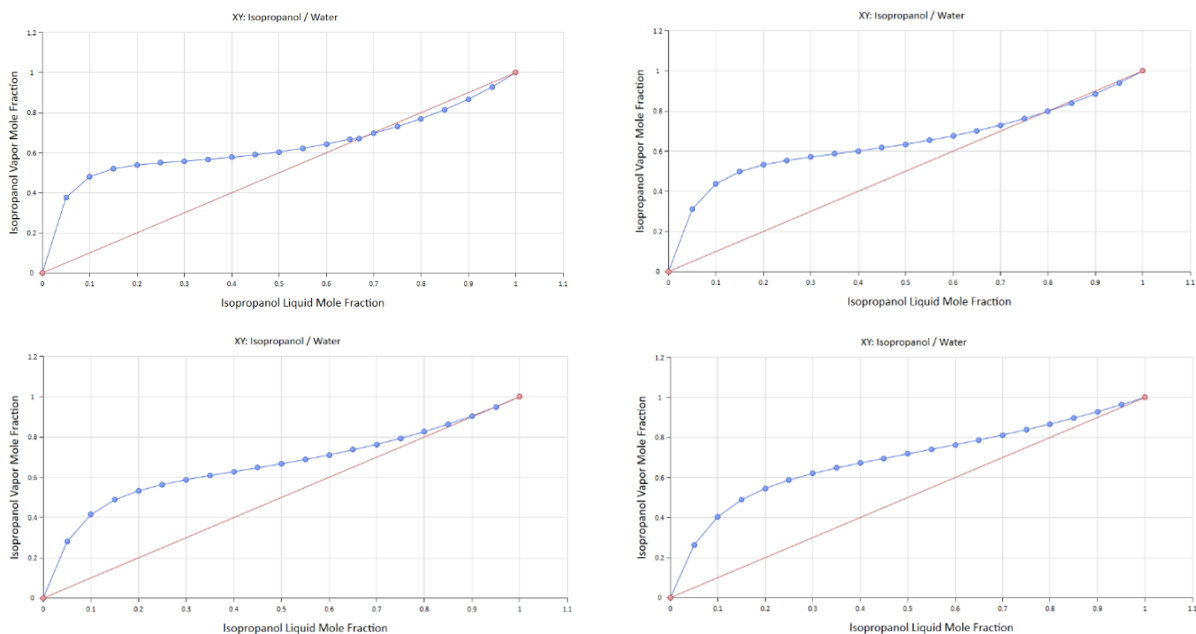


Figure 3.2: Effect of the DMSO concentration in the charge composition on the azeotrope, at 0.01, 0.05, 0.1 and 0.2 mole fraction, respectively.

The VLE diagrams show that the addition of DMSO to a solvent-free concentration changes the phase-equilibrium behavior of the IPA-water system. Without an entrainer, the IPA-water mixture forms an azeotrope, which limits the maximum IPA purity attainable by ordinary distillation. When DMSO is added, the equilibrium curve is shifted away from the diagonal, meaning that the vapour and liquid compositions are no longer equal at the original azeotropic point. This confirms that DMSO successfully changes the relative volatility of IPA and water, allowing IPA to be concentrated beyond the azeotropic composition. The effect becomes more visible as the DMSO mole fraction increases, showing that the entrainer is essential for achieving the required high-purity IPA distillate.

Chapter 4

Parameter Effects

4.1 Targets

In this chapter, only the IPA recovery step, or the dehydration of isopropanol, is studied. The distillate's solution must be ≥ 99.75 mol% IPA, ≤ 0.25 mol% water, and ≤ 0.15 mol% dimethyl sulfoxide (DMSO). In the water recovery step, the water will be withdrawn as distillate. In the water obtained, less than 0.05 mol% IPA should be present, and less than 0.035 mol% DMSO. After the whole process is finished, the remaining DMSO from the bottom will be collected from the column; the following purity requirements will be used: ≥ 99.75 mol% DMSO, ≤ 0.25 mol% water, and ≤ 0.15 mol% IPA.

4.2 Parameters studied

- Feed stage [-]
- Reflux ratio [-]
- Number of stages [-]
- Feed flow rate [kmol/h]
- Feed Temperature [$^{\circ}\text{C}$]

Dependent variables:

- Mass of the IPA product [kg]
- Time [h]
- Molar fraction (purity) of the product [-]

Goals:

- Maximum mass
- Minimum time
- Meeting purity requirements

Constant values:

- Volume of charge: 3500 L
- Charge temperature: 80°C
- Reboiler duty: 1000 MJ/h
- Charge composition:

- 0.66 mole fraction IPA
- 0.34 mole fraction water
- Feed composition: 1 mole fraction DMSO

We have adhered to an interval of 15-30 for the number of stages, 5-10 kmol/h for the feed rate, 4-8 for the feed stage, 50-70 °C for the feed temperature, and 4-6 for the reflux ratio, as these are common industrial values that have proved to have a good efficiency of separation in literature [2], [3]. A quite difficult, azeotropic separation calls for several stages and a high reflux ratio.

The stopping criteria selected was a purity of 0.9998 mol fraction IPA in the accumulator, trending downwards; this ensures that once the purity of IPA in the distillate contained in the accumulator tank starts to fall below 98 mol%, the process will terminate.

The IPA purity results and total elapsed time of the last successful simulation attempts tried with different values for the feed stage, reflux ratio, feed rate, stage number and feed temperature are shown in Tables 4.1, 4.2, 4.3, 4.4 and 4.5, respectively. Data was obtained from sensitivity analyses. The final parameters were chosen through an iterative process, in which a sensitivity analysis is conducted, parameters were changed to achieve optimal results, and another sensitivity analysis is carried out until results are satisfactory.

Table 4.1: Feed Stage Data

Trial	Feed Stage	Mole Fraction of IPA	Time (h)	Mass (kg)
1	1	0.727319	8.45001	3509.90
2	3	0.997528	9.20002	2380.31
3	6	0.997400	9.20002	2381.18
4	9	0.997040	9.20002	2380.57
5	11	0.997273	9.15002	2370.28
6	14	0.995347	8.50001	2225.38
7	17	0.986126	8.15001	2132.24
8	19	0.970703	7.90001	2051.91
9	22	0.904141	8.20001	2021.87
10	25	0.717659	13.55	2696.94

Which can be plotted as:

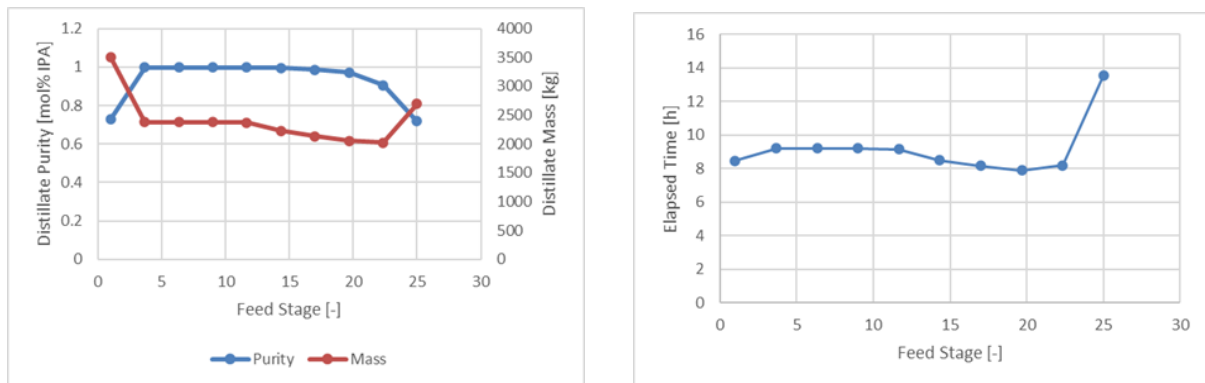


Figure 4.1: Feed stage effect on purity, mass and time, respectively.

Table 4.2: Reflux Ratio Data

Trial	Reflux Ratio	Mole Fraction of IPA	Time (h)	Mass (kg)
1	3	0.996872	16.15	2393.41
2	4	0.997535	18.05	2406.52
3	5	0.997973	19.95	2413.92
4	6	0.997836	21.90	2424.64
5	7	0.997399	23.7999	2426.82

Which can be plotted as:

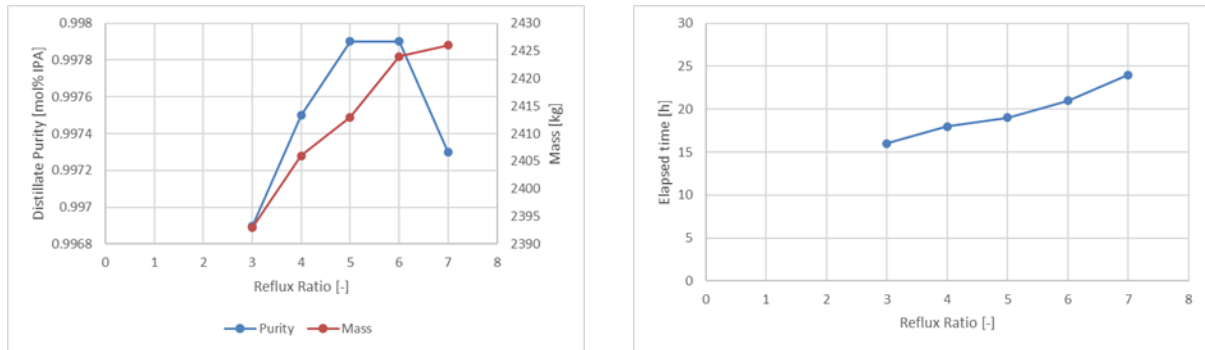


Figure 4.2: Reflux ratio effect on purity, mass and time, respectively.

Table 4.3: Feed Rate Data

Trial	Feed Rate (kmol/h)	Mole Fraction of IPA	Time (h)	Mass (kg)
1	5	0.987313	8.60001	2375.32
2	6.11111	0.995814	8.65001	2356.56
3	7.22222	0.997105	8.95002	2381.33
4	8.33334	0.996503	9.20002	2387.23
5	9.44445	0.997112	9.40002	2380.46
6	10.5556	0.996768	9.65002	2379.43
7	11.6667	0.996776	9.90002	2375.50
8	12.7778	0.997171	10.15	2368.91
9	13.8889	0.996996	10.45	2367.03
10	15	0.997244	10.75	2362.07

Which can be plotted as:

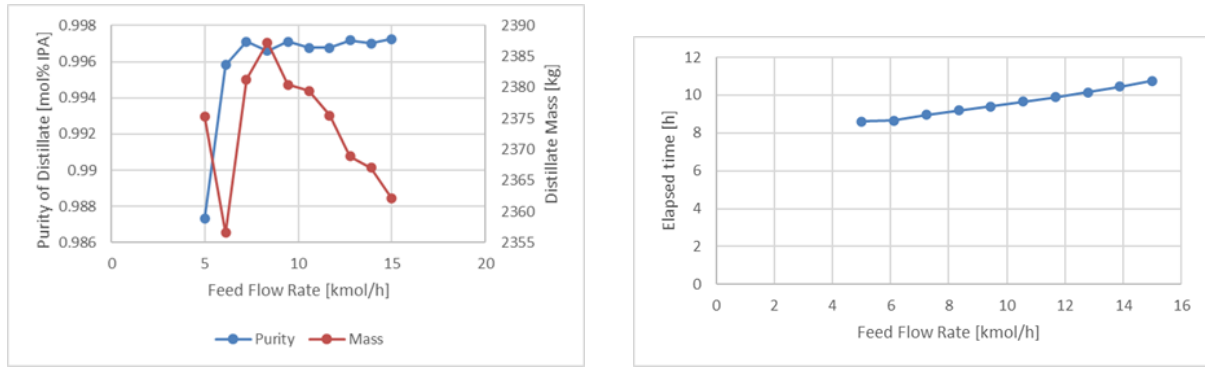


Figure 4.3: Feed rate effect on purity, mass and time, respectively.

Table 4.4: Stage Number Data

Trial	Stage Number	Mole Fraction of IPA	Time (h)	Mass (kg)
1	10	0.907955	8.15001	2016.42
2	12.2222	0.958189	7.85001	2022.11
3	14.4444	0.980373	8.05001	2100.75
4	16.6667	0.990602	8.30001	2173.13
5	18.8889	0.995448	8.50001	2225.54
6	21.1111	0.997299	9.15002	2370.32
7	23.3333	0.997048	9.20002	2380.59
8	25.5556	0.997349	9.20002	2381.05
9	27.7778	0.997496	9.20002	2381.28
10	30	0.997590	9.20002	2381.42

Which can be plotted as:

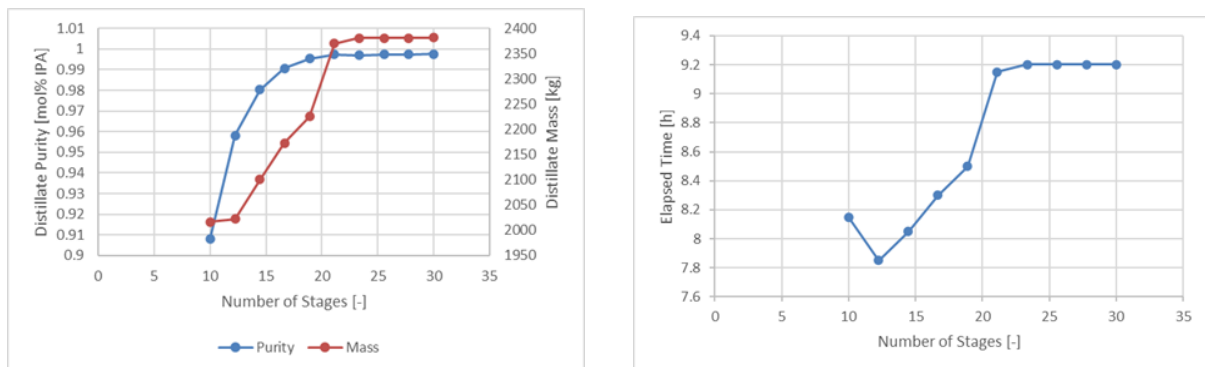


Figure 4.4: Stage number effect on purity, mass and time, respectively.

Table 4.5: Feed Temperature Data

Trial	Feed Temperature (°C)	Mole Fraction of IPA	Time (h)	Mass (kg)
1	20	0.996916	9.90002	2391.57
2	32	0.996842	9.70002	2390.15

Trial	Feed Temperature (°C)	Mole Fraction of IPA	Time (h)	Mass (kg)
3	44	0.996943	9.50002	2387.46
4	56	0.997194	9.30002	2383.51
5	68	0.997534	9.10002	2378.27
6	80	0.996955	8.95002	2381.23

Which can be plotted as:

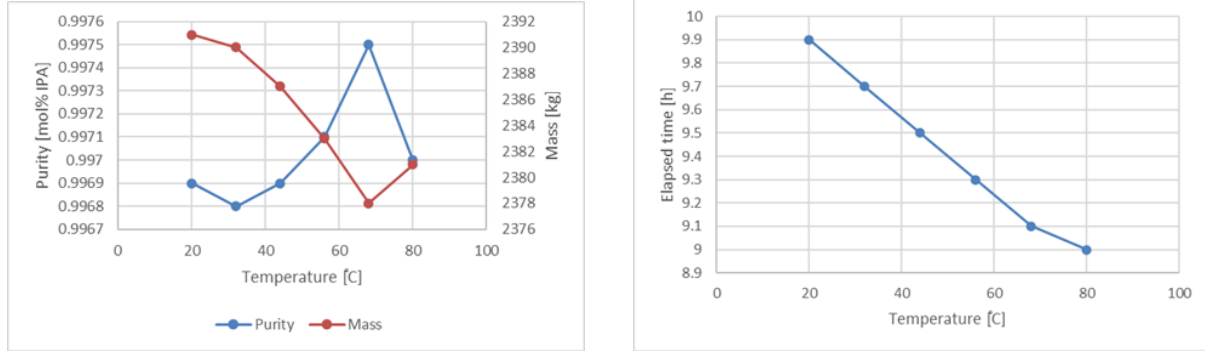


Figure 4.5: Feed temperature effect on purity, mass and time, respectively.

4.3 Conclusion on Parameter Effects

The sensitivity analyses show that each parameter affects the IPA distillate purity, recovered mass, and elapsed time through different mechanisms: The feed stage strongly influences the contact between the DMSO feed and the internal vapour-liquid flows (Figure 4.1). If the feed is introduced in too high of a stage, it starts appearing in the distillate; if it is introduced too low, it cannot modify the relative volatility across enough of the column. For this reason, adequate results occur at intermediate feed stages, while the extremes show clear drops in purity and, in one case, a strong increase in time. The reflux ratio improves separation by returning more condensed liquid to the column, increasing vapour-liquid contact. However, after a certain point, higher reflux gives only small purity improvements while greatly increasing the operating time. The purity peak at the middle reflux values is an indicator of the fittest option (Figure 4.2). The feed flow rate mainly controls how much DMSO is available to break the IPA-water azeotropic behavior. Too low a flow rate gives insufficient solvent action, while too high a flow rate increases the elapsed time and can slightly reduce the recovered mass (Figure 4.3); therefore, an intermediate feed rate is preferred. Increasing the number of stages improves purity and mass because more equilibrium contacts are available, but the effect reaches a plateau at higher stage numbers, meaning that additional stages give diminishing returns and would mainly increase column size and cost (Figure 4.4). Finally, increasing feed temperature reduces the elapsed time because the feed enters closer to column conditions, but excessive heating can disturb the internal temperature and composition profiles (Figure 4.5). The purity peak occurs at a moderately high feed temperature, while the recovered mass tends to decrease as temperature rises. Overall, the selected parameter values should represent a compromise between high purity, high recovered mass, and low elapsed time, rather than the maximum value of only one dependent variable.

Table 4.6: Selected parameter values.

Feed Stage	Reflux Ratio	Feed Flow Rate	Number of Stages	Feed Temperature
4	4	7.5 kmol/h	24	62 °C

These final parameters displayed in Table 4.6 were chosen taking maximum efficiency into consideration; they produce balanced results regarding purity, elapsed time and mass. We selected 24 stages for the column because, even though the elapsed time is high, it will result in sufficient purity and a great amount of mass. A feed stage of 4 gives us a good balance between distillate purity and mass, while keeping time at a minimum. As for the selected feed flow rate, a value between 7 and 8 kmol/h will be suitable, since it was possible to complete the process at a relatively short amount of time and obtain a satisfactory amount of distillate mass, with high purity. Therefore, 7.5 kmol/h is acceptable, or 586 kg/h. A reflux ratio of 6 would give us the best results according to the graph; however, the time and the energy it needs are significantly greater. A reflux ratio of 4 gives sufficient results, while making up for time. Feed temperature is chosen to be 62°C because, even though it will not produce the highest amount of distillate possible, it will reduce the elapsed time and produce a high IPA purity.

Chapter 5

Distillation Results

A maximum purity of 99.856 mol% IPA was achieved, meeting the requirements; $0.14 \leq 0.25$ mol% water, $0.0012 \leq 0.15$ mol% DMSO; these results calculated by ChemCAD are shown in Figure 5.1. The total elapsed time was 9.05 hours.

Stream No.	3
Stream Name	distillate
Temp C	82.4424
Pres bar	1.013
Vapor Fraction	0.001214817
Enthalpy MJ/h	-12265.24
Total flow	2397.456
Total flow unit	kg/h
Comp unit	Mole %
Isopropanol	99.85659
Water	0.142225
DiMth Sulfoxide	0.001175972

Figure 5.1: Calculated properties of the IPA product.

In Figure 5.2, it is possible to observe how the composition of the distillate changed with time, under the chosen parameter values. This plot was produced by the ChemCAD simulation.

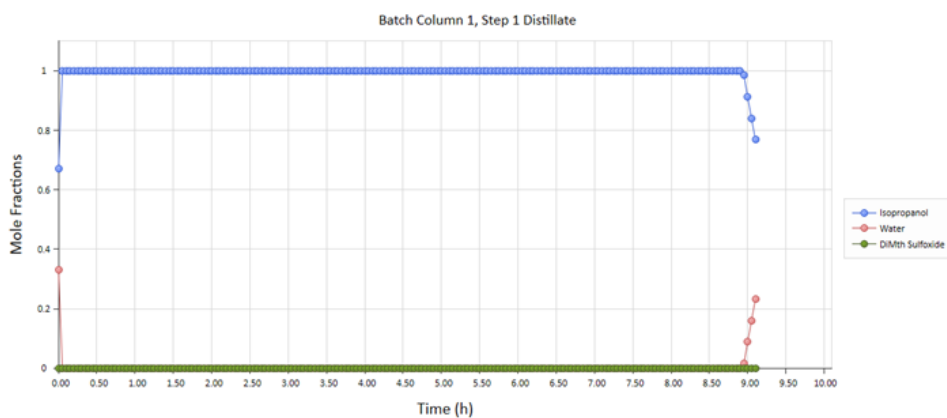


Figure 5.2: Depiction of the distillate composition's change with time.

The distillate composition graph shows that the IPA mole fraction remains very high during most of the main distillation step, while the water and DMSO mole fractions remain low. This confirms that the selected operating parameters allow IPA to be removed as the main overhead product with only small amounts of impurities. Near the end of the step, the IPA mole fraction begins to decrease while the impurity content increases, indicating that the useful IPA-rich distillation period is ending. Therefore, the stopping point was selected before a significant decrease in distillate purity occurs, in order to maintain the required product specification.

5.1 Additional Steps

After obtaining the desired product, the distillation column will still have liquid at the bottom, whose composition is still rich in all three components: water, DMSO and IPA.

In an attempt to recover as much water as possible, as well as solvent, three more steps to the process were added: a water-IPA run-off step, a water distillation step, and a water-DMSO run-off step. The new simulation layout, after adding the necessary equipment for these additional steps, is illustrated in Figure 5.3.

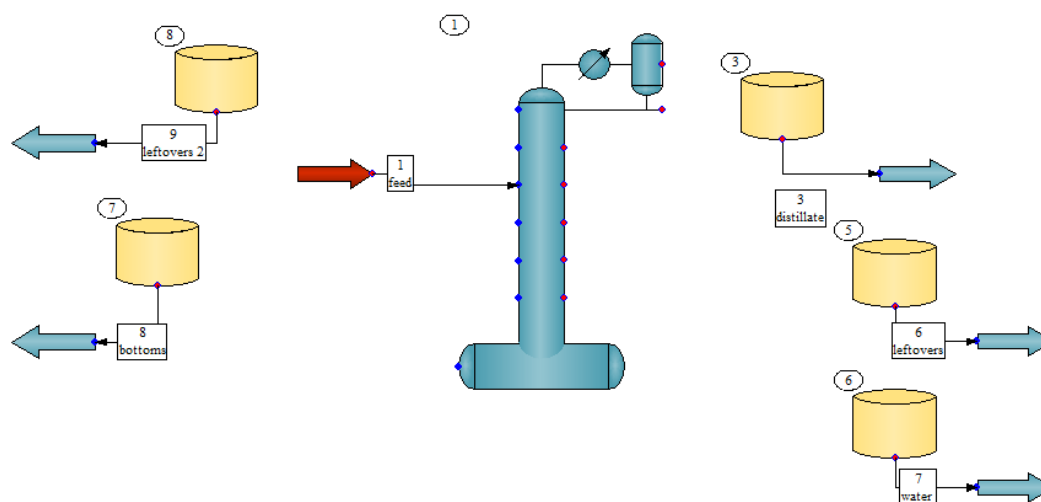


Figure 5.3: New ChemCAD simulation layout.

In the water-IPA run-off step, the goal is to dispose of all leftover IPA in the bottom, in order to produce water more efficiently in the next step. In the water distillation step, as much water as possible will be separated from the leftover mixture (less than 0.05 mol% IPA should be present, and less than 0.035 mol% DMSO). Finally, for the water-DMSO run-off step, it is attempted to leave as little water in the bottom as possible in order to recover the solvent.

In this case, extensive sensitivity analyses were not carried out, since the investigated parameters were now set to the chosen values for the whole process. Instead, the stop values for each step were set as the following:

- stop value for step 2: 0.8000 mole fraction water in the accumulator
- stop value for step 3: 0.9995 mole fraction water in the accumulator
- stop value for step 4: 0.9995 mole fraction DMSO in the still pot

The stop values were chosen according to the purpose of each additional distillation step. In step 2, the objective is not to obtain a final pure product, but to remove the remaining IPA-rich fraction before the main water recovery step. Therefore, the step is stopped when the accumulator reaches a high water content, indicating that most of the remaining IPA has already been removed from the bottom mixture. In step 3, the goal is to recover water as a final product, so a much stricter stop value is used: the accumulator must reach approximately pure water. In step 4, the objective is to recover the DMSO remaining in the still pot, so the process is continued until the bottom product reaches the required DMSO purity.

The reflux ratios for each step were chosen in a small iteration process, using values ranging from $R = 2$ to 5 [4]. After four simulations (corresponding to the four integer values selected) were carried out for each step, the following values yielded the best results in the least possible amount of time:

- R for step 2: 5
- R for step 3: 4
- R for step 4: 2

A higher reflux ratio was used in step 2 because this step handles a transition mixture containing IPA and water, and stronger internal liquid-vapour contact is needed to remove the remaining IPA effectively. Step 3 uses a moderate reflux ratio because water is the main volatile component and can be recovered at high purity without excessive reflux. Step 4 uses the lowest reflux ratio because the target product is the DMSO-rich bottom stream rather than the distillate; therefore, very high reflux is not necessary and would only increase the operating time.

This will allow us to carry out the simulation until the desired purity for both water and DMSO are reached.

The resulting graphs of the additional distillation steps are showcased in Figures 5.4, 5.5 and 5.6.

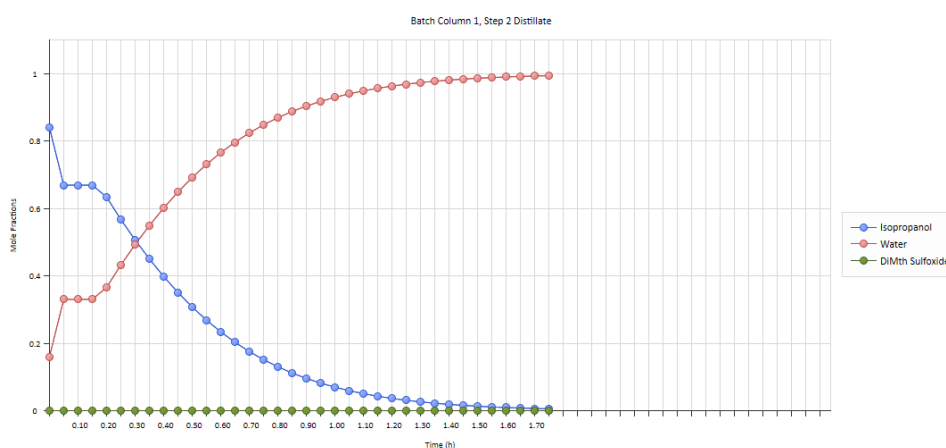


Figure 5.4: Distillate composition as a function of time for step 2.

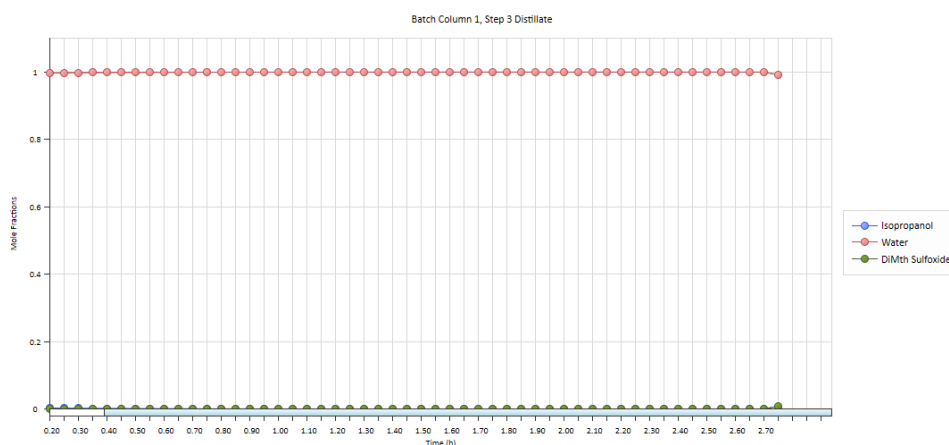


Figure 5.5: Distillate composition as a function of time for step 3.

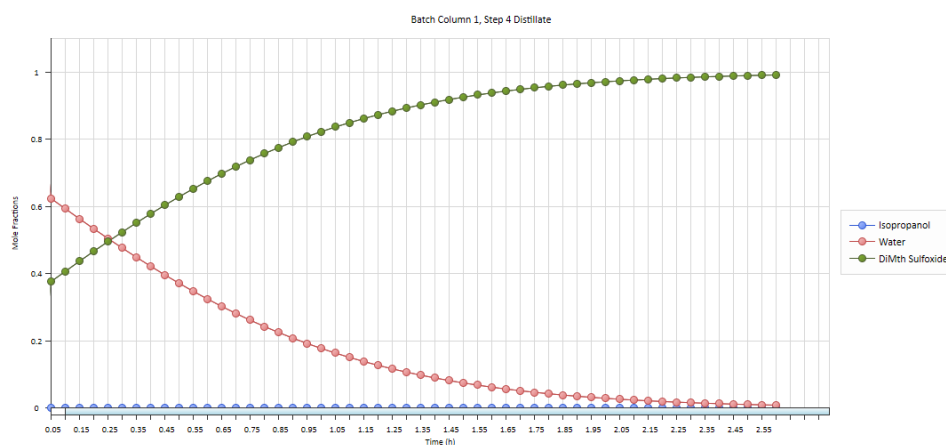


Figure 5.6: Distillate composition as a function of time for step 4.

After the process was finished, the results were:

- Water distillate:
 - 226.52 kg, 0.9994 mol fraction water, 0.0004432 mol fraction IPA, 0.0001375 mol fraction DMSO
- DMSO bottom:
 - 4060.02 kg, 0.9995 mol fraction DMSO, 0.0004904 mol fraction water, 7.226e-18 mol fraction IPA

The results are satisfying and the liquids are usable for other applications. The total elapsed time was 16.15 hours, still under the 22 hour goal. Run-off steps will later be re-integrated into the system, further explained in Chapter 9, in order to recover as much material as possible. These results support the inclusion of material recovery steps in the overall process design, as they reduce waste formation and improve the reuse potential of both water and DMSO.

Chapter 6

Column Geometry

Our next step is to calculate the main dimensions of the column; that is, its column diameter, tank diameter, volume, height and thickness. In order to do this, we will need to obtain the flow rate of the vapour phase and the density of both the vapour and liquid phases in every stage throughout the column. This data was obtained from the ChemCAD software, given by the tray analysis report.

6.1 F-factor and HETP value

The F factor is a parameter that represents the momentum of the vapour phase. From the Sulzer packing catalogue, the structured MellapakPlus 252.Y packing was chosen, as it offers a high operational efficiency and minimum hold-up for atmospheric pressure distillation [12]. It is widely used in the pharmaceutical and food industry. In the same catalogue, it is possible to see the HETP-F factor graphs for each packing type. The F-factor will be picked so that it will not cause flooding or weeping in the column; $2 \sqrt{\text{Pa}}$ was selected (See Figure 6.1).

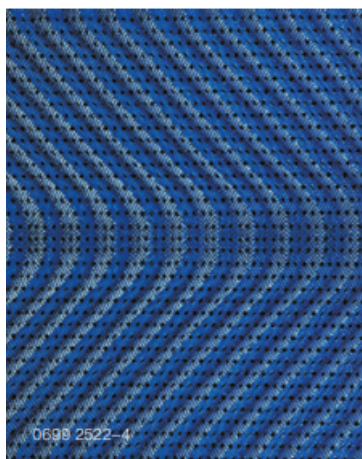


Figure 6.1: Zoomed-in view of the structured packing MellapakPlus 252.Y.

In the same catalogue section, we have a graph indicating what HETP (Height Equivalent to one Theoretical Plate) value corresponds to an F factor of 2: 0.4 m (see Figure 6.2). The blue line refers to the distillation pressure 960 mbar.

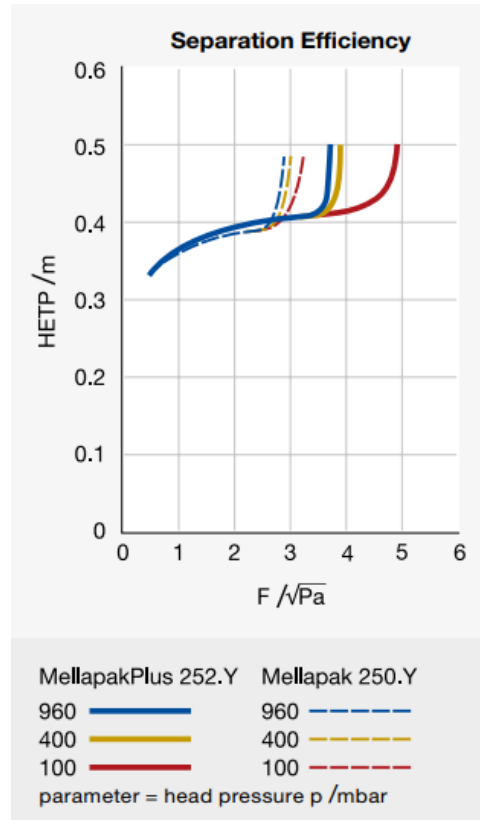


Figure 6.2: HETP as a function of the F-factor for MellapakPlus 252.Y and 250.Y.

6.2 Column Height

Since the HETP value is known, the calculation of the column height can be carried out directly. The HETP represents the height of packing required to achieve the separation effect of one theoretical stage.

The column height is calculated using the following equation [4]:

$$H = N \times HETP \quad (6.1)$$

where:

- H = packed column height, in m
- N = number of theoretical stages
- $HETP$ = height equivalent to one theoretical plate, in m

The column was chosen to have 24 theoretical stages, and the selected HETP value for the chosen packing is $HETP = 0.4$ m. Therefore, the required packed height is:

$$H = 24 \times 0.4 = 9.6 \text{ m} \quad (6.2)$$

The selected column height is 9.6 m, which can be rounded up to 10 m to allow for some extra safety space. This height is only representative of the column alone, without accounting for the reboiler, bottom tank or condenser.

6.3 Pressure Drop

The pressure drop is the decrease in pressure that occurs as the vapour phase flows through the packing of the distillation column. In a packed column, this pressure loss is mainly caused by friction between the rising vapour, the descending liquid, and the surface of the packing. It is an important design parameter because excessive pressure loss can increase the pressure and temperature at the bottom of the column, which may negatively affect the separation and increase the operating requirements of the process. For a packed distillation column, the total pressure drop can generally be expressed as [12]:

$$\Delta P_{total} = \left(\frac{\Delta P}{H} \right) \cdot H$$

where:

- ΔP_{total} = total pressure drop across the packed section, in mbar.
- $\frac{\Delta P}{H}$ = pressure drop per unit height of packing, in mbar/m.
- H = total packed height of the column, in m.

For a distillation column, the bottom pressure can then be estimated from the top pressure and the total column pressure drop:

$$P_{bottom} = P_{top} + \Delta P_{total}$$

However, in the Sulzer catalogue's MellapakPlus 252.Y packing section, a relationship between the pressure drop and the F-factor is already provided:

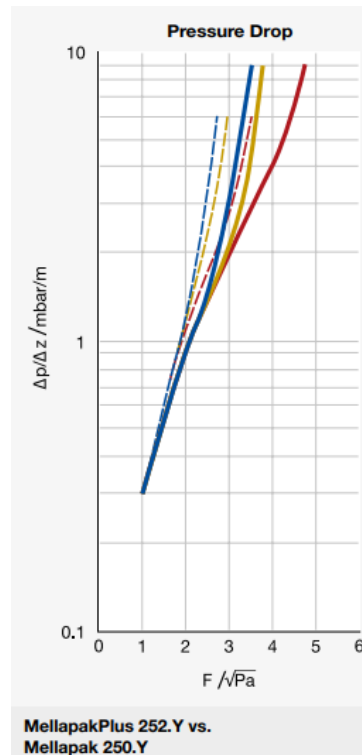


Figure 6.3: Pressure drop per meter of the column (mbar/m) as a function of the F-factor for MellapakPlus 252.Y and 250.Y.

$$\frac{\Delta P}{H} = 1 \text{ mbar/m} \quad (6.3)$$

Since the packed height of the column is $H = 10 \text{ m}$, the total pressure drop is:

$$\Delta P_{total} = 1 \text{ mbar/m} \cdot 10 \text{ m} = 10 \text{ mbar}$$

Therefore, the pressure at the bottom of the packed section is expected to be approximately 10 mbar higher than the pressure at the top of the column. The calculated pressure drop is small compared with the operating pressure of the distillation column. Therefore, the selected MellapakPlus 252.Y structured packing is considered suitable, since it provides the required separation height while maintaining a low pressure drop. The 10 mbar pressure increase across the 10 m packed section is not expected to significantly affect the vapour-liquid equilibrium or the overall column operation.

6.4 Column Diameter

In order to calculate the diameter of the column, the vapour phase velocity on each stage must first be evaluated by rearranging the F-value equation, whose value we selected: [12]:

$$F = v\sqrt{\rho_v} \quad (6.4)$$

Solving for the vapour velocity:

$$v = \frac{F}{\sqrt{\rho_v}} \quad (6.5)$$

where:

- F = F-factor, in $\sqrt{Pa}$
- v = maximum vapour velocity, in m/s
- ρ_v = vapour phase density, in kg/m^3

With the vapour velocity, the cross-sectional area of the column can be calculated by rearranging the continuity equation:

$$A = \frac{\dot{V}}{v} \quad (6.6)$$

where:

- A = cross-sectional area of the column, in m^2
- $\dot{V}$ = vapour volumetric flow rate, in m^3/s
- v = vapour velocity, in m/s

The column diameter is then calculated from the circular area equation:

$$A = \frac{\pi D^2}{4} \quad (6.7)$$

Solving for the diameter:

$$D = \sqrt{\frac{4A}{\pi}} \quad (6.8)$$

For each step of the process, the highest vapour volumetric flow rate and the lowest vapour density were considered in order to calculate the largest required column diameter, and these were obtained from ChemCAD's tray properties' reports (see Table 6.1). This approach was used for safety, since the column must be able to operate under the most demanding vapour-flow conditions.

Table 6.1: Diameter calculation results.

Simulation Steps	Vapour vol. flow rate [m ³ /s]	Vapour density [kg/m ³]	Vapour velocity [m/s]	Area [m ²]	Diameter [m]
0	0.492	1.625	1.5689	0.314	0.632
1	0.174	0.853	2.1655	0.0803	0.320
2	0.210	0.595	2.5928	0.0810	0.321
3	0.215	0.593	2.5972	0.0828	0.324
4	0.240	2.100	1.3801	0.1739	0.471

As shown in Table 6.1, the highest calculated diameter value is 0.632 m. To provide an additional safety margin, a safety factor of 20% is applied:

$$D = 0.632 \times 1.20 = 0.758 \text{ m} \quad (6.9)$$

Therefore, the final selected column diameter is 0.76 m after rounding up the second decimal, for the sake of simplicity.

6.5 Column Thickness

The wall thickness of the distillation column was calculated by treating the column as a cylindrical pressure vessel under internal pressure. For safety purposes, the design pressure was selected as 6 bar, which is higher than the expected operating pressure of the column. The column is assumed to be made of carbon steel, with an allowable stress of $S = 120 \times 10^6$ Pa and a weld joint efficiency of $E = 0.85$. [13]

The minimum required wall thickness was calculated using the cylindrical vessel equation [13]:

$$x = \frac{PR}{SE - 0.6P} \quad (6.10)$$

where:

- x = minimum required wall thickness, in m
- P = design pressure, in Pa

- R = internal radius of the column, in m
- S = allowable stress of the construction material, in Pa
- E = weld joint efficiency, dimensionless

For this calculation, the following values used can be found in Table 6.2.

Table 6.2: Parameters relevant to wall thickness.

Parameter	Symbol	Value	Unit
Design pressure	P	600000	Pa
Column radius	R	0.38	m
Allowable stress of carbon steel	S	120×10^6	Pa
Weld joint efficiency	E	0.85	-

Substituting the values:

$$x = \frac{600000 \times 0.38}{(120 \times 10^6)(0.85) - 0.6(600000)} = 0.00224 \text{ m} = 2.24 \text{ mm} \quad (6.11)$$

The calculated theoretical wall thickness is therefore 2.24 mm. However, this value only represents the minimum thickness required to withstand the selected internal design pressure. A corrosion allowance must also be included, especially because carbon steel is used as the construction material. Assuming a corrosion allowance of 3.2 mm [14]:

$$x_{\text{final}} = t + C_A = 2.24 + 3.2 = 5.44 \text{ mm}$$

The practical wall thickness is rounded up to 6 mm in order to guarantee safety.

The column is later planned to operate at a pressure of 0.1 bar. Since the outside of the column is assumed to be at atmospheric pressure, this condition creates an external pressure difference of approximately $P_{\text{external}} = 1.013 - 0.1 = 0.913 \text{ bar} = 91300 \text{ Pa}$

A preliminary external pressure check shows that the selected 6 mm wall thickness can withstand the vacuum condition. To check for shell buckling, the following expression was used [15]:

$$P_{cr} = \frac{2E}{\sqrt{3(1-\nu^2)}} \left(\frac{x}{R} \right)^3 \quad (6.12)$$

where P_{cr} is the critical external pressure, E is Young's modulus, ν is Poisson's ratio, t is the wall thickness, and R is the column radius.

Using carbon steel properties [16]:

$$E = 200 \cdot 10^9 \text{ Pa}$$

$$\nu = 0.30$$

and the selected column dimensions:

$$x = 0.006 \text{ m}$$

$$R = 0.38 \text{ m}$$

the critical external pressure is:

$$P_{cr} = \frac{2(200 \cdot 10^9)}{\sqrt{3(1 - 0.30^2)}} \left(\frac{0.006}{0.38} \right)^3 = 9.82 \cdot 10^5 \text{ Pa} = 9.82 \text{ bar} \quad (6.13)$$

Since the estimated critical external pressure is higher than the vacuum pressure difference, the selected 6 mm wall thickness is acceptable for this preliminary vacuum check.

6.6 Tank Geometry

The tank geometry was determined based on the maximum liquid volume obtained from the ChemCAD simulation reports. From the tank's liquid volume values, the maximum required liquid volume was found at the end of step 1, due to the addition of DMSO feed. This value is approximately $V_{\max} = 6 \text{ m}^3$.

In order to provide an additional safety margin for bubbling and avoid operating the tank exactly at its maximum capacity, a larger geometrical volume was selected. The tank was designed as a vertical cylindrical vessel with a height of 2.5 m and a diameter of 2.0 m.

The volume of a cylindrical tank is calculated as:

$$V = \frac{\pi D^2}{4} H \quad (6.14)$$

where:

- V = tank volume, in m^3
- D = tank diameter, in m
- H = tank height, in m

Substituting with the selected dimensions:

$$V = \frac{\pi(2.0)^2}{4}(2.5) = 7.85 \text{ m}^3 \quad (6.15)$$

Since this value is greater than the maximum required process volume of 6 m^3 , the tank geometry provides sufficient additional capacity for safe operation. The final selected tank dimensions are displayed in Table 6.3.

Table 6.3: Final Dimensions of the selected tank.

Parameter	Symbol	Value	Unit
Maximum required volume	$V_{\max}$	6.00	m^3
Selected tank diameter	D	2.00	m
Selected tank height	H	2.50	m
Selected tank volume	V_{tank}	7.85	m^3

Chapter 7

Heat Exchangers

7.1 Column Condenser Design

The condenser was designed as a counter-current, one-pass shell-and-tube heat exchanger used to condense the IPA-rich vapour leaving the top of the distillation column. A one-pass shell-and-tube heat exchanger is suitable for the column condenser because it avoids unnecessary mechanical complexity, reduces the number of flow reversals, and helps keep the pressure drop low. Since the required heat transfer area can be achieved with a simple tube bundle, this arrangement is sufficient [14].

The total heat transfer area listed in the calculation corresponds to the total available outside tube surface area. The IPA properties used for the condensation-side calculation were obtained from ChemCAD's reports, while the cooling-water properties were taken from literature [17]. These are listed below in Table 7.1.

Table 7.1: Material properties of isopropanol and cooling water.

Property	Value	Unit
thermal conductivity of IPA $\lambda_{IPA, film}$	0.1235	W/(m*K)
density of IPA $\rho_{IPA, film}$	730.616	kg/m ³
dynamic viscosity of IPA $\mu_{IPA, film}$	0.0005851	N*s/m ²
latent heat of IPA $r_{IPA, film}$	680200	J/kg
specific heat capacity of IPA, const. pressure $c_{p, IPA}$	3296.2	J/(kg*K)
temperature of IPA T_{dist}	80	°C
thermal conductivity of water λ_w	0.624	W/(m*K)
density of water ρ_w	994.1	kg/m ³
dynamic viscosity of water μ_w	0.000719	N*s/m ²
specific heat capacity of water, const. pressure $c_{p, w}$	4175.5	J/(kg*K)
temperature of inlet water $T_{w, in}$	25	°C
temperature of outlet water $T_{w, out}$	50	°C

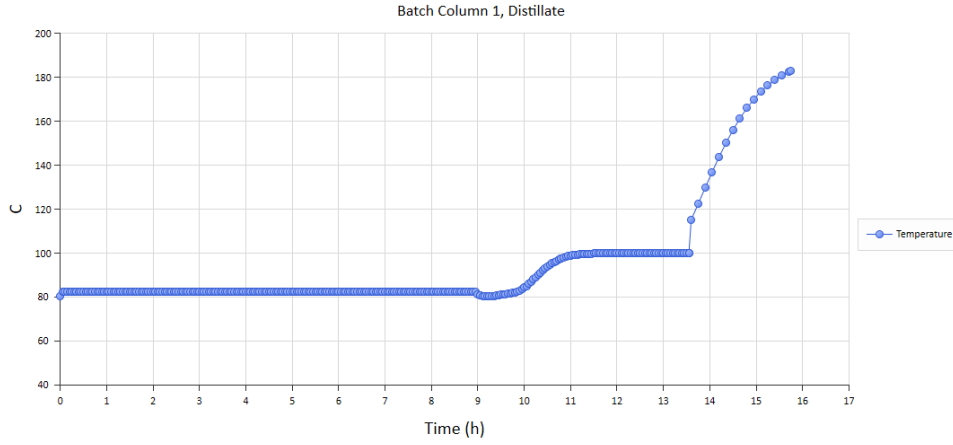


Figure 7.1: Minimum temperature of the distillate T_{dist} (80°C), which was used in the condenser calculations.

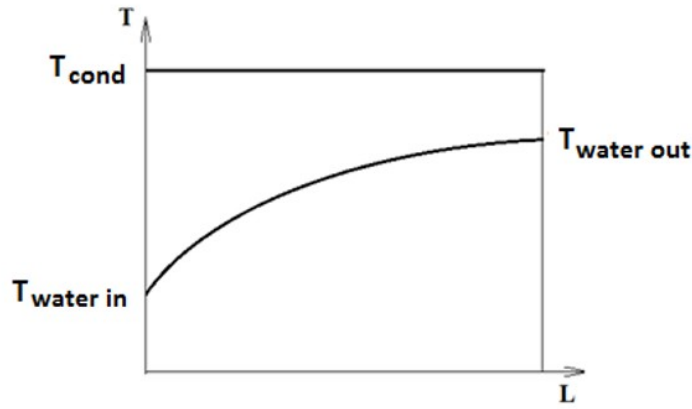


Figure 7.2: Temperature profiles in the case of a condenser, where T_{cond} is T_{dist} in this case, and the x-axis represents the heat transfer surface length [18]. Since the distillate vapour changes phase, the temperature can be assumed to be constant (non-sensible heat transfer).

The main design equation used for the condenser was [19]:

$$Q = kAF\Delta T_m \quad (7.1)$$

Rearranging for the required heat transfer area:

$$A = \frac{Q}{kF\Delta T_m} \quad (7.2)$$

where Q is the condenser heat duty, k is the overall heat transfer coefficient, A is the total heat transfer area, F is the correction factor and ΔT_m is the mean temperature difference. Since the temperature difference between the condensing vapour and cooling water changes along the exchanger, the logarithmic mean temperature difference was used:

$$\Delta T_{log} = \frac{\Delta T_1 - \Delta T_2}{\ln \left(\frac{\Delta T_1}{\Delta T_2} \right)} \quad (7.3)$$

The IPA vapour was assumed to condense at constant temperature since we are dealing with a phase change, while the cooling water was assumed to be heated up from 25°C (room temperature) to 50°C. Therefore:

$$\Delta T_1 = T_{IPA} - T_{w,in} = 353.15 - 298.15 = 55 \text{ K}$$

$$\Delta T_2 = T_{IPA} - T_{w,out} = 353.15 - 323.15 = 30 \text{ K}$$

Thus:

$$\Delta T_{log} = \frac{55-30}{\ln \left(\frac{55}{30} \right)} = 41.24 \text{ K}$$

The correction factor was calculated:

$$F = 1$$

From the formulas indicated in the correction factor chart for a single shell pass heat exchanger (Figure 7.3).

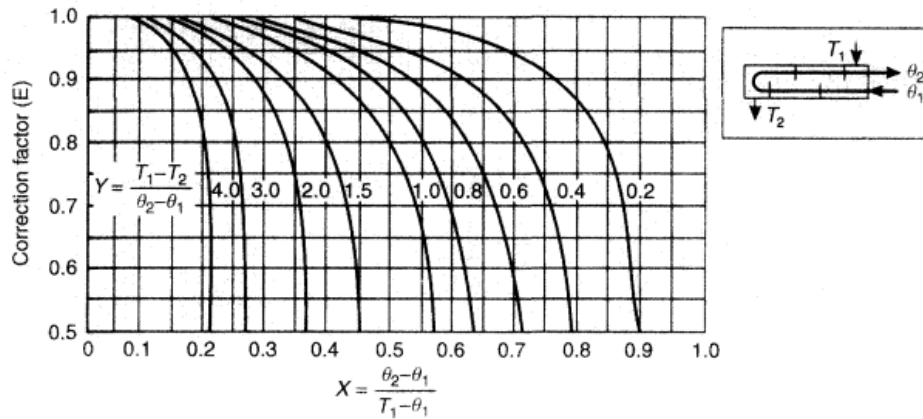


Figure 7.3: Correction factor for a single shell pass heat exchanger [14].

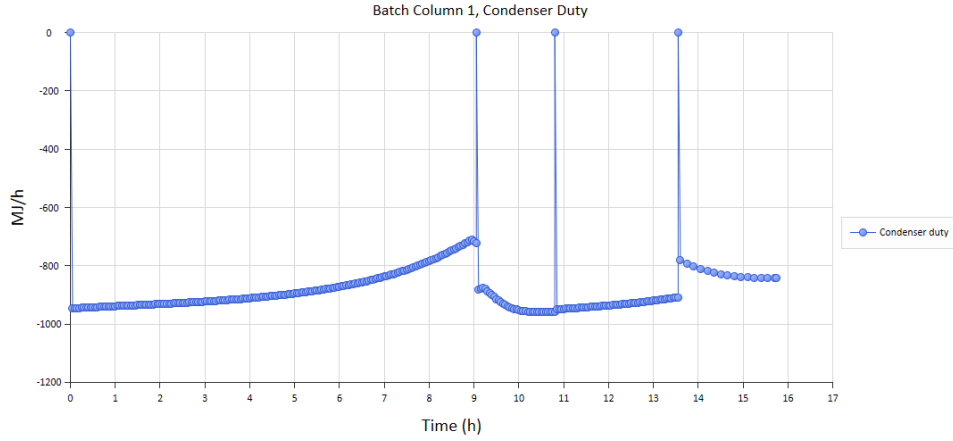


Figure 7.4: Graph indicating the maximum condenser heat duty (minimum value in the plot, since heat duty is negative) during the process (957.7 MJ/h).

The highest condenser heat duty obtained from the simulation was $Q = 957.7$ MJ/h. Converting to watts:

$$Q = \frac{957.7 \times 10^6}{3600} = 266027.78 \text{ W}$$

Using the initially assumed overall heat transfer coefficient for phase change $k = 850 \text{ W/m}^2\text{K}$ [20], the first estimated heat transfer area was:

$$A = \frac{266027.78}{850 \times 41.24} = 7.59 \text{ m}^2$$

This value was used only as an initial estimate. A corrected design was then obtained by calculating the individual heat transfer coefficients, the wall resistance, and the fouling resistance.

The clean overall heat transfer coefficient was calculated from the thermal resistance relationship [18]:

$$\frac{1}{k_{clean}} = \frac{1}{\alpha_{w,out}} + \frac{x_{tube}}{\lambda_{wall}} + \frac{1}{\alpha_{cond}} \quad (7.4)$$

where $\alpha_{w,out}$ is the cooling-water heat transfer coefficient converted to the outside tube area, x_{tube} is the tube wall thickness, λ_{wall} is the thermal conductivity of the carbon-steel wall, and α_{cond} is the condensation-side heat transfer coefficient.

As it is a widely used, affordable material, it was decided to use carbon steel as the main material for the heat exchangers' tubes and shell walls. The wall thermal conductivity (the thermal conductivity of carbon steel) was then taken to be $\lambda_{wall} = 45 \text{ W/mK}$ [18].

The water-side Reynolds number was calculated using [19]:

$$Re = \frac{\rho_w v D_{in}}{\mu} \quad (7.5)$$

The Prandtl number was calculated using [19]:

$$Pr = \frac{\mu C_{p,w}}{\lambda} \quad (7.6)$$

The Nusselt number used was [19]:

$$Nu = 0.3Re^{0.5}Pr^{1/3} \quad (7.7)$$

The water-side heat transfer coefficient was then calculated from [19]:

$$\alpha_{w,in} = \frac{Nu\lambda_w}{D_{in}} \quad (7.8)$$

Since the total heat transfer area was based on the outside tube surface, the water-side coefficient was converted to the outside area basis:

$$\alpha_{w,out} = \alpha_{w,in} \frac{D_{in}}{D_{out}} \quad (7.9)$$

The calculated condenser heat transfer coefficients can be found in Table 7.2.

Table 7.2: Condenser heat transfer coefficients.

Parameter	Symbol	Value	Unit
Condensation-side coefficient	α_{cond}	1995.01	W/m ² K
Water-side coefficient, inside basis	$\alpha_{w,in}$	1514.09	W/m ² K
Water-side coefficient, outside basis	$\alpha_{w,out}$	1211.27	W/m ² K
Clean overall coefficient	k_{clean}	729.25	W/m ² K
Fouling resistance	R_f	0.0005	m ² K/W
Dirty overall coefficient	k_{dirty}	534.40	W/m ² K

Fouling resistance was obtained from literature [18].

The dirty overall heat transfer coefficient was calculated by adding the fouling resistance:

$$\frac{1}{k_{dirty}} = \frac{1}{k_{clean}} + R_f \quad (7.10)$$

$$k_{dirty} = 534.40 \text{ W/m}^2\text{K}$$

Using the dirty overall heat transfer coefficient, the final required heat transfer area was:

$$A_{calc} = \frac{Q}{k_{dirty}\Delta T_{log}}$$

$$A_{calc} = \frac{266027.78}{534.40 \times 41.24} = 12.07 \text{ m}^2$$

Therefore, the required total condenser heat transfer area is 12.07 m^2

The selected condenser geometry is shown in Table 7.3.

Table 7.3: Selected and calculated condenser geometry.

Parameter	Symbol	Value	Unit
Number of tubes	N	48	-
Tube outside diameter	D_{out}	0.020	m
Tube inside diameter	D_{in}	0.016	m
Tube wall thickness	x_{tube}	0.002	m
Tube pitch	p_t	0.025	m
Tube length	L	4.002	m
Bundle diameter	D_{bundle}	0.232	m
Shell clearance	C_{shell}	0.011	m
Shell inside diameter	$D_{shell,in}$	0.243	m
Shell wall thickness	x_{shell}	0.0041	m
Shell length	L_{shell}	4.028	m
Baffle spacing	-	0.049	m

The tube outside diameter, tube thickness, tube pitch and shell clearance values were within those recommended by literature [14]. Similarly, the formulas for the bundle diameter, shell thickness, shell inside diameter and shell length were given, respectively:

$$D_{bundle} = D_{out} \left(\frac{N}{a} \right)^{1/b} \quad (7.11)$$

For the selected one-pass square-pitch arrangement, the constants used were [14]:

- $a = 0.215$
- $b = 2.207$

$$x_{shell} = \frac{PD_{shell,in}}{2S} + CA \quad (7.12)$$

where:

- P is the operating pressure (selected to be 600000 Pa)
- S is the allowable stress of the shell material, (120 MPa for carbon steel [16])
- CA is the corrosion allowance (0.0035 m, recommended by literature [14])

$$D_{shell,in} = D_{bundle} + C_{shell} \quad (7.13)$$

$$L_{shell} = L + 2C_{shell} + x_{shell} \quad (7.14)$$

The number of tubes was arbitrarily selected to comply with length recommendations and the chosen array. Inversely, tube length was calculated with the following formula [19]:

$$L = \frac{A}{N\pi D_{out}} \quad (7.15)$$

The selected tube layout for both heat exchangers is a square, 90° tube array [14] which is displayed in Figure 7.5. This arrangement is suitable for both the condenser and later, the reboiler because it gives a simple and compact tube bundle geometry, while also allowing easier mechanical cleaning between tube rows compared with triangular layouts. This is useful for a preliminary design, since the condenser may require periodic inspection or cleaning due to possible solvent impurities.

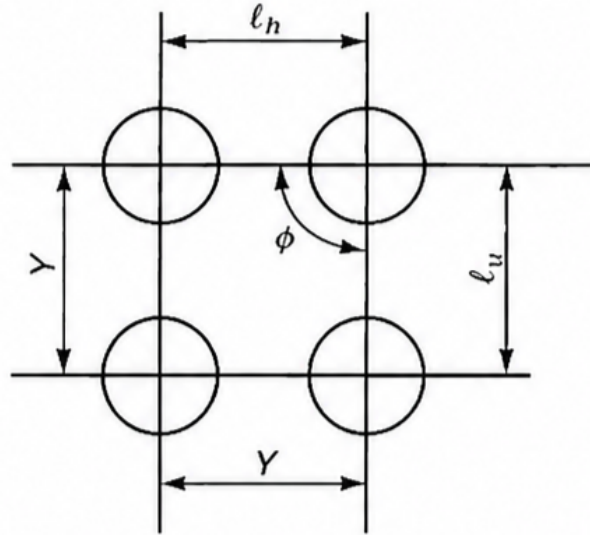


Figure 7.5: Selected tube array.

The available outside tube surface area was checked using:

$$A = N\pi D_{out}L = 48\pi(0.020)(4.002) = 12.07 \text{ m}^2$$

Therefore, the selected condenser geometry provides the required total heat transfer area.

The initially estimated condenser heat transfer area was 7.59 m², while the corrected calculated area was 12.07 m². The difference is mainly caused by the lower calculated dirty overall heat transfer coefficient, which includes the individual heat-transfer resistances, tube-wall resistance, and fouling. Therefore, a larger area was required than in the first estimate.

The cooling-water mass flow rate was calculated from the sensible heat balance equation [19]:

$$Q = \dot{m}_w c_{p,w} \Delta T_w \quad (7.16)$$

Where ΔT_w is the temperature difference of the cooling water (25°C). The calculated cooling-water mass flow rate was, after rearranging the previous equation:

$$\dot{m}_w = \frac{Q}{c_{p,w} \Delta T_w} = 2.55 \text{ kg/s}$$

The main hydraulic results for the cooling water are given in Table 7.4.

Table 7.4: Cooling water hydraulic results.

Parameter	Symbol	Value	Unit
Water mass flow rate	$\dot{m}_w$	2.55	kg/s
Tube-side flow area	A_{tube}	0.00965	m ²
Water velocity	v_w	0.270	m/s
Reynolds number	Re_w	5876.23	-
Prandtl number	Pr_w	4.81	-
Nusselt number	Nu_w	38.82	-

As the IPA vapour changes phase, it will form a saturated liquid film along the surface of the tube walls. In order to accurately estimate the required temperature of the tube walls to carry out this heat transfer operation, the temperature difference across the film must be accounted for. The calculated wall and film temperatures can be observed in Table 7.5.

Table 7.5: Condenser wall and film temperatures.

Parameter	Symbol	Value [°C]
Real mean wall temperature	$\bar{T}_{wall,real}$	68.62
Calculated mean wall temperature	$\bar{T}_{wall,calc}$	70.00
Mean film temperature	$\bar{T}_{film}$	75.00
Film temperature difference	ΔT_{film}	10.00

The value of the film temperature difference was chosen by iteration, performed until the calculated wall temperature was close to the real wall temperature. A normal temperature difference interval between the film and wall temperatures is of around 5 to 15°C.

The calculated mean wall temperature could be obtained from the mean temperature of the condensed vapour or distillate $\bar{T}_{dist}$, and the film temperature difference:

$$\bar{T}_{wall,calc} = \bar{T}_{dist} - \Delta T_{film} \quad (7.17)$$

The real, mean wall temperature was calculated to be [19]:

$$\bar{T}_{wall,real} = T_{dist} - \frac{k_{dirty}}{\alpha_{cond}}(T_{dist} - \bar{T}_w) \quad (7.18)$$

Where $\bar{T}_w$ is the mean temperature of the water.

The small difference between the real and calculated wall temperatures shows that the assumed film temperature difference is acceptable for the preliminary condenser design. All other calculated parameters are realistic for the scale of the process and adequate for manufacturing.

7.2 Column Reboiler Design

The reboiler was designed as a kettle reboiler. Although it is not a traditional shell-and-tube heat exchanger in the same way as the condenser, it still uses a tube bundle to transfer heat from condensing steam to the process liquid. The reboiler was designed as a two-pass heat exchanger arrangement, with the listed heat transfer area representing the total heat transfer area.

A kettle reboiler is more suitable for this project than a jacketed tank or internal coils since they are mainly used for heating agitated vessels, where the liquid temperature changes with time. In this case, the required duty is continuous vaporization at the bottom of the column. A kettle reboiler provides a submerged tube bundle with a large heat transfer surface, allowing the liquid to boil directly on the shell side while also providing enough volume for vapour-liquid disengagement [18].

The DMSO-rich liquid properties were obtained from ChemCAD using the relevant column-bottom composition, while the steam-side properties were obtained from a saturated steam table [17]. The obtained values are showcased below, in Table 7.6.

Table 7.6: Material properties of DMSO and steam.

Property	Value	Unit
vapor density of DMSO $\rho_{DMSO,vap}$	2.1	kg/m ³
liquid density of DMSO ρ_{DMSO}	927.4	kg/m ³
latent heat of DMSO r_{DMSO}	605000	J/kg
specific heat capacity of DMSO $c_{p,DMSO}$	2323	J/(kg*K)
surface tension of DMSO σ_{DMSO}	0.022	N/m
thermal conductivity of DMSO λ_{DMSO}	0.1	W/(m*K)
dynamic viscosity of DMSO μ_{DMSO}	0.00042	N*s/m ²
temperature of DMSO T_{DMSO}	190	°C
thermal conductivity of steam $\lambda_{st,film}$	0.65455	W/(m*K)
liquid density of steam $\rho_{st,film}$	848.7	kg/m ³
vapor density of steam $\rho_{st,v}$	10.286	kg/m ³
dynamic viscosity of steam $\mu_{st,film}$	0.0001262	N*s/m ²
latent heat of steam r_{st}	1887700	J/kg
specific heat capacity of steam $c_{p,st}$	4571.5	J/(kg*K)
temperature of steam T_{st}	220	°C

In this case, it was not necessary to calculate the logarithmic temperature difference, since the steam temperature was chosen and the temperature of the bottoms is known (see Figure 7.6). Furthermore, the heat duty is known.

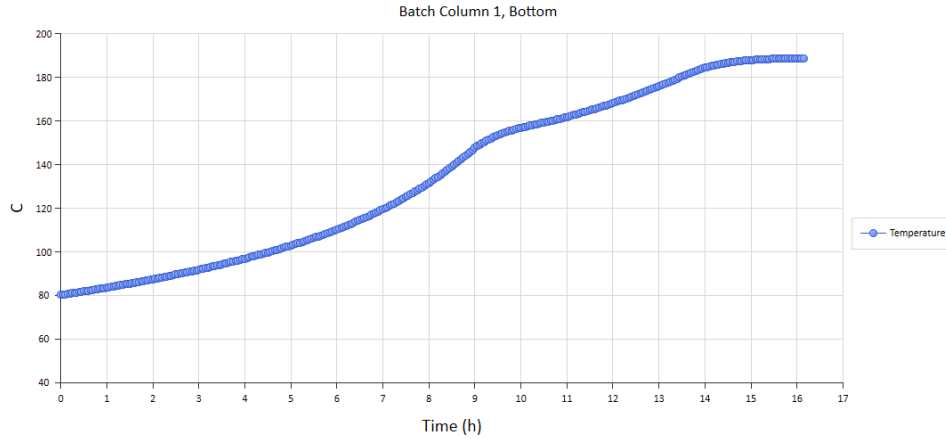


Figure 7.6: Bottom temperature as a function of time, highest temperature was selected (190°C).

TABLE A-9

Properties of saturated water

Temp. <i>T</i> , °C	Saturation Pressure <i>P</i> _{sat} , kPa	Density <i>ρ</i> , kg/m ³		Enthalpy of Vaporization <i>h</i> _{fg} , kJ/kg	Specific Heat <i>c</i> _p , J/kg·K		Thermal Conductivity <i>k</i> , W/m·K		Dynamic Viscosity <i>μ</i> , kg/m·s		Prandtl Number Pr		Volume Expansion Coefficient <i>β</i> , 1/K
		Liquid	Vapor		Liquid	Vapor	Liquid	Vapor	Liquid	Vapor	Liquid	Vapor	
100	101.33	957.9	0.5978	2257	4217	2029	0.679	0.0251	0.282 × 10 ⁻³	1.227 × 10 ⁻⁵	1.75	1.00	0.750 × 10 ⁻³
110	143.27	950.6	0.8263	2230	4229	2071	0.682	0.0262	0.255 × 10 ⁻³	1.261 × 10 ⁻⁵	1.58	1.00	0.798 × 10 ⁻³
120	198.53	943.4	1.121	2203	4244	2120	0.683	0.0275	0.232 × 10 ⁻³	1.296 × 10 ⁻⁵	1.44	1.00	0.858 × 10 ⁻³
130	270.1	934.6	1.496	2174	4263	2177	0.684	0.0288	0.213 × 10 ⁻³	1.330 × 10 ⁻⁵	1.33	1.01	0.913 × 10 ⁻³
140	361.3	921.7	1.965	2145	4286	2244	0.683	0.0301	0.197 × 10 ⁻³	1.365 × 10 ⁻⁵	1.24	1.02	0.970 × 10 ⁻³
150	475.8	916.6	2.546	2114	4311	2314	0.682	0.0316	0.183 × 10 ⁻³	1.399 × 10 ⁻⁵	1.16	1.02	1.025 × 10 ⁻³
160	617.8	907.4	3.256	2083	4340	2420	0.680	0.0331	0.170 × 10 ⁻³	1.434 × 10 ⁻⁵	1.09	1.05	1.145 × 10 ⁻³
170	791.7	897.7	4.119	2050	4370	2490	0.677	0.0347	0.160 × 10 ⁻³	1.468 × 10 ⁻⁵	1.03	1.05	1.178 × 10 ⁻³
180	1,002.1	887.3	5.153	2015	4410	2590	0.673	0.0364	0.150 × 10 ⁻³	1.502 × 10 ⁻⁵	0.983	1.07	1.210 × 10 ⁻³
190	1,254.4	876.4	6.388	1979	4460	2710	0.669	0.0382	0.142 × 10 ⁻³	1.537 × 10 ⁻⁵	0.947	1.09	1.280 × 10 ⁻³
200	1,553.8	864.3	7.852	1941	4500	2840	0.663	0.0401	0.134 × 10 ⁻³	1.571 × 10 ⁻⁵	0.910	1.11	1.350 × 10 ⁻³
220	2,318	840.3	11.60	1859	4610	3110	0.650	0.0442	0.122 × 10 ⁻³	1.641 × 10 ⁻⁵	0.865	1.15	1.520 × 10 ⁻³
240	3,344	813.7	16.73	1767	4760	3520	0.632	0.0487	0.111 × 10 ⁻³	1.712 × 10 ⁻⁵	0.836	1.24	1.720 × 10 ⁻³
260	4,688	783.7	23.69	1663	4970	4070	0.609	0.0540	0.102 × 10 ⁻³	1.788 × 10 ⁻⁵	0.832	1.35	2.000 × 10 ⁻³
280	6,412	750.8	33.15	1544	5280	4835	0.581	0.0605	0.094 × 10 ⁻³	1.870 × 10 ⁻⁵	0.854	1.49	2.380 × 10 ⁻³
300	8,581	713.8	46.15	1405	5750	5980	0.548	0.0695	0.086 × 10 ⁻³	1.965 × 10 ⁻⁵	0.902	1.69	2.950 × 10 ⁻³

Figure 7.7: Steam table.

The steam table used can be seen in Figure 7.7. A saturated steam pressure of 23.2 bar corresponds to the selected saturated steam temperature. A design steam pressure of 24 bar was selected for wall thickness selection, to provide an extra margin of safety.

The design equation used for the reboiler was:

$$Q = kAF\Delta T \quad (7.19)$$

Rearranging for heat transfer area:

$$A = \frac{Q}{kF\Delta T} \quad (7.20)$$

The DMSO-rich liquid temperature was taken to be $T_{DMSO} = 190^\circ\text{C}$. The heating steam temperature was then chosen as $T_{st} = 220^\circ\text{C}$, for an adequate temperature difference of $\Delta T =$

30°C. This allows proper heat transfer without overheating the DMSO. DMSO properties were taken as the reference because DMSO has the highest boiling point in the mixture. For stages where DMSO does not make up most of the bottom liquid, the pressure of steam (and therefore the temperature) is to be decreased with a control system. At higher temperatures, a peak and then a rapid decrease in the value of the heat transfer coefficient can be observed [21]. An illustration of this phenomenon with water as an example is shown in Figure 7.8.

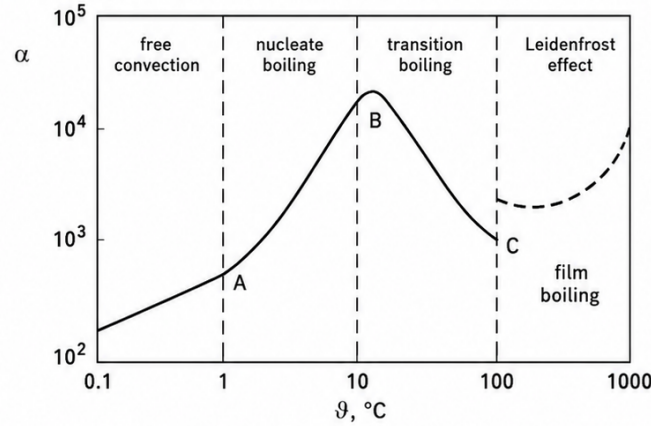


Figure 7.8: Heat-transfer coefficient during the boiling of water. Small values of T (θ in the plot) correspond to free-convective boiling up to point A in the figure. For larger values of T , nucleate boiling occurs. The maximum point marked as B in the figure occurs at a critical value of T characteristic of the liquid, which is around 25°C for water.

The reboiler heat duty was $Q = 1000$ MJ/h. Converting to watts:

$$Q = \frac{1000 \times 10^6}{3600} = 277777.78 \text{ W}$$

Using the initially assumed overall heat transfer coefficient for phase change $k = 850 \text{ W/m}^2\text{K}$, the preliminary heat transfer area was:

$$A = \frac{277777.78}{850 \times 30} = 10.89 \text{ m}^2$$

This initial estimate was corrected by calculating the individual boiling-side and steam-condensation-side heat transfer coefficients.

The boiling-side heat transfer coefficient for the process was calculated using the pool-boiling correlation [18]:

$$\alpha_{boiling} = 0.225 \left[\left(\frac{Q \cdot c_p}{A \cdot r} \right)^{0.69} \left(\frac{P \cdot \lambda}{\sigma} \right)^{0.31} \left(\frac{\rho_l}{\rho_v} \right)^{0.33} \right] \quad (7.21)$$

where:

- $\alpha_{boiling}$ = boiling-side heat transfer coefficient, W/m²K
- Q = heat duty, W
- A = heat transfer area, m²
- c_p = liquid specific heat capacity, J/kgK
- r = latent heat of vaporization, J/kg
- P = operating pressure, Pa

- λ = liquid thermal conductivity, W/mK
- σ = surface tension, N/m
- ρ_l = liquid density, kg/m³
- ρ_v = vapour density, kg/m³

These values for DMSO were obtained from ChemCAD reports and listed at the beginning of the subchapter.

$$\alpha_{shell} = 2261.06 \text{ W/m}^2\text{K}$$

The steam-side condensation coefficient was calculated using the filmwise-condensation method. The inside coefficient was first calculated and then converted to the outside tube surface basis [19]:

$$\alpha_{st,out} = \alpha_{st,in} \frac{D_{in}}{D_{out}}$$

The calculated reboiler heat transfer coefficients are displayed in Table 7.7.

Table 7.7: Reboiler heat transfer coefficients.

Parameter	Symbol	Value	Unit
Boiling-side coefficient	α_{shell}	2261.06	W/m ² K
Steam condensation coefficient, inside basis	$\alpha_{st,in}$	12152.33	W/m ² K
Steam condensation coefficient, outside basis	$\alpha_{st,out}$	10194.74	W/m ² K
Clean overall coefficient	k_{clean}	1708.92	W/m ² K
Fouling resistance	R_f	0.00026	m ² K/W
Dirty overall coefficient	k_{dirty}	1183.20	W/m ² K

The clean overall heat transfer coefficient was calculated as:

$$\frac{1}{k_{clean}} = \frac{1}{\alpha_{st,out}} + \frac{x_{tube}}{\lambda_{wall}} + \frac{1}{\alpha_{shell}}$$

The dirty overall heat transfer coefficient was calculated by including the fouling resistance [18]:

$$\frac{1}{k_{dirty}} = \frac{1}{k_{clean}} + R_f$$

$$k_{dirty} = 1183.20 \text{ W/m}^2\text{K}$$

The correction factor for the two-pass heat exchanger was possible to compute:

$$F = 1$$

From the correction factor diagram for a two-pass heat exchanger (see Figure 7.9).

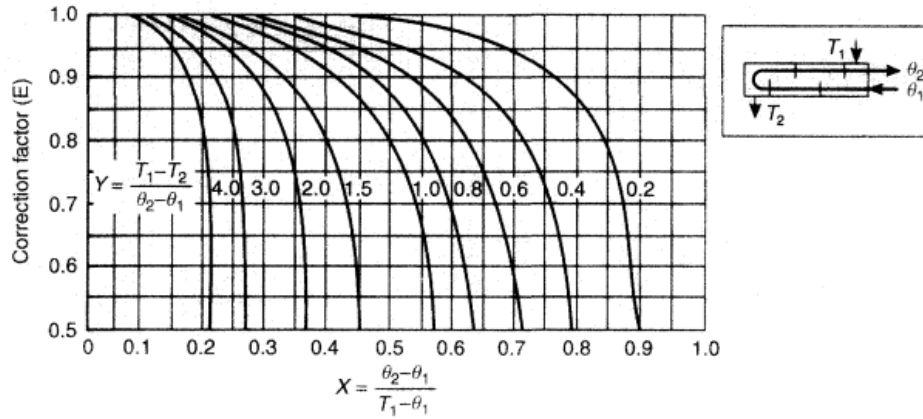


Figure 7.9: Correction factor for a two-pass heat exchanger.

Using the dirty overall coefficient, the corrected reboiler heat transfer area was:

$$A_{calc} = \frac{Q}{k_{dirty} F \Delta T} = \frac{277777.78}{1183.20 \times 30} = 7.83 \text{ m}^2$$

Therefore, the required total reboiler heat transfer area is 7.83 m^2

The initially estimated reboiler heat transfer area was 10.89 m^2 , while the corrected calculated area was 7.83 m^2 . The calculated area is lower because the detailed calculation gave a higher dirty overall heat transfer coefficient than the value assumed in the first estimate. As a result, the required heat duty could be transferred through a smaller surface area.

The selected reboiler geometry is shown below, in Table 7.8. The same formulas and values obtained from literature used in subchapter 7.1 were applied here [14]. The tube wall thickness was selected to be compatible with high pressure steam.

Table 7.8: Selected and calculated reboiler geometry.

Parameter	Symbol	Value	Unit
Number of tubes	N	28	-
Tube outside diameter	D_{out}	0.0250	m
Tube inside diameter	D_{in}	0.0210	m
Tube wall thickness	x_{tube}	0.00202	m
Tube pitch	p_t	0.0313	m
Tube length	L	1.78	m
Bundle diameter	D_{bundle}	0.227	m
Shell clearance	C_{shell}	0.011	m
Shell inside diameter	$D_{shell,in}$	0.477	m
Shell wall thickness	x_{shell}	0.00493	m

The tube length was calculated from the total required heat transfer area, dividing the result by 2 since the configuration will have two tube passes (U-tubes). The required straight tube length is halved compared with a single-pass arrangement, while the shell diameter will double [19]. The formula used to compute the tube length was:

$$L = \frac{A}{2N\pi D_{out}} = \frac{7.83}{2(28)\pi(0.0250)} = 1.78 \text{ m} \quad (7.22)$$

The steam mass flow rate was calculated from the latent heat balance formula [19]:

$$\dot{m}_{st} = \frac{Q}{r_{st}} = \frac{277777.78}{1887700} = 0.147 \text{ kg/s} \quad (7.23)$$

The main steam-side values related to flow rate were also computed, using the same formula for flow cross-sectional area and velocity as in the previous chapter. Results are listed in Table 7.9

Table 7.9: Steam flow rate parameters.

Parameter	Symbol	Value	Unit
Steam mass flow rate	$\dot{m}_{st}$	0.147	kg/s
Tube-side flow area	A_{tube}	0.00485	m ²
Steam velocity	v_{st}	2.95	m/s

Similarly, film and wall mean temperature results could be obtained by utilizing the same formulas as in the case of the condenser, but assuming that this time it will be steam forming a film around the tubes as it condenses. The calculated and selected film and wall temperatures are found in Table 7.10.

Table 7.10: Reboiler wall and film temperatures.

Parameter	Symbol	Value [°C]
Real wall temperature	$\overline{T}_{wall,real}$	204.30
Calculated wall temperature	$\overline{T}_{wall,calc}$	205.00
Film temperature	$\overline{T}_{film}$	212.50
Film temperature difference	ΔT_{film}	15.00

The small difference between the real and calculated wall temperatures confirms that the assumed film temperature is acceptable for the preliminary kettle reboiler calculation.

All calculated parameters are realistic for the scale of the process and adequate for the manufacturing of parts for this reboiler.

7.3 Selection of Heat Exchanger Construction Types

The condenser was selected as a conventional one-pass shell-and-tube heat exchanger. Since the exchanger is relatively small and the temperature difference is moderate, a fixed-tube-sheet construction is suitable. For the front-end stationary head, a removable channel and cover is selected so that the tube side can be accessed for inspection and cleaning. The selected front-end type is therefore TEMA type A [14]. The shell is selected as a one-pass shell, corresponding to TEMA shell type E. The rear-end head is selected as a fixed-tube-sheet rear channel with

removable cover, corresponding to TEMA rear-end type L. Therefore, the selected condenser construction is:

AEL

where:

- *A* = front-end stationary head with removable channel and cover
- *E* = one-pass shell
- *L* = fixed-tube-sheet rear end with removable cover

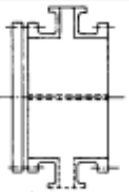
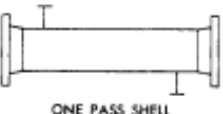
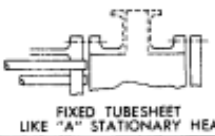
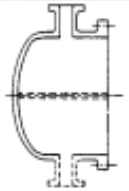
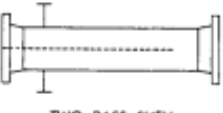
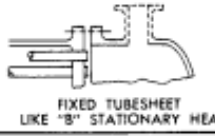
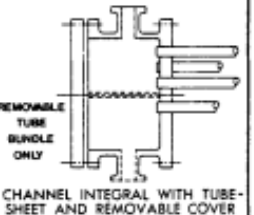
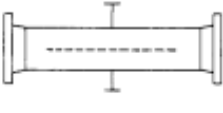
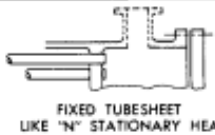
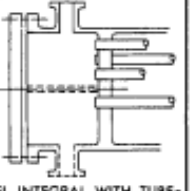
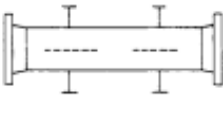
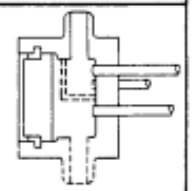
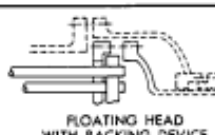
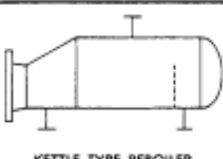
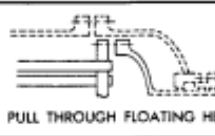
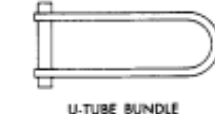
	FRONT END STATIONARY HEAD TYPES		SHELL TYPES		REAR END HEAD TYPES
A	 CHANNEL AND REMOVABLE COVER	E	 ONE PASS SHELL	L	 FIXED TUBESHEET LIKE "A" STATIONARY HEAD
B	 BONNET (INTEGRAL COVER)	F	 TWO PASS SHELL WITH LONGITUDINAL BAFFLE	M	 FIXED TUBESHEET LIKE "B" STATIONARY HEAD
C	 REMOVABLE TUBE BUNDLE ONLY CHANNEL INTEGRAL WITH TUBE- SHEET AND REMOVABLE COVER	G	 SPLIT FLOW	N	 FIXED TUBESHEET LIKE "N" STATIONARY HEAD
N	 CHANNEL INTEGRAL WITH TUBE- SHEET AND REMOVABLE COVER	H	 DOUBLE SPLIT FLOW	P	 OUTSIDE PACKED FLOATING HEAD
D	 SPECIAL HIGH PRESSURE CLOSURE	J	 DIVIDED FLOW	S	 FLOATING HEAD WITH BACKING DEVICE
		K	 KETTLE TYPE REBOILER	T	 PULL THROUGH FLOATING HEAD
		X	 CROSS FLOW	U	 U-TUBE BUNDLE
				W	 EXTERNALLY SEALED FLOATING TUBESHEET

Figure 7.10: TEMA-type designations for shell-and-tube heat exchangers.

This construction is appropriate for a normal one-pass condenser because it is mechanically simple, allows tube-side access, and provides the required heat transfer area.

The reboiler was selected as a kettle-type reboiler rather than a conventional shell-and-tube heat exchanger. For this reason, the shell type must correspond to a kettle arrangement. The

selected shell type is therefore TEMA type K [14]. Since the reboiler uses a two-pass tube arrangement and must allow thermal expansion of the tube bundle, a U-tube rear-end construction is suitable. The selected rear-end type is TEMA type U. For the front-end stationary head, a bonnet-type head is selected because the tube-side fluid is condensing steam, which is relatively clean compared with process liquids. The selected front-end type is therefore TEMA type B. Therefore, the selected reboiler construction is:

BKU

where:

- *B* = bonnet-type front-end stationary head
- *K* = kettle-type shell
- *U* = U-tube rear-end construction

This construction is appropriate for the kettle reboiler because the kettle shell provides space for boiling and vapour-liquid disengagement, while the U-tube arrangement allows thermal expansion and provides the required two-pass tube-side flow path.

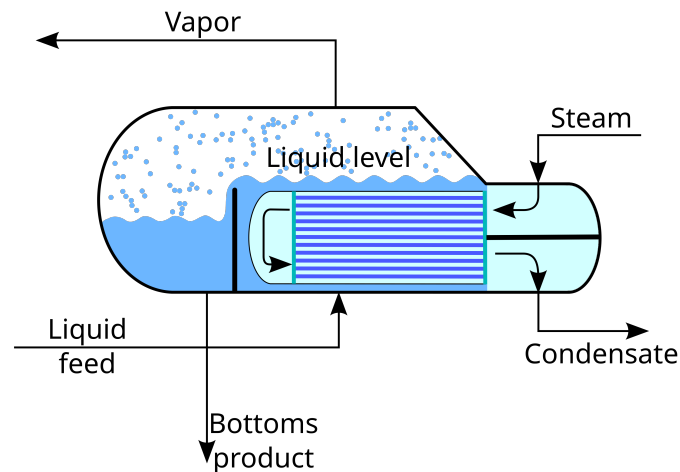


Figure 7.11: Diagram of a kettle-type reboiler [22].

Chapter 8

Heat Recycling

Distillation is an energy-intensive separation process because heat must be supplied to the reboiler while it is simultaneously removed from the overhead vapour in the condenser. In the original configuration, a significant part of the thermal energy contained in the top vapour is rejected during condensation. For this reason, heat recycling was studied as a way to reduce the external heating demand of the column and improve the energetic efficiency of the process.

In this chapter, two heat recovery methods are investigated. The first method is the pre-heater method, in which the vapour leaving the top of the column during the IPA distillation is used as a heat source in a heat exchanger. This recovered heat is transferred to the DMSO feed stream before it enters the column, eliminating external energy required to pre-heat the feed. This method was simulated in ChemCAD using a simple heat exchanger model, as can be seen in Figure 8.1.

The second method is the heat pump method, in which the top vapour from the column is compressed in order to increase its temperature and make it suitable for heat recovery at the bottom of the column. The simulated arrangement includes two compressors with an inter-cooler, a heat exchanger, an expansion valve, and a second heat exchanger. In this configuration, heat is extracted from the compressed overhead vapour and used to help heat the column bottom, reducing the required reboiler heat load. This method was studied using a dynamic ChemCAD simulation. The top-vapour composition was introduced into the model at time steps of 0.05 h over the distillation operation.

The purpose of comparing these two methods is to determine how effectively the available overhead-vapour heat can be reused in the process. The pre-heater method represents a simpler and more direct heat-recovery option, while the heat pump method represents a more complex but potentially more powerful heat-integration strategy. Both methods are evaluated based on their ability to reduce external heat requirements and improve the energetic performance of the distillation process.

8.1 Pre-heater Method

In the pre-heater method, the heat contained in the vapour leaving the top of the distillation column is used to pre-heat the DMSO feed before it enters the column. This reduces the amount of external heat required to bring the solvent feed to the selected inlet temperature. The simulation was carried out in ChemCAD using a heat exchanger, where the hot side corresponds to

the overhead vapour and the cold side corresponds to the feed stream.

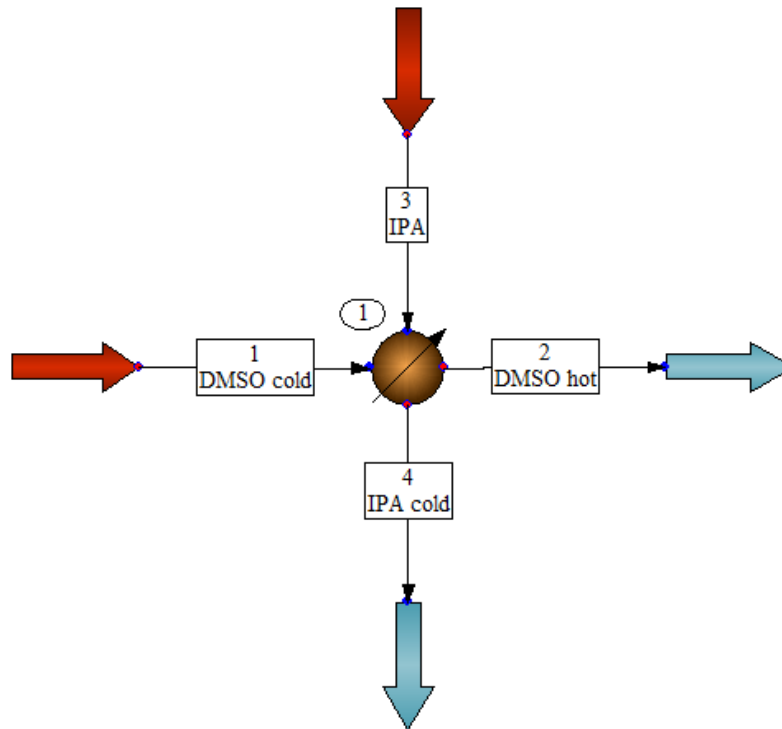


Figure 8.1: ChemCAD simulation arrangement for the pre-heater calculations.

We used the main design equation for the heat exchangers is, and the logarithmic mean temperature difference was calculated from the two terminal temperature differences:

$$\Delta T_{log} = \frac{\Delta T_1 - \Delta T_2}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)}$$

where:

$$\Delta T_1 = T_{1,1} - T_{2,1}$$

$$\Delta T_2 = T_{1,2} - T_{2,2}$$

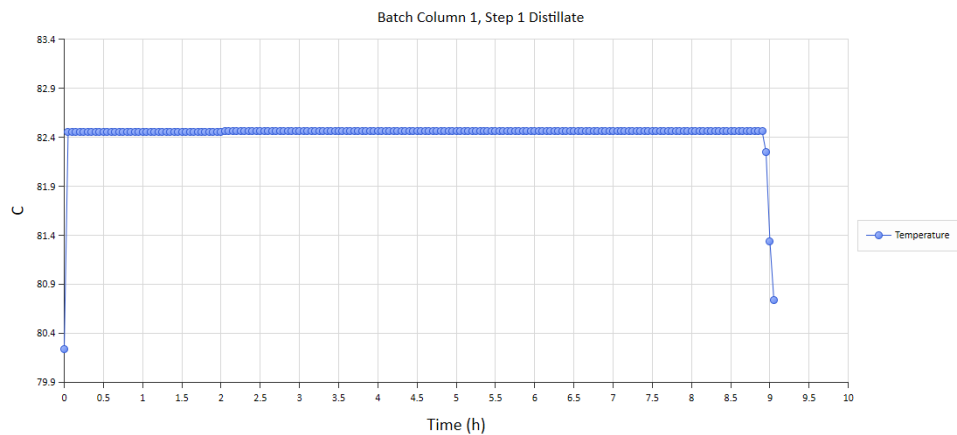


Figure 8.2: ChemCAD plot displaying the top vapour temperature during step 1. It was determined to be 82.4 °C.

We need to heat up the feed to increase its temperature from 20°C (room temperature) to 62°C. Since the vapour, which is almost pure IPA, experiences a phase change (condensation), its temperature can be assumed to remain constant [23]. The temperatures used in the pre-heater calculation, in Kelvin, are shown in Table 8.1.

Table 8.1: Pre-heater temperatures.

Parameter	Symbol	Value	Unit
Hot inlet temperature	$T_{1,1}$	355.55	K
Hot outlet temperature	$T_{1,2}$	355.55	K
Cold inlet temperature	$T_{2,1}$	293.15	K
Cold outlet temperature	$T_{2,2}$	335.15	K

Replacing the logarithmic mean temperature difference formula with these values:

$$\Delta T_{log} = 37.57 \text{ K}$$

The value of the total heat duty of this process, obtained from the ChemCAD heat-exchanger simulation, was:

$$Q = 45.1324 \text{ MJ/h}$$

This value was converted into watts:

$$Q = \frac{45.1324 \times 10^6}{3600} = 12536.78 \text{ W}$$

The calculated temperature differences and heat duty are summarized in Table 8.2.

Table 8.2: Pre-heater heat duty.

Parameter	Symbol	Value	Unit
Terminal T difference 1	ΔT_1	62.40	K
Terminal T difference 2	ΔT_2	20.40	K
LMTD	ΔT_{log}	37.57	K
Heat duty	Q	45.13	MJ/h

Three possible heat transfer areas were calculated using different overall heat transfer coefficients. The first value, $k_{vap} = 850 \text{ W/m}^2\text{K}$, represents the assumed overall heat transfer coefficient for a change of phase in a heat exchanger. The second value, $k_{dist} = 570 \text{ W/m}^2\text{K}$ [20], represents an estimation of the required heat transfer coefficient for the condensation of IPA vapour. The third value, $k_{calc} = 534.4 \text{ W/m}^2\text{K}$, represents the calculated overall heat transfer coefficient used for the final conservative sizing. This is the heat transfer coefficient that was calculated for the main condenser of this column.

The highest calculated heat transfer area is obtained using the lowest overall heat transfer coefficient:

For $k_{calc} = 534.4 \text{ W/m}^2\text{K}$:

$$A_{calc} = \frac{12536.78}{534.4 \times 37.57}$$

$$A_{calc} = 0.6245 \text{ m}^2$$

Therefore, the selected required heat transfer area for the pre-heater is 0.6245 m^2 .

The pre-heater therefore allows the feed to enter the distillation column at the selected feed temperature of 62°C using recovered heat from the overhead vapour. This decreases the amount of additional external energy that would otherwise be required to heat the feed before it enters the column. Not only would it efficiently save on energy, but also on costs. It provides a simple and practical way to recycle part of the heat contained in the overhead vapour, while its required size area is relatively small.

8.2 Heat Pump Method

The second heat recycling method studied was the heat pump method. In this configuration, the vapour leaving the top of the distillation column is compressed. This makes the overhead vapour suitable for heat recovery at a higher temperature level. The recovered heat is then transferred to the bottom of the column, helping the reboiler supply the energy required for vaporization.

Before the final arrangement was selected, different compression ratios were investigated using a single compressor. The purpose of these trials was to determine whether the overhead vapour could be compressed sufficiently in one step to reach a useful temperature level for reboiler heat recovery. Several pressure ratios were tested, but the results showed that a single compressor was not suitable. Lower compression ratios did not increase the temperature enough for useful heat recovery over a sufficient part of the operation, while higher compression ratios produced excessively high outlet temperatures and unsuitable vapour-fraction behavior (the stream does not remain in the vapour phase during compression; condensation should occur only in the heat exchanger where useful heat is released to the reboiler.). Therefore, the final selected configuration used two compressors with an inter-cooler between them. Each compressor was operated with a compression ratio of 4:1.

The simulated heat pump arrangement in ChemCAD is displayed in Figure 8.3. It included two compressors, a first heat exchanger that acted as an inter-cooler between them (unit 1 in the figure), a second heat exchanger (unit 3 in the figure) with a set heat duty of -1000 MJ/h to represent the heat taken out by the reboiler (the heat duty of the reboiler for the all distillation simulations was set to 1000 MJ/h ; therefore, theoretically, this amount of energy per hour that will be demanded), an expansion valve, and a third heat exchanger (unit 5 in the figure) that was used to calculate the amount of energy that could be recovered. After supplying heat to the reboiler, the remaining heat indicated by the heat exchanger could theoretically be used for secondary heating duties, such as pre-heating process streams, feed streams, or utility water. The inter-cooler was required to reduce the vapour temperature between compression stages and to avoid excessively high temperatures after the second compressor. It was set to superheat the compressed vapour stream in order to keep the working fluid in gaseous state, and cooling was assumed to be done with room-temperature cooling water (25°C). The heat transfer coefficient of the IPA-rich fluid was given as $k = 150 \text{ W/m}^2\text{K}$ by literature [18], which was applied as the inter-cooler heat transfer coefficient.

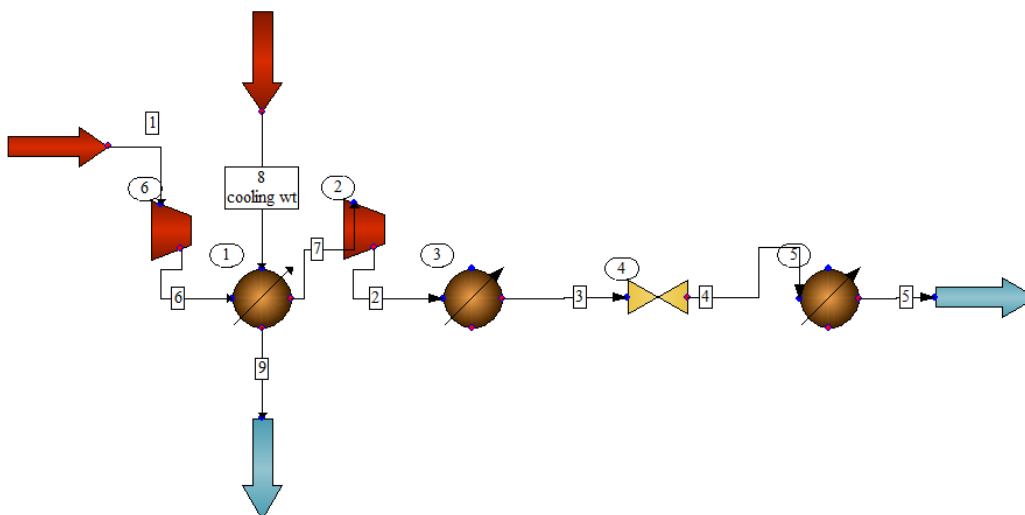


Figure 8.3: ChemCAD heat pump simulation arrangement.

The simulation was carried out dynamically using ChemCAD data maps and Excel coding. The input vapour composition and temperature at each time step were obtained from ChemCAD reports for the top vapour stream, these values were outputted as a large Excel table. Our Excel file was programmed to transfer the data values to a specific cell in the document every 0.05 h. ChemCAD then read these values through a data map and used them as inputs for the heat pump simulation. In the same way, the output values from ChemCAD were exported back into Excel every 0.05 h and stored in a second table.

The main outputs recorded were the temperature after compression, vapour fraction after compression, temperature after the first heat exchanger, vapour fraction after the first heat exchanger, heat duty of the second heat exchanger, inter-cooler heat duty, compressor duties, cooling-water outlet temperature, and logarithmic mean temperature difference. Although the full simulation was carried out over the complete operation, the heat pump was found to be suitable only during the first 4.5 h of the distillation. As the batch progressed, the recoverable heat decreased, and later the temperature difference became too small for practical heat transfer; in some later intervals, heat input would even be required instead of useful heat recovery. Therefore, only this time interval was considered for the useful heat recovery evaluation.

8.2.1 Input Vapour Stream

The input vapour stream composition changes continuously during the batch distillation. The dynamic heat pump simulation therefore required the vapour composition to be updated at each time step. Table 8.3 shows representative values from the input table, using 0.5 h intervals up to 4.5 h. After 4.5 hours had passed, the output results from the simulation were not good enough for efficient heat recycling anymore.

Table 8.3: Heat pump input values.

Time [h]	Bottom Temperature [°C]	IPA vapour flow rate [kmol/h]	H ₂ O vapour flow rate [kmol/h]	DMSO vapour flow rate [kmol/h]
0.50	81.82	23.6395	0.0129	0.000263
1.00	83.57	23.5455	0.0116	0.000263
1.50	85.44	23.4585	0.0106	0.000263
2.00	87.43	23.3680	0.0098	0.000263
2.50	89.56	23.2690	0.0092	0.000263
3.00	91.82	23.1535	0.0085	0.000263
3.50	94.26	23.0215	0.0080	0.000262
4.00	96.89	22.8661	0.0075	0.000262
4.50	99.75	22.6821	0.0070	0.000262

As shown in Table 8.3, the IPA vapour flow rate slowly decreases during the first 4.5 h, while the bottom temperature increases from approximately 80°C to nearly 100°C. The DMSO vapour flow rate remains very small compared with the IPA vapour flow rate. This is expected because DMSO has a much higher boiling point and remains mainly in the liquid phase.

8.2.2 Heat Pump Output Results

Table 8.4 shows the main output values obtained from the ChemCAD heat pump simulation. The table includes only the useful operating region up to 4.5 h. The abbreviations shown in the column titles stand for:

- VF: Vapour fraction
- IC: inter-cooler
- HX1: second heat exchanger (first if the inter-cooler is not counted)
- HX2: third heat exchanger (second if the inter-cooler is not counted)
- C1: first compressor
- C2: second compressor

Table 8.4: Heat pump output values.

Time [h]	T after comp. [°C]	VF after C2	T after HX1 [°C]	VF after HX1	HX2 duty [MJ/h]	IC duty [MJ/h]	C1 duty [MJ/h]	C2 duty [MJ/h]	LMTD [K]
0.50	177.21	1.00	114.36	0.00	164.31	23.97	97.18	99.29	89.36
1.00	177.21	1.00	113.62	0.00	159.62	23.87	96.79	98.89	89.41
1.50	177.21	1.00	112.93	0.00	155.30	23.77	96.43	98.52	89.46
2.00	177.20	1.00	112.20	0.00	150.79	23.68	96.05	98.13	89.51
2.50	177.20	1.00	111.39	0.00	145.88	23.57	95.64	97.71	89.57
3.00	177.20	1.00	110.45	0.00	140.18	23.45	95.17	97.23	89.64
3.50	177.20	1.00	109.36	0.00	133.65	23.32	94.62	96.67	89.71
4.00	177.20	1.00	108.05	0.00	125.97	23.16	93.98	96.01	89.80
4.50	177.20	1.00	106.47	0.00	116.89	22.97	93.22	95.24	89.90

The results show that the vapour fraction after compression remains equal to 1.00 during the useful period, meaning that the stream remains fully vapour after compression. After the first heat exchanger, the vapour fraction becomes 0.00, meaning that the compressed vapour is condensed and heat is successfully extracted. This confirms that the heat pump arrangement works properly during the first 4.5 h.

The time-zero value is treated as a start-up point and is not considered representative of stable heat pump operation, therefore 0.05 h is the initial time value. The heat duty of the second heat exchanger decreases in magnitude during the useful operating period, from approximately 169.40 MJ/h at 0.05 h to 116.89 MJ/h at 4.5 h. This shows that the amount of useful recoverable heat decreases as the batch distillation progresses, and that the method would be most efficient when only applied during the first 4.5 hours. During the time interval 7.5 h - 9.5 h, the heat duty becomes negative, meaning that the expanded working fluid remains sub-cooled liquid, and heating would be required to obtain a saturated liquid reflux. This would make recovery definitely not viable during this range. The compressor heat duties also decrease slightly over time, while the logarithmic mean temperature difference remains almost constant, close to 89–90 °C. Furthermore, the temperature difference between the input vapour and output liquid after heat recovery decreases to less than 10 °C, which is not suitable for efficient heat transfer applications [20], [10].

8.2.3 Useful Heat Pump Operation

The useful heat pump operating period and the energy demand of the whole configuration can be summarized in Table 8.5.

Table 8.5: Useful heat pump period.

Parameter	Value	Unit
Start time	0.05	h
End time	4.50	h
Max. HX2 duty	169.40	MJ/h
Max. inter-cooler duty	24.08	MJ/h
Max. compressor 1 duty	97.61	MJ/h
Max. compressor 2 duty	99.72	MJ/h
Avg. LMTD	89.57	K

Using the maximum compressor duties as a conservative estimate, the energy needed to operate the heat pump for 4.5 h is:

$$(97.61 + 99.72) \times 4.5 = 888 \text{ MJ} \quad (8.1)$$

However, the heat pump will replace the reboiler duty during this time period, which was assumed to be 1000 MJ/h throughout the entire process. Therefore, the total energy demand in this period is:

$$1000 \times 4.5 = 4500 \text{ MJ} \quad (8.2)$$

The net energy saving is therefore approximately:

$$4500 - 887.985 = 3612.015 \text{ MJ} \quad (8.3)$$

This means that, from an energy-balance point of view, the heat pump is viable since the energy recovered for the reboiler is much greater than the compressor energy required. This method can be considered energetically beneficial as a partial heat-recovery option, but its final practical viability must also consider its higher equipment complexity and capital cost, since it requires two compressors, an inter-cooler, an expansion valve, and additional heat exchangers.

For future cost estimation, the inter-cooler's maximum heat duty and heat transfer area were used.

Conversion to watts:

$$Q_{max} = \frac{24.08 \times 10^6}{3600} = 6688.01 \text{ W}$$

Assuming cooling water is heated from 25°C to 50°C:

$$\dot{m}_w = \frac{Q}{c_p \Delta T}$$

Using $c_p = 4180 \text{ J/kgK}$ and $\Delta T = 25 \text{ K}$:

$$\dot{m}_w = \frac{6688.01}{4180 \times 25} = 0.0640 \text{ kg/s}$$

$$\dot{m}_w = 230.40 \text{ kg/h}$$

The logarithmic mean temperature difference calculated was:

$$\Delta T_{log} = 88.46 \text{ K}$$

Assuming the inter-cooler is a single pass heat exchanger, the correction factor F was calculated to be 1 from correction factor chart for a single-pass heat exchanger [14].

Using $k = 150 \text{ W/m}^2\text{K}$ as the heat transfer coefficient for this application [18], The heat transfer area was calculated using the heat transfer area equation for heat exchangers:

$$A = \frac{6688.01}{150 \times 88.46} = 0.501 \text{ m}^2$$

8.3 Method Comparison

For the pre-heater method, the recovered overhead-vapour heat is used to heat the feed before it enters the column. The total recovered energy is $45 \times 9.05 = 407.25 \text{ MJ}$. Therefore, only about $(50 - 45) \times 9.05 = 45.25 \text{ MJ}$ of additional external feed-heating energy would still be required. In comparison, the heat pump method can replace $1000 \times 4.5 = 4500 \text{ MJ}$ of reboiler energy during its useful 4.5 h operating period. It requires more energy to operate, but the net energy saving is around 3600 MJ. Therefore, the heat pump method is clearly superior from a net energy-saving perspective. However, the heat pump method is much more complex and requires additional equipment, including two compressors, an inter-cooler, an expansion valve, and additional heat exchangers. Its main disadvantage is that it is only useful for a limited part of the batch operation and requires significant compressor work. For this reason, the heat pump method may be useful as a partial heat-recovery option, but it is more complicated and less broadly applicable than the pre-heater method, as well as more costly.

Chapter 9

Material Recovery

After the main batch extractive distillation step, two secondary off-cut streams are produced: an IPA/water off-cut and a water/DMSO off-cut. These streams do not meet the purity requirements to be considered final products, but they still contain recoverable amounts of IPA, water, and DMSO. Therefore, instead of treating them as waste, their re-integration into later batches was studied in order to reduce material losses and improve the overall efficiency of the process.

Two recycling strategies were compared using ChemCAD simulations. In both cases, the results of one batch were used as input for the next batch. Batch 1 corresponds to the original reference batch, while Batches 2–6 represent five consecutive recycle batches. For each batch, the total moles, mass, purity of the off-cut streams, IPA distillate purity, distillate mass, elapsed time, and DMSO bottom-product purity were recorded.

In the first method, only the IPA/water off-cut is added to the pot charge of the next batch, while the recovered DMSO bottom product from the previous distillation is used as the solvent feed for the next batch. Since the amount of recovered DMSO is lower than the amount required for a complete new batch, the missing amount would be replaced in practice by externally supplied, theoretically pure DMSO. However, in the simulation, the additional pure DMSO was not included in the purity calculation. Instead, the purity of the recovered bottom DMSO was used as the next solvent-feed purity, representing the worst-case scenario. The water/DMSO off-cut is not directly recycled in this method and is instead kept for later recovery distillation.

In the second method, both the IPA/water off-cut and the water/DMSO off-cut are added directly to the pot charge of the next batch. This method has the advantage of immediately reusing all off-cut material, but it also introduces more water and DMSO into the next batch, which can influence the charge composition, separation time, and simulation convergence.

9.1 Method 1, Water/IPA off-cut Recycling

In the first recycling method, the IPA/water off-cut from each batch was added to the pot charge of the following batch. The DMSO bottom product from the previous batch was reused as the next batch's solvent feed. The water/DMSO off-cut was not added directly to the following batch, but was instead collected for later recovery. This approach aims to recover IPA-containing material without continuously increasing the amount of DMSO in the pot charge.

The initial charge molar amounts used for this recycle sequence were 41.18 kmol IPA and

21.21 kmol water. During the ChemCAD simulations, some operating adjustments were required to maintain convergence and product purity. For Batch 2, the reflux ratio in step 3 was changed from 4 to 5. For Batch 5, the reflux ratio in step 3 was further changed from 5 to 6. These changes were necessary to keep the distillate within the required purity range as the recycled material influenced the batch composition. The meaning of the abbreviations used in the tables' column titles are:

- B: batch number
- Chg.: charge
- Dist.: distillate
- OC: off-cut
- x: molar fraction
- m: mass

9.1.1 IPA/Water off-cut and IPA Distillate Results

Table 9.1: Results of the first material recovery method.

B	IPA OC	H ₂ O OC	xIPA OC	m OC	nIPA Chg.	nH ₂ O Chg.	xIPA Chg.	xH ₂ O Chg.	xIPA Dist.	mDist.	t
	kmol	kmol	mol frac.	kg	kmol	kmol	mol frac.	mol frac.	mol frac.	kg	h
1	1.30	5.29	0.1975	173.6	42.48	26.51	0.6158	0.3842	0.9985	2397.4	16.1
2	1.71	7.04	0.1957	229.9	42.90	28.26	0.6029	0.3971	0.9989	2477.9	18.2
3	1.68	6.71	0.1999	221.6	42.86	27.92	0.6055	0.3945	0.9985	2477.2	18.5
4	1.65	6.73	0.1971	220.6	42.83	27.94	0.6052	0.3948	0.9985	2477.1	18.8
5	1.80	7.36	0.1963	240.6	42.98	28.57	0.6007	0.3993	0.9990	2466.7	19.5
6	1.74	7.04	0.1977	231.1	42.92	28.25	0.6030	0.3970	0.9987	2479.4	19.8

Results are listed in Table 9.1. The IPA distillate purity remained consistently above the required value in all six batches. The average IPA distillate purity was 0.9987 mol fraction, while the total recovered IPA distillate mass was 14775.70 kg. The total operating time for the six batches was approximately 111 h. A slight increase in operating time can be observed from Batch 1 to Batch 6, which is expected because the recycled IPA/water off-cut changes the initial pot composition of each new batch.

9.1.2 Water/DMSO off-cut Obtained During Method 1

Table 9.2: Water/DMSO off-cut obtained during method 1.

B	H ₂ O	DMSO	xH ₂ O	xDMSO	mTotal
	kmol	kmol	mol frac.	mol frac.	kg
1	3.29	15.92	0.1713	0.8287	1302.9
2	2.47	15.75	0.1356	0.8644	1275.1
3	2.57	16.01	0.1382	0.8618	1297.5
4	2.60	15.98	0.1398	0.8602	1295.3
5	2.06	15.48	0.1176	0.8824	1246.8
6	2.00	15.56	0.1139	0.8861	1252.1

The water/DMSO off-cut contains a significant amount of DMSO, with an average DMSO mole fraction of 0.8639, as indicated in Table 9.2. Since this material is not directly returned to the next batch in Method 1, it remains available for a later recovery distillation. This prevents additional DMSO accumulation in the main IPA recovery process, but requires an additional recovery step.

9.1.3 DMSO Bottom Product Obtained During Method 1

Table 9.3: DMSO bottom product obtained during method 1.

B	DMSO	H ₂ O	xDMSO	xH ₂ O	mTotal
	kmol	kmol	mol frac.	mol frac.	kg
1	51.96	0.0255	0.9995	0.00049	4060.02
2	53.20	0.0253	0.9995	0.00047	4157.48
3	53.70	0.0252	0.9995	0.00047	4195.97
4	53.73	0.0256	0.9995	0.00048	4198.89
5	53.85	0.0253	0.9995	0.00047	4207.86
6	54.14	0.0248	0.9995	0.00046	4230.33

The DMSO bottom product remained highly pure throughout the recycle sequence, with an average DMSO mole fraction of 0.9995 according to Table 9.3. This confirms that the recovered bottom DMSO can be reused as solvent feed in the following batch. However, the recovered amount is not enough to supply a full new batch, so external pure DMSO would still be required in practice.

The water/IPA off-cut re-integration method is stable and effective. It maintains the IPA distillate purity above the required specification, gives a high total recovered distillate mass, and produces a pure DMSO bottom product suitable for reuse. Its main disadvantage is that the water/DMSO off-cut is not immediately recycled and must be treated separately later. However, this also makes the main distillation sequence easier to control, since DMSO does not accumulate as strongly in the next batch charge.

9.2 Method 2, Total off-cut Recycling

In the second recycling method, both the IPA/water off-cut and the water/DMSO off-cut were added directly to the next pot charge. This means that all off-cut material from the previous batch was immediately reused in the following batch. The purpose of this method was to determine whether direct recycling of all secondary material streams could improve material recovery and reduce the need for a separate recovery step.

The same initial charge molar values were used as in Method 1: 41.18 kmol IPA and 21.214 kmol water. Since both off-cut streams were recycled, the DMSO content in the pot charge increased compared with Method 1. This made the simulation more difficult to converge in some cases. In Batch 2, the reflux ratio in step 3 was changed from 4 to 5. In Batch 3, the stop value in step 3 was changed from 0.9995 to 0.9997. In Batch 4, the reflux ratio in step 3 was increased from 5 to 6. In Batch 5, the reflux ratio was changed back from 6 to 5. These adjustments show that Method 2 required more operational changes than Method 1.

9.2.1 Combined IPA/Water/DMSO off-cut and IPA Distillate Results

Table 9.4: Results of the second material recovery method.

B	IPA OC	H ₂ O OC	DMSO OC	mOC	nIPA Chg.	nH ₂ O Chg.	xIPA Chg.	xH ₂ O Chg.	xDMSO Chg.	xIPA Dist.	mDist.	t
	kmol	kmol	kmol	kg	kmol	kmol	mol frac.	mol frac.	mol frac.	mol frac.	kg	h
1	1.30	8.58	15.92	1476.5	42.48	29.80	0.4817	0.3378	0.1805	0.9985	2397.4	16.1
2	1.98	11.11	19.26	1824.0	43.17	32.32	0.4556	0.3411	0.2033	0.9989	2468.5	19.7
3	2.10	11.9	20.43	1936.6	43.28	33.08	0.4471	0.3417	0.2111	0.9988	2468.5	20.7
4	2.06	11.23	20.26	1909.2	43.24	32.44	0.4507	0.3381	0.2112	0.9985	2478.1	21.9
5	2.03	11.77	20.93	1969.6	43.21	32.98	0.4449	0.3396	0.2155	0.9985	2477.4	20.8
6	2.19	12.33	21.00	1994.2	43.37	33.54	0.4429	0.3426	0.2145	0.9989	2465.9	21.0

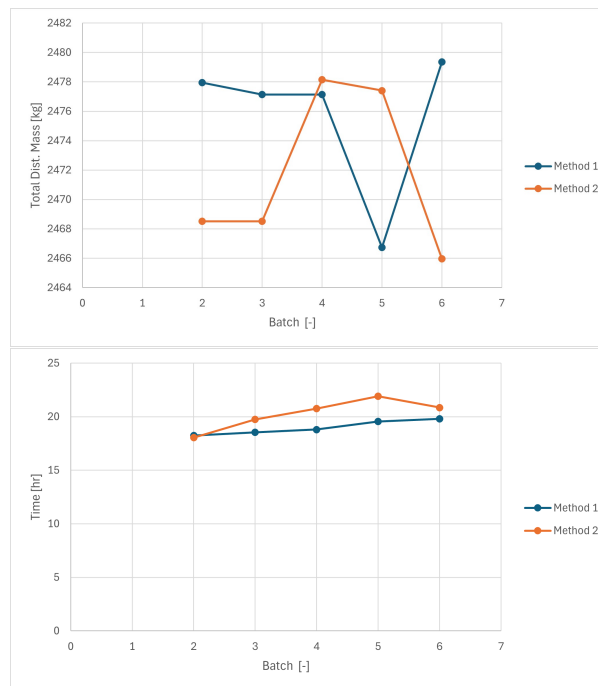


Figure 9.1: Comparison between the total distillate purity and total distillate mass results, respectively, of Method 1 and Method 2.

Results are listed in Table 9.4. Both methods provided a distillate with sufficient purity, meaning that all batches met the required product specification. Figure 9.1 compares the total distillate purity and total distillate mass obtained with Method 1 and Method 2.

The IPA distillate purity remained above the target in all batches, with an average value of 0.9987 mol fraction. However, the total distillate mass was slightly lower than in Method 1, with 14755.93 kg recovered over the six batches. The total operating time increased to 120.20 h, which is 9.20 h longer than Method 1.

9.2.2 DMSO Bottom Product Obtained During Method 2

Table 9.5: DMSO bottom product obtained during method 2.

B	DMSO	H ₂ O	xDMSO	xH ₂ O	mTotal
	kmol	kmol	mol frac.	mol frac.	kg
1	51.96	0.0255	0.9995	0.00049	4060.02
2	66.36	0.0323	0.9995	0.00049	5185.60
3	69.66	0.0331	0.9995	0.00048	5443.17
4	71.38	0.0343	0.9995	0.00048	5578.05
5	70.54	0.0330	0.9995	0.00047	5512.53
6	70.77	0.0328	0.9995	0.00046	5530.48

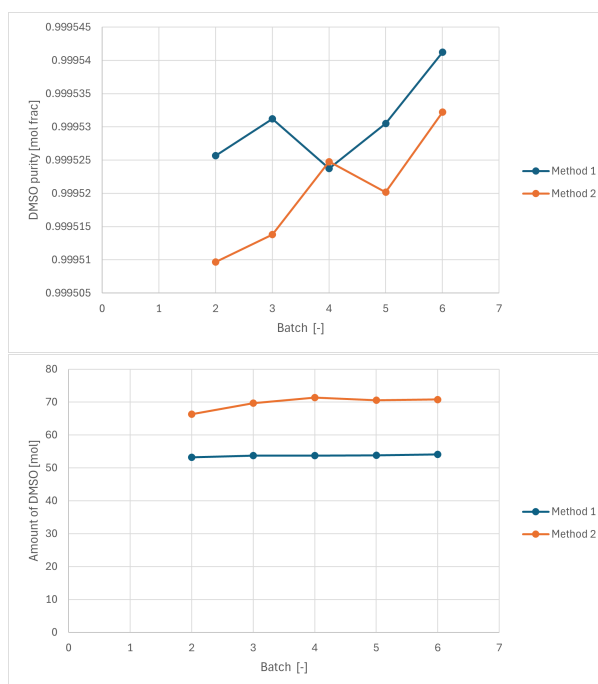


Figure 9.2: Comparison between the total bottom DMSO purity and total amount of DMSO recovered, respectively, with Method 1 and Method 2. Peaks can be seen when the reflux ratio is adjusted.

As showcased in Figure 9.2, DMSO bottom purity remained high, with an average value of 0.9995 mol fraction. However, the total DMSO bottom mass increased significantly compared with Method 1. This is expected because the water/DMSO off-cut is recycled directly into the pot charge, causing more DMSO to remain within the main batch sequence (see Table 9.5).

9.3 Comparison of the Two Recycling Methods

Table 9.6: Comparison between Method 1 and Method 2.

Parameter	Method 1	Method 2
Avg. xIPA Dist. [mol frac.]	0.9987	0.9987

Parameter	Method 1	Method 2
Total mDist. [kg]	14775.70	14755.93
Total t [h]	111.00	120.20
Total Bottom m [kg]	25050.56	31309.85
OC handled [kg]	7669.81 H ₂ O/DMSO OC	11110.08 combined OC
Recycle batches studied	5	5
Operating adjustments	Fewer	More

Table 9.6 shows that both recycling strategies achieved the required IPA product purity. The average IPA distillate purity was practically the same for both methods, at approximately 0.9987 mol fraction. However, Method 1 produced a slightly larger total IPA distillate mass and required less total operating time. Method 2 required 120.20 h compared with 111.00 h for Method 1, which means that total off-cut re-integration increased the total process time by 9.20 h over the six simulated batches.

Method 2 also required more changes to the simulation settings to maintain convergence and product quality. This suggests that adding both off-cut streams directly to the next pot charge makes the process more sensitive to composition changes. The larger DMSO bottom mass in Method 2 also shows that DMSO accumulates more strongly in the batch sequence when the water/DMSO stream is directly recycled.

Total off-cut re-integration is technically possible, since it still achieves the required IPA purity and produces a high-purity DMSO bottom stream, and it actually allows us to recover more total DMSO (moles) from the bottom of the column. However, it is less favorable than Method 1 because it increases the operating time, requires more reflux ratio adjustments, and causes greater DMSO accumulation in the system; furthermore, even though more DMSO is recovered overall, each batch produces lower quality bottom compared to Method 1. Based on this comparison, the preferred material recovery strategy is Method 1: water/IPA off-cut re-integration combined with separate later treatment of the water/DMSO off-cut. This approach gives similar product purity, slightly higher IPA recovery, shorter operating time, and better process stability. Since the reboiler's heat duty is directly correlated to the elapsed time due to it being set at 1000 MJ/h, less time needed to complete the process means that it is more environmentally and financially affordable, not only less time consuming.

9.4 Water/DMSO off-cut Distillation

The water/DMSO off-cut distillation serves as a following step to the water/IPA re-integration. The same column is to be used for this process. The goal of this extra step is to recover as much DMSO and water as possible, for it to be repurposed later. The obtained, total molar amount of water and DMSO from the water/DMSO off-cut were calculated after the 5 batches for Method 1, and this result was the pot charge for the distillation. There will be no feed or external entrainer. We created a separate ChemCAD simulation with the same column parameters and initial conditions, presented in Figure 9.3.

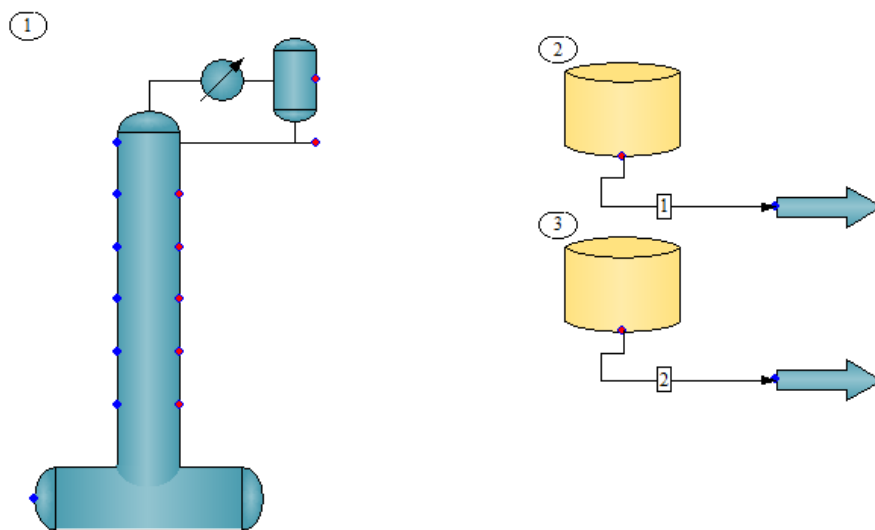


Figure 9.3: Simulation layout for the Water-DMSO distillation.

Different stop values, reflux ratios, and internal column values were investigated for this process. The final selected stop value was 0.998 mole fraction water in the accumulator, since water is more volatile than DMSO. The reflux ratio was varied in the range $R = 2$ to $R = 10$ (see Figure 9.4), and the pressure was studied between 1 bar and 0.1 bar (see Figure 9.5) [4].

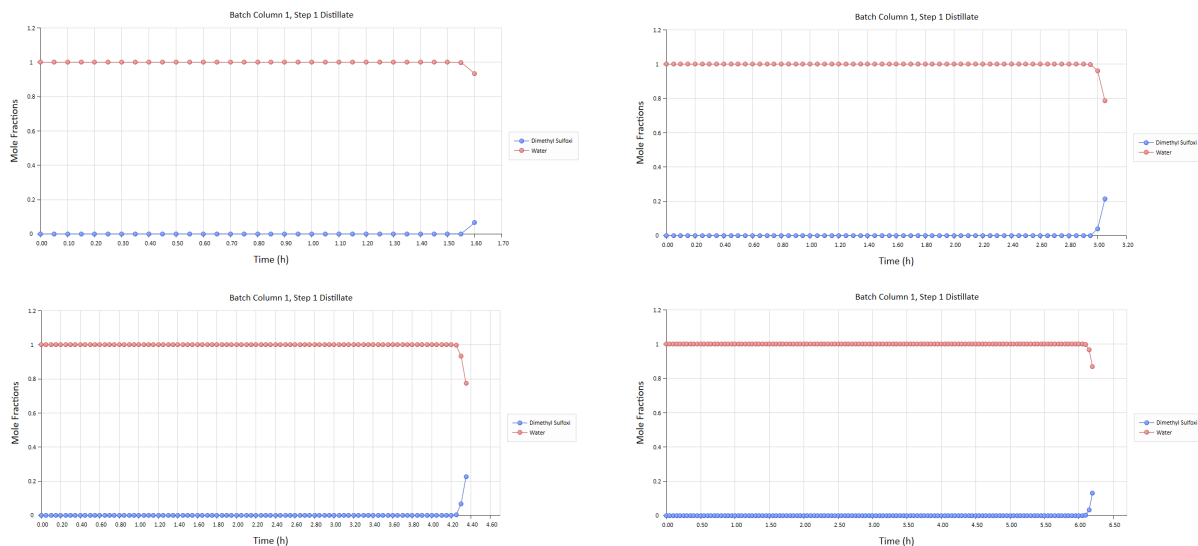


Figure 9.4: Distillate graphs demonstrating the effect of the reflux ratio, first parameter studied at 1 bar (at $R = 2, 4, 6$ and 9 , respectively).

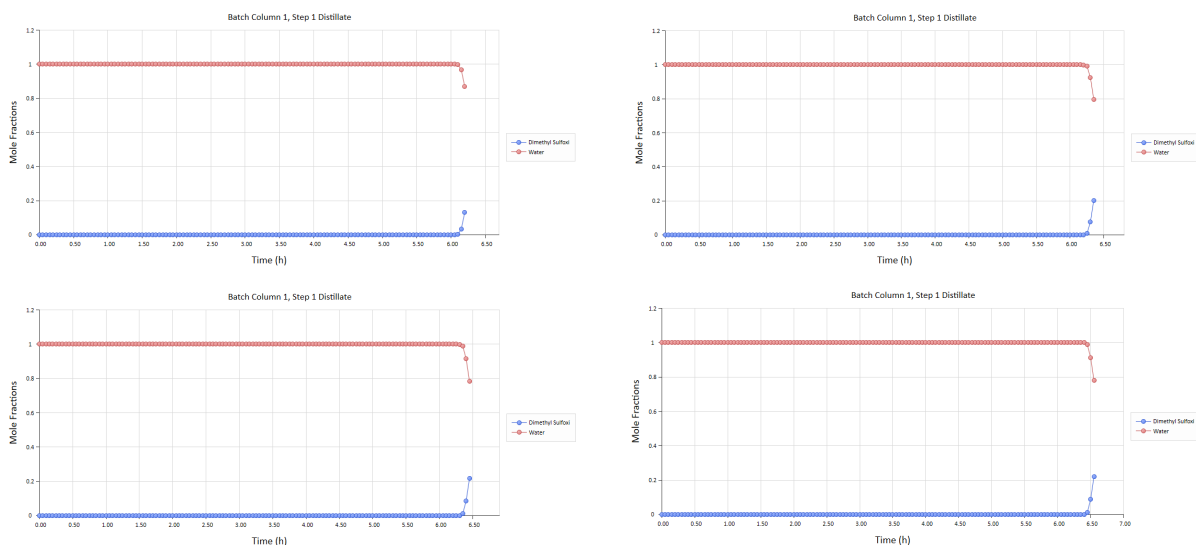


Figure 9.5: Distillate graphs demonstrating the effect of pressure when the reflux ratio is set to 9 (at 1, 0.5, 0.25 and 0.1 bar, respectively).

The reflux-ratio graphs must be interpreted together with the elapsed time and the actual distillate composition. At low reflux ratios, the separation may appear sharper because the operation is completed in a shorter time and the transition between composition regions is compressed on the time axis. However, this does not correspond to better separation, since the distillate obtained at lower reflux ratios contains a higher amount of DMSO. In contrast, at $R = 9$, the operation lasts longer, but the water-rich and DMSO-rich regions are more clearly separated, and DMSO carry-over into the distillate is reduced. Therefore, the high reflux ratio is required to provide sufficient rectification, not simply because of the longer operating time.

The pressure-variation graphs must be interpreted in the same way. DMSO has a non-zero vapour pressure throughout the distillation, so the relevant factor is not whether DMSO starts evaporating, but how the water/DMSO ratio in the vapour and distillate changes as the batch progresses. The profiles below 1 bar appear almost unchanged at first, but the slightly longer operating times give a clearer transition between the water-rich and DMSO-rich regions. The length and slope of the decreasing section remain approximately similar, while the total elapsed time increases, indicating a more gradual and better-resolved separation. Based on these results, the selected conditions were $R = 9$ and $P = 0.1$ bar, which gave the most suitable separation between the water-rich distillate and the DMSO-rich bottom product.

However, the ChemCAD simulation needed additional aid in order to converge. Hold-up values were added: 5 dm³ for the condenser and 3 dm³ for the stages [20]. These represent the liquid volume that is retained due to the geometry of the condenser and the type of packing. Adding hold-up values also makes the process more accurate and realistic.

In order to maintain realism, several batches for both recycling methods were performed again, with the same hold-up values for the column as in the water/DMSO distillation. Our products (composition, mass, time) only slightly changed, affecting the values only after 2-3 decimal points. Purity of the distillate products remained acceptable. It was then concluded that it is not necessary to add hold-up values in a more complex ChemCAD simulation for accurate results.

The recovered product is summarized below:

- Mass of recovered water: 268.72 kg
- Purity of recovered water: 0.9975 mole fraction water
- Mass of recovered DMSO: 7401.16 kg
- Purity of recovered DMSO: 0.9975 mole fraction DMSO
- Total elapsed time: 6.55 h

As can be seen, Method 1 still gives better yields than Method 2, and the new elapsed time is still shorter: 117.55 h compared with 120.20 h. Therefore, even though it includes more steps, Method 1 can be considered the more efficient option.

Since this distillation process is under vacuum, a proper vacuum pump was selected for this application. A breakdown of the selection is carried out in the next sub-chapter.

9.5 Vacuum Pump Selection

A vacuum pump was required to evacuate the distillation system prior to the start of each batch. The objective was to reduce the system pressure from atmospheric pressure (1.0 bar absolute) to the operating pressure of 0.1 bar absolute before heating commenced. Since the vacuum pump is only used during the initial evacuation stage and not during normal distillation operation, the pump was sized based on the required pump-down time and the total gas volume contained within the equipment.

The total volume to be evacuated consists of the vapour space in the feed tank and the internal volume of the distillation column. The feed tank has a total volume of 7.85 m³ and contains approximately 6 m³ of liquid before the start of the distillation, resulting in a vapour space of 1.85 m³. The distillation column has a height of 10 m and a diameter of 0.76 m, corresponding to an internal volume of approximately 4.54 m³. Therefore, the total gas volume is approximately 6.39 m³. To allow for additional system volume, the total gas volume to be evacuated was approximated as 6.5 m³.

The required pumping speed was calculated using the standard vacuum pump-down equation [24]:

$$S = \frac{V}{t} \ln \left(\frac{P_1}{P_2} \right) \quad (9.1)$$

where:

- S = required pumping speed (m³/s)
- V = gas volume to be evacuated (m³)
- t = desired pump-down time (s)
- P_1 = initial pressure (bar abs)
- P_2 = final pressure (bar abs)

Substituting the design values:

$$V = 6.5 \text{ m}^3$$

We want the pump-down time to be approximately 20 minutes:

$$t = 20 \times 60 = 1200 \text{ s}$$

This gives:

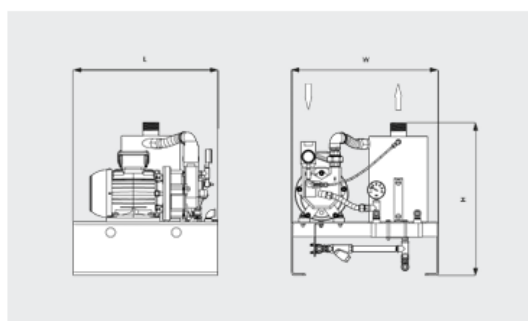
$$S = \frac{6.5}{1200} \ln\left(\frac{1.0}{0.1}\right) = 0.01247 \text{ m}^3/\text{s} = 44.9 \text{ m}^3/\text{h}$$

Therefore, the vacuum system must provide a minimum pumping speed of approximately 45 m³/h to achieve the required vacuum level under the desired time.

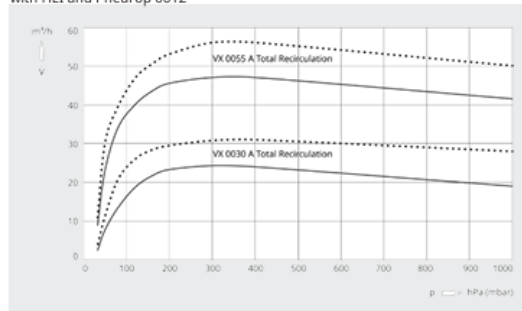
A liquid ring vacuum pump was selected due to its suitability for handling condensable vapors and traces of process fluids. Compared to oil-sealed rotary vane pumps, liquid ring pumps are more tolerant to water vapor and solvent carryover, making them well suited for vacuum distillation applications. Since the process involves water and dimethyl sulfoxide (DMSO), a liquid ring design provides increased operational robustness and reliability [25].

The selected vacuum pump is the Busch DOLPHIN VX 0055, a total recirculation liquid ring vacuum pump (see Figure 9.6). The manufacturer specifies a nominal pumping speed of 47 m³/h at 50 Hz and an ultimate pressure of 33 mbar absolute. Since the required operating pressure is 100 mbar absolute and the required pumping speed is 45 m³/h, the selected pump satisfies both requirements.

Dimensional drawing



Pumping speed
Dry air at 20 °C. Operating fluid water
15°C. Tolerance: - 10%, in accordance
with HEI and Pneurop 6612



	DOLPHIN VX 0030 A Total Recirculation	DOLPHIN VX 0055 A Total Recirculation
Nominal pumping speed	24 / 31 m ³ /h (50 / 60 Hz)	47 / 56 m ³ /h (50 / 60 Hz)
Ultimate pressure	33 hPa (mbar) (50 / 60 Hz)	33 hPa (mbar) (50 / 60 Hz)
Nominal motor rating	1.1 / 1.5 kW (50 / 60 Hz)	2.2 / 1.5 kW (50 / 60 Hz)
Weight approx.	73 / 81 kg (50 / 60 Hz)	83 / 91 kg (50 / 60 Hz)
Dimensions (L x W x H)	540 x 550 x 570 mm	540 x 550 x 570 mm
Gas inlet / outlet	G 1" / R 2"	G 1" / R 2"

Figure 9.6: Dimensional drawing, pumping speed vs flow rate plot, and specifications of the selected pump [25].

Chapter 10

Cost Estimation

The cost estimation of the distillation system was carried out using preliminary equipment cost correlations. This type of estimation is suitable for conceptual design because it allows the main process equipment costs to be estimated from the principal sizing parameters, such as heat transfer area, column diameter, column height, tank dimensions, and compressor horsepower. The objective of this chapter is not to obtain a final vendor quotation, but to estimate the relative investment required for the main distillation system and for the two heat recycling alternatives. However, the costs of cooling water will be neglected in this estimation.

The equipment cost correlations were taken from literature [26], where purchased equipment costs are estimated from equipment size and then corrected to installed costs using installation factors. Since the original correlations are based on an older cost index, the Marshall and Swift index correction was applied. The Marshall & Swift index used in this calculation, for the year 2020, was [27]:

$$MS = 2171.6$$

Two tanks were assumed: one tank for the pot charge and one tank for the distillate.

10.1 Energy Cost Estimation

The heating energy cost was estimated from the reboiler duty and the additional energy required to heat the DMSO feed from 20°C to 62°C. Since the process heating duties are supplied by steam, the steam cost was used for these calculations instead of the electricity cost. Electricity is only used later for the compressor work required by the heat pump method. As of 2013 in the United States, literature reports high-pressure steam and electricity costs of approximately [20]:

$$C_{\text{HPS}} = \$ 9.88/\text{GJ}$$

$$C_{\text{electricity}} = \$ 16.8/\text{GJ}$$

where C_{HPS} is the cost of high-pressure steam and $C_{\text{electricity}}$ is the cost of electricity. The heating cost was calculated by multiplying the required heat input by the steam cost:

$$C = Q_{\text{total}} \cdot C_{\text{HPS}} \quad (10.1)$$

where C is the utility cost and Q_{total} is the total heating energy demand. The reboiler operated for 16.15 h at 1000 MJ/h:

$$Q_{\text{reb,main}} = 1000 \cdot 16.15 = 16150 \text{ MJ} = 16.15 \text{ GJ} \quad (10.2)$$

The DMSO feed-heating duty was approximately 50 MJ/h, this value was obtained from simulation measurements. Since the feed is heated during the main distillation step, which lasts 9.05 h:

$$Q_{\text{feed}} = 50 \cdot 9.05 = 452.5 \text{ MJ} = 0.4525 \text{ GJ}$$

Therefore, the total heating energy required for the complete base distillation sequence was:

$$Q_{\text{main,total}} = 16.15 + 0.4525 = 16.6025 \text{ GJ}$$

$$C_{\text{main}} = 16.6025 \cdot 9.88 = \$ 164.03/\text{batch}$$

The estimated operating cost of the complete base distillation heating duty is therefore \$ 164.03/batch.

10.2 Heat Exchanger Cost Estimation

The heat transfer areas used for the cost estimation were obtained from the previously calculated exchanger duties and temperature differences.

The resulting areas were converted from square meters to square feet using:

$$1 \text{ m}^2 = 10.764 \text{ ft}^2$$

The converted heat transfer areas used in the cost estimation are shown in Table 10.1, their values converted from meters to feet, in order to follow the calculations from the selected literature.

Table 10.1: Converted heat transfer areas used for cost estimation.

Equipment	Area [ft ²]
Condenser	129.93
Reboiler	84.04
Inter-cooler	5.39
Pre-heater	6.65

The purchased cost of a heat exchanger was calculated as [26]:

$$C_{p,HX} = \frac{MS}{280} (101.3A^{0.65}F_c) \quad (10.3)$$

where:

- $C_{p,HX}$ = purchased heat exchanger cost, in \$
- A = heat transfer area, in ft²
- F_c = correction factor

- MS = Marshall and Swift index

The installed cost was calculated using [26]:

$$C_{i,HX} = \frac{MS}{280} (101.3A^{0.65}(F_c + 2.29)) \quad (10.4)$$

where $C_{i,HX}$ is the installed heat exchanger cost, in \$. The operating steam cost was calculated separately in the energy-cost estimation section. The cost results for all heat exchangers used are shown in Table 10.2.

Table 10.2: Heat exchanger cost estimation results.

Equipment	Area [ft ²]	F_c	Purchased [\$]	Installed [\$]
Condenser	129.93	0.80	14867.20	57424.56
Reboiler	84.04	1.60	22401.64	54463.99
Inter-cooler	5.39	0.80	1879.36	7259.03
Pre-heater	6.65	0.80	2154.16	8320.45

The condenser and reboiler are part of the base distillation system. The inter-cooler is only required for the heat pump method, while the pre-heater is only required for the pre-heater heat recycling method. The installed reboiler cost is therefore only the installed capital cost of the reboiler.

10.3 Tank Cost Estimation

Two tanks were included in the cost estimation: one for the pot charge and one for the distillate. The tank dimensions used in the calculation are shown in Table 10.3.

Table 10.3: Tank sizing values.

Parameter	Symbol	Value	Unit
Tank height	H	8.20	ft
Tank diameter	D	6.56	ft
Number of tanks	N	2	-
Correction factor	F_c	1.00	-

The purchased cost of the two tanks was calculated with [26]:

$$C_{p,tank} = 2 \frac{MS}{280} (101.9D^{1.066}H^{0.802}F_c) \quad (10.5)$$

where $C_{p,tank}$ is the purchased cost of the two tanks, D is the tank diameter, H is the tank height, F_c is the correction factor, and MS is the Marshall and Swift index. Substituting the selected tank dimensions:

$$C_{p,tank} = \$ 63463.52$$

The installed cost of the two tanks was calculated as well [26]:

$$C_{i,tank} = 2 \frac{MS}{280} (101.9D^{1.066}H^{0.802}(F_c + 2.18)) \quad (10.6)$$

where $C_{i,tank}$ is the installed cost of the two tanks. Substituting the selected values:

$$C_{i,tank} = \$ 201814.01$$

10.4 Column Cost Estimation

The cost of the distillation column was estimated using the column height and diameter. The column dimensions used in the cost estimation are shown in Table 10.4, converted to feet:

Table 10.4: Column sizing values.

Parameter	Symbol	Value	Unit
Column height	H	32.80	ft
Column diameter	D	2.49	ft
Correction factor	F_c	1.05	-

The purchased cost of the column shell was calculated using [26]:

$$C_{p,col} = \frac{MS}{280} (101.9D^{1.066}H^{0.802}F_c) = \$ 36217.64 \quad (10.7)$$

where $C_{p,col}$ is the purchased cost of the column shell, D is the column diameter, H is the column height, F_c is the correction factor, and MS is the Marshall and Swift index. The installed cost of the column shell was calculated to be [26]:

$$C_{i,col} = \frac{MS}{280} (101.9D^{1.066}H^{0.802}(F_c + 2.18)) = \$ 111412.36 \quad (10.8)$$

where $C_{i,col}$ is the installed cost of the column shell.

10.5 Compressor Cost Estimation

The compressor costs correspond to the heat pump method. Two compressors are required, one for each compression stage. The compressor powers were obtained from the heat pump simulation and converted into horsepower, as shown in Table 10.5.

Table 10.5: Compressor power values.

Equipment	Power [hp]
Compressor 1	36.36
Compressor 2	37.15

The purchased compressor cost was calculated using the formula [26]:

$$C_{p,comp} = \frac{MS}{280} (51.75HP^{0.82}F_t) \quad (10.9)$$

where $C_{p,comp}$ is the purchased compressor cost, HP is the compressor power in horsepower, F_t is the compressor correction factor, and MS is the Marshall and Swift index. Since two compressors are required:

$$C_{p,comp,total} = \frac{MS}{280} (51.75HP_1^{0.82}F_t) + \frac{MS}{280} (51.75HP_2^{0.82}F_t)$$

where:

- HP_1 = horsepower of compressor 1
- HP_2 = horsepower of compressor 2
- F_t = compressor correction factor

Using $F_t = 1$, the total purchased compressor cost is:

$$C_{p,comp,total} = \$ 15420.85$$

The installed compressor cost was calculated using:

$$C_{i,comp,total} = \frac{MS}{280} (51.75HP_1^{0.82}(F_t + 2.11)) + \frac{MS}{280} (51.75HP_2^{0.82}(F_t + 2.11)) \quad (10.10)$$

where $C_{i,comp,total}$ is the installed cost of the two compressors. The resulting installed compressor cost for the heat pump method is \$ 47958.84.

10.6 Packing Cost Estimation

The selected column internal is MellapakPlus 252.Y structured packing. Since no vendor quotation was available, the packing cost was estimated from the random-packing cost data given by literature and then corrected with a structured-packing multiplier.

Stainless steel Pall rings are listed at approximately \$ 1750 – 3200/m³ [20]. Since the selected packing is metallic, the average value was used as the reference random-packing cost (after EUR to USD conversion, as of June 2026):

$$C_{random,unit} = \$ 2475/\text{m}^3$$

The column has a selected diameter of 0.76 m and a packed height of 10 m, giving a packing volume of:

$$V_{packing} = 4.54 \text{ m}^3$$

Therefore, the estimated equivalent random-packing cost is:

$$C_{random} = 4.54 \times 2475 = \$ 11236.50$$

Structured packing is reported to cost approximately 30 – 60% more per cubic meter than comparable metal random packing [28]. Therefore, the midpoint (45%) was used. The estimated structured-packing cost is therefore:

$$C_{packing} = 11236.5 \times 1.45 = \$ 16292.92 \quad (10.11)$$

where $C_{packing}$ is the estimated cost of the selected structured packing.

10.7 Solvent Cost Estimation

This subchapter estimates the cost of purchasing the DMSO solvent and compares it with the steam cost required to recover DMSO from the water/DMSO off-cut stream. Recovering it reduces the amount of fresh solvent that must be purchased.

For the DMSO purchase price, a recent European market value was used [29]. The price of DMSO in Germany was reported as, after converting the price from EUR to USD as of June 2026:

$$C_{DMSO} = \$ 5.322/\text{kg}$$

From the material recovery chapter, the water/DMSO off-cut distillation recovered 7401.16 kg of usable DMSO. The estimated cost of purchasing an equivalent amount of fresh DMSO would be:

$$C_{DMSO,purchase} = 7401.16 \times 5.322 = \$ 39388.97 \quad (10.12)$$

The water/DMSO recovery distillation required a total elapsed time of 6.55 h. Since the reboiler duty was set at 1000 MJ/h, the total energy required for this recovery step was:

$$E_{recovery} = 1000 \times 6.55 = 6550 \text{ MJ} = 6.55 \text{ GJ}$$

where $E_{recovery}$ is the energy required for the recovery distillation step. Since this is a reboiler heating duty, the cost was calculated using the high-pressure steam cost [20]:

$$C_{HPS} = \$ 9.88/\text{GJ}$$

The steam cost of the DMSO recovery step is:

$$C_{recovery} = 6.55 \times 9.88 = \$ 64.71$$

Table 10.6 compares the cost of purchasing fresh DMSO with the steam cost required for DMSO recovery.

Table 10.6: Comparison of DMSO purchase cost and recovery energy cost.

Parameter	Value
DMSO purchase cost [\$]	39388.97
Steam cost [\$/GJ]	9.88
Recovery steam cost [\$]	64.71
Net saving by recovery [\$]	39324.26

The comparison shows that recovering DMSO is much cheaper than purchasing the same amount of fresh solvent. The steam cost required to recover 7401.16 kg of DMSO is approximately \$ 64.71, while purchasing the same amount of DMSO would cost approximately

\$ 39388.97. Therefore, the estimated net saving is \$ 39324.26. From a feed-cost perspective, DMSO recovery is financially advisable. In addition, it reduces solvent waste and improves the sustainability of the material recovery process.

10.8 Base Distillation System Cost

The base distillation system includes the two tanks, the column shell, the column internals, condenser, and reboiler. The column internals correspond to the selected structured packing, with an estimated packing cost of:

$$C_{packing} = \$ 16292.92$$

The installed costs used to calculate the base system cost are summarized in Table 10.7.

Table 10.7: Base distillation system installed cost.

Equipment	Installed Cost [\$]
Two tanks	201814.01
Column shell	111412.36
Packing	16292.92
Condenser	57424.56
Reboiler	54463.99

The total installed cost of the base system is:

$$C_{base} = 201814.01 + 111412.36 + 16292.92 + 57424.56 + 54463.99 = \$ 441407.84 \quad (10.13)$$

where C_{base} is the total installed cost of the base distillation system. This is a capital cost only and does not include the batch operating cost of steam.

10.9 Cost of the Pre-heater Method

The pre-heater method only requires one additional heat exchanger; its installed cost was calculated earlier in this chapter:

$$C_{i,pre} = \$ 8320.45$$

where $C_{i,pre}$ is the installed cost of the pre-heater. Therefore, the total installed cost of the system with the pre-heater method is \$ 449728.29.

10.10 Cost of the Heat Pump Method

The heat pump method requires two compressors and an inter-cooler. The compressor cost was estimated earlier in this chapter using the compressor cost correlation [26]. The total purchased compressor cost is:

$$C_{p,comp,total} = \$ 15420.85$$

The installed cost of the two compressors is:

$$C_{i,comp,total} = \$ 47958.84$$

The installed cost of the inter-cooler is:

$$C_{i,int} = \$ 7259.03$$

where $C_{i,int}$ is the installed cost of the inter-cooler. Therefore, the total additional installed cost of the heat pump system is \$ 55217.87, making the grand total \$ 496625.71.

10.11 Comparison of Heat Recycling Alternatives

The two heat recycling alternatives were compared by considering both their additional installed cost and their operating utility saving. The additional installed costs calculated in the previous subchapters are summarized in Table 10.8.

Table 10.8: Cost comparison of heat recycling methods.

Method	Additional Equipment	Add. Installed Cost [\$]
Pre-heater	heat exch.	8320.45
Heat pump	2 compressors + inter-cooler	55217.87

For the operating comparison, one batch per operating day and 330 operating days per year were assumed. Therefore, over a 10-year period:

$$N_{cycles} = 330 \cdot 10 = 3300 \text{ batches} \quad (10.14)$$

where N_{cycles} is the number of batches operated during the 10-year period. This value is used to calculate the total 10-year saving from the saving per batch.

For the pre-heater method, the recovered heat duty from the ChemCAD heat-exchanger simulation was 45.1324 MJ/h. Since this heat is recovered during the 9.05 h DMSO feed-heating period:

$$E_{pre} = 45.1324 \cdot 9.05 = 408.45 \text{ MJ} = 0.40845 \text{ GJ/batch} \quad (10.15)$$

where E_{pre} is the heat recovered by the pre-heater per batch. Since this replaces external steam heating, the saving is calculated using the high-pressure steam cost:

$$C_{pre,saving} = 0.40845 \cdot 9.88 = \$ 4.04/\text{batch} \quad (10.16)$$

For the heat pump method, the useful operating period was 4.5 h. During this time, the heat pump replaces the reboiler steam duty of 1000 MJ/h:

$$E_{HP,steam} = 1000 \cdot 4.5 = 4500 \text{ MJ} = 4.5 \text{ GJ/batch} \quad (10.17)$$

The value of this saved steam is:

$$C_{HP,steam\ saved} = 4.5 \cdot 9.88 = \$ 44.46/\text{batch} \quad (10.18)$$

However, the heat pump also requires electricity for the two compressors. Their duties were 97.61 MJ/h and 99.72 MJ/h, both operating for 4.5 h:

$$E_{HP,electricity} = (97.61 + 99.72) \cdot 4.5 = 887.99 \text{ MJ} = 0.888 \text{ GJ/batch} \quad (10.19)$$

The electricity cost is:

$$C_{HP,electricity} = 0.888 \cdot 16.8 = \$ 14.92/\text{batch} \quad (10.20)$$

Therefore, the net saving of the heat pump method is:

$$C_{HP,saving} = 44.46 - 14.92 = \$ 29.54/\text{batch} \quad (10.21)$$

The 10-year utility saving was calculated by multiplying the saving per batch by N_{cycles} . The net 10-year saving was then obtained by subtracting the additional installed cost. The results are shown in Table 10.9.

Table 10.9: Ten-year comparison of heat recycling methods.

Method	Steam Saved [GJ/batch]	Electricity Used [GJ/batch]	Net Saving [\$ /batch]	10-year Saving [\$]	Add. Installed Cost [\$]	Net 10-year Saving [\$]	Payback [years]
Pre-heater method	0.408	0.000	4.04	13317.10	8320.45	4996.65	6.25
Heat pump method	4.500	0.888	29.54	97488.11	55217.87	42270.24	5.66

The pre-heater method has the lower installed cost, because it only requires one additional heat exchanger. It also gives a smaller utility saving, because it only recovers part of the DMSO feed-heating duty. The heat pump method has a higher installed cost, because it requires two compressors and an inter-cooler, but it gives a larger utility saving by replacing part of the reboiler steam duty. The electricity required by the compressors was subtracted from the saved steam cost because steam and electricity have different unit costs. Therefore, both methods are economically justifiable over the assumed 10-year operating period, while the heat pump method gives the larger net saving.

Chapter 11

Summary

This paper studied the regeneration of a pharmaceutical waste solvent by batch extractive distillation. The objective was to recover an IPA-rich distillate with at least 99.75 mol% IPA from an IPA–water mixture using DMSO as an extractive solvent. Since IPA and water form a minimum-boiling azeotrope, conventional batch distillation cannot reach the required purity. ChemCAD simulations were used to study the IPA–water–DMSO system and evaluate the effects of solvent feed stage, reflux ratio, feed rate, number of stages, and feed temperature. Under the selected operating conditions, the main distillation step reached a maximum IPA purity of 99.856 mol% and lasted 9.05 hours.

Additional distillation stages were considered to recover remaining IPA, water, and DMSO from the bottom mixture, improving material recovery and reducing waste. Two recycling strategies were compared. The preferred option was the reintegration of the water/IPA run-off into the following batch, while collecting the water/DMSO run-off separately. This method provided similar product purity, slightly higher IPA recovery, shorter operating time, and better process stability.

The heating energy requirement of the complete base distillation sequence was approximately 16.60 GJ, corresponding to an estimated steam cost of \$ 164.03/batch. The DMSO recovery distillation required an additional 6.55 GJ, corresponding to a steam cost of \$ 64.71. Since recovering 7401.16 kg of DMSO avoids an estimated fresh solvent purchase cost of \$ 39388.97, DMSO recovery gives an estimated net saving of \$ 39324.26.

Two heat-recycling alternatives were also studied. The pre-heater method was simpler and required much lower investment, giving an estimated net saving of \$ 4996.65 over 10 years. The heat pump method involved more equipment but offered greater utility savings, giving an estimated net saving of \$ 42270.24 over 10 years after subtracting the additional installed cost.

The main process equipment was preliminarily sized, including the packed column, tanks, condenser, reboiler, heat-recovery equipment, and vacuum pump. The selected tank geometry had a diameter of 2.00 m, a height of 2.50 m, and a total volume of 7.85 m³. The estimated installed cost of the base distillation system was \$ 441407.84.

Overall, the results show that batch extractive distillation with DMSO is a feasible method for regenerating high-purity IPA from pharmaceutical waste solvent streams. The proposed process supports solvent recovery, run-off reuse, waste reduction, and improved energy efficiency, making it a technically viable and potentially sustainable regeneration solution.

Declaration about the independent work

I, Camille Cardoze (Y3GTYB), hereby declare that the Thesis submitted for assessment and defence, exclusively contains the results of my own work assisted by my supervisor. Further to it, it is also stated that all other results taken from the technical literature or other sources are clearly identified and referred to according to copyright (footnotes/references are chapter and verse, and are placed appropriately).

I accept that the scientific results presented in my Thesis can be utilised by the Department of the supervisor for further research or teaching purposes.

Budapest, 6 July 2026

A handwritten signature in black ink, appearing to read 'Camille', with a large, sweeping loop at the end.

Cardoze Camille

Declaration about the acceptability of the thesis

This BSc thesis of Cardoze Camille, “Regeneration of a pharmaceutical waste solvent by batch extractive distillation” fulfills every formal and content requirements of the regulation of the Budapest University of Technology and Economics, moreover it fulfills the assignment of the final project. This thesis is suitable for a review and an open defense.

Budapest, 6 July 2026

Dr László Hégyel
Supervisor

Appendix A

Batch Results ChemCAD Report of the Distillation Process

Simulation: Batch distillation

BATCH DISTILLATION 1 CALCULATION RESULTS

Operation Step 1:

STREAM PROPERTIES

Name	Pot Charge	Accumulator	Pot Residue	Distillate
- - Overall - -				
Molar flow kmol	71.5527	41.2935	100.0092	0.1841
Mass flow kg	3097.6099	2479.3572	6066.2036	10.3529
Temp C	80.2389	82.4446	142.0314	81.2891
Pres bar	1.0130	1.0130	1.0130	1.0130
Vapor mole fraction	0.000	1.105E-05	0.000	0.000
Enth MJ	-21268.	-12685.	-25329.	-56.153
Tc C	257.5353	235.2141	440.3150	239.0848
Pc bar	58.6826	47.7569	117.6872	53.9094
Std. sp gr. wtr = 1	0.822	0.794	1.087	0.799
Std. sp gr. air = 1	1.495	2.073	2.094	1.942
Degree API	40.6544	46.7489	-1.3678	45.6866
Average mol wt	43.2913	60.0423	60.6564	56.2480
Actual dens kg/m3	756.8923	608.4261	964.8813	729.2065
Actual vol m3	4.0925	4.0750	5.9978	0.0142
Std liq m3	3.7687	3.1233	5.5789	0.0130
Std vap 0 C m3	1603.7571	925.5381	2241.5715	4.1254
- - Vapor only - -				
Molar flow kmol		0.0227		
Mass flow kg		1.3630		
Average mol wt		60.0083		
Actual dens kg/m3		2.1118		
Actual vol m3		0.6454		
Std liq m3		0.0017		
Std vap 0 C m3		0.5091		
Cp kJ/kg-K		1.7026		
Z factor		0.9736		
Visc N-s/m2		9.301e-06		
Th cond W/m-K		0.0224		
- - Liquid only - -				
Molar flow kmol	71.5527	41.2708	100.0092	0.1841
Mass flow kg	3097.6099	2477.9939	6066.2036	10.3529
Average mol wt	43.2913	60.0423	60.6564	56.2480
Actual dens kg/m3	756.8923	722.5254	964.8813	729.2065
Actual vol m3	4.0925	3.4296	6.2870	0.0142
Std liq m3	3.7687	3.1216	5.5789	0.0130
Std vap 0 C m3	1603.7571	925.0291	2241.5715	4.1254
Cp kJ/kg-K	3.5054	3.3993	2.3643	3.4065
Z factor	0.0024	0.0033	0.0026	0.0031
Visc N-s/m2	0.0004483	0.0005015	0.0003773	0.0004961
Th cond W/m-K	0.1506	0.1219	0.1335	0.1261
Surf. tens. N/m	0.0202	0.0164	0.0287	0.0170
Component mole fractions				
Isopropanol	0.600658	0.998707	0.017385	0.908548
Water	0.399342	0.001281	0.285512	0.091445
DiMth Sulfoxide	0.000000	0.000012	0.697103	0.000007

Operation Step 2:

STREAM PROPERTIES

Name	Accumulator	Pot Residue	Distillate
- - Overall - -			
Molar flow kmol	8.7748	91.2344	0.1966
Mass flow kg	231.0943	5835.1094	3.5674
Temp C	81.6701	155.1982	98.5217
Pres bar	1.0130	1.0130	1.0130
Vapor mole fraction	0.01001	0.000	0.000
Enth MJ	-2510.9	-22693.	-55.098
Tc C	310.1375	446.5912	372.7422
Pc bar	100.4756	106.9352	217.9982
Std. sp gr. wtr = 1	0.895	1.097	0.997
Std. sp gr. air = 1	0.909	2.208	0.626
Degree API	26.5887	-2.4750	10.3711
Average mol wt	26.3362	63.9573	18.1432
Actual dens kg/m3	83.0079	959.8750	955.9047
Actual vol m3	2.7840	5.8501	0.0037
Std liq m3	0.2582	5.3207	0.0036
Std vap 0 C m3	196.6749	2044.8966	4.4070
- - Vapor only - -			
Molar flow kmol	0.0879	0.0001	
Mass flow kg	3.5833	0.0039	
Average mol wt	40.7794	33.9610	
Actual dens kg/m3	1.4254	0.9757	
Actual vol m3	2.5139	0.0040	
Std liq m3	0.0043	0.0000	
Std vap 0 C m3	1.9695	0.0026	
Cp kJ/kg-K	1.7368	1.6254	
Z factor	0.9824	0.9900	
Visc N-s/m2	1.015e-05	1.244e-05	
Th cond W/m-K	0.0217	0.0232	
- - Liquid only - -			
Molar flow kmol	8.6869	91.2343	0.1966
Mass flow kg	227.5110	5835.1055	3.5674
Average mol wt	26.1901	63.9573	18.1432
Actual dens kg/m3	842.3177	960.5107	955.9048
Actual vol m3	0.2701	6.0750	0.0037
Std liq m3	0.2539	5.3207	0.0036
Std vap 0 C m3	194.7054	2044.8940	4.4070
Cp kJ/kg-K	3.8367	2.3332	4.2114
Z factor	0.0013	0.0027	0.0008
Visc N-s/m2	0.0003738	0.0003747	0.0002862
Th cond W/m-K	0.2905	0.1246	0.6637
Surf. tens. N/m	0.0329	0.0273	0.0580
Component mole fractions			
Isopropanol	0.197742	0.000039	0.003047
Water	0.802258	0.235812	0.996953
DiMth Sulfoxide	0.000000	0.764150	0.000000

Operation Step 3:

STREAM PROPERTIES

Name	Accumulator	Pot Residue	Distillate
- - Overall - -			
Molar flow kmol	19.5114	71.7231	0.1574
Mass flow kg	352.7416	5482.3677	3.5762
Temp C	99.8998	183.8788	101.5942
Pres bar	1.0130	1.0130	1.0130
Vapor mole fraction	0.000	0.000	0.000
Enth MJ	-5463.6	-16957.	-43.691
Tc C	374.3753	452.2162	392.4992
Pc bar	221.3230	62.0678	237.2394
Std. sp gr. wtr = 1	1.000	1.104	1.026
Std. sp gr. air = 1	0.624	2.639	0.784
Degree API	9.9681	-3.2756	6.4059
Average mol wt	18.0788	76.4380	22.7152
Actual dens kg/m3	958.4564	931.7993	973.3488
Actual vol m3	0.3680	5.7836	0.0037
Std liq m3	0.3527	4.9680	0.0035
Std vap 0 C m3	437.3209	1607.5757	3.5287
- - Vapor only - -			
Molar flow kmol		0.0001	
Mass flow kg		0.0079	
Average mol wt		68.6628	
Actual dens kg/m3		1.8680	
Actual vol m3		0.0042	
Std liq m3		0.0000	
Std vap 0 C m3		0.0026	
Cp kJ/kg-K		1.5138	
Z factor		0.9799	
Visc N-s/m2		1.087e-05	
Th cond W/m-K		0.0207	
- - Liquid only - -			
Molar flow kmol	19.5114	71.7230	0.1574
Mass flow kg	352.7416	5482.3594	3.5762
Average mol wt	18.0788	76.4380	22.7152
Actual dens kg/m3	958.4564	932.4653	973.3488
Actual vol m3	0.3680	5.8794	0.0037
Std liq m3	0.3527	4.9680	0.0035
Std vap 0 C m3	437.3209	1607.5729	3.5287
Cp kJ/kg-K	4.2100	2.3162	3.6354
Z factor	0.0008	0.0031	0.0011
Visc N-s/m2	0.0002821	0.0004125	0.0002971
Th cond W/m-K	0.6729	0.1025	0.4664
Surf. tens. N/m	0.0585	0.0226	0.0508
Component mole fractions			
Isopropanol	0.000180	0.000000	0.000000
Water	0.998885	0.028227	0.921819
DiMth Sulfoxide	0.000934	0.971773	0.078181

Operation Step 4:

STREAM PROPERTIES

Name	Accumulator	Pot Residue	Distillate
- - Overall - -			
Molar flow kmol	17.5626	54.1605	0.3886
Mass flow kg	1252.0337	4230.3335	30.1852
Temp C	171.1395	188.7203	187.4501
Pres bar	1.0130	1.0130	1.0130
Vapor mole fraction	0.008973	0.000	0.000
Enth MJ	-4240.6	-12700.	-91.317
Tc C	450.1214	452.8400	452.6825
Pc bar	79.7714	56.5894	57.9808
Std. sp gr. wtr = 1	1.101	1.104	1.104
Std. sp gr. air = 1	2.461	2.697	2.682
Degree API	-2.9800	-3.3631	-3.3411
Average mol wt	71.2897	78.1074	77.6796
Actual dens kg/m3	179.2986	896.0194	928.7385
Actual vol m3	6.9830	4.7213	0.0325
Std liq m3	1.1372	3.8308	0.0273
Std vap 0 C m3	393.6422	1213.9333	8.7096
- - Vapor only - -			
Molar flow kmol	0.1576	0.0043	
Mass flow kg	7.8036	0.3373	
Average mol wt	49.5159	77.9652	
Actual dens kg/m3	1.3771	2.1062	
Actual vol m3	5.6669	0.1601	
Std liq m3	0.0072	0.0003	
Std vap 0 C m3	3.5324	0.0970	
Cp kJ/kg-K	1.5496	1.5052	
Z factor	0.9861	0.9764	
Visc N-s/m2	1.164e-05	1.061e-05	
Th cond W/m-K	0.0216	0.0206	
- - Liquid only - -			
Molar flow kmol	17.4050	54.1561	0.3886
Mass flow kg	1244.2301	4230.0000	30.1852
Average mol wt	71.4869	78.1074	77.6796
Actual dens kg/m3	945.3953	927.4034	928.7385
Actual vol m3	1.3161	4.5611	0.0325
Std liq m3	1.1300	3.8305	0.0273
Std vap 0 C m3	390.1099	1213.8363	8.7096
Cp kJ/kg-K	2.3094	2.3231	2.3211
Z factor	0.0029	0.0031	0.0031
Visc N-s/m2	0.0003955	0.0004191	0.0004174
Th cond W/m-K	0.1107	0.0997	0.1004
Surf. tens. N/m	0.0246	0.0220	0.0221
Component mole fractions			
Isopropanol	0.000000	0.000000	0.000000
Water	0.113860	0.000459	0.007575
DiMth Sulfoxide	0.886140	0.999541	0.992425

Appendix B

Tray Properties ChemCAD Report of the Distillation Process

24	0	0.00	0.00	0.00	0.0000	0.000	0.000
0.000							

Vapor		Average	Actual	Actual		Thermal		Vap H
Stg	kg/h	mol wt	vol rate	density	viscosity	conduct.	Compr.	
			m3/h	kg/m3	N-s/m2	W/m-K	factor	MJ/h
1	0	0.00	0	0.0000	0.0000	0.000	0.000	
0.000								
2	956	53.39	509	1.8804	0.0000	0.022	0.977	
-4712.979								
3	982	54.76	510	1.9268	0.0000	0.022	0.976	
-4733.087								
4	976	55.73	513	1.9022	0.0000	0.023	0.978	
-4604.353								
5	978	51.53	559	1.7507	0.0000	0.023	0.980	
-4939.203								
6	822	44.01	560	1.4692	0.0000	0.023	0.983	
-4765.934								
7	628	33.83	571	1.0997	0.0000	0.023	0.988	
-4603.256								
8	536	28.41	588	0.9125	0.0000	0.024	0.990	
-4607.943								
9	515	27.02	595	0.8652	0.0000	0.024	0.990	
-4632.278								
10	510	26.73	596	0.8556	0.0000	0.024	0.990	
-4639.069								
11	509	26.68	597	0.8538	0.0000	0.024	0.990	
-4640.456								
12	509	26.67	597	0.8534	0.0000	0.024	0.990	
-4640.724								
13	509	26.67	597	0.8534	0.0000	0.024	0.990	
-4640.775								
14	509	26.67	597	0.8534	0.0000	0.024	0.990	
-4640.783								
15	509	26.67	597	0.8534	0.0000	0.024	0.990	
-4640.781								
16	509	26.67	597	0.8534	0.0000	0.024	0.990	
-4640.784								
17	509	26.67	597	0.8534	0.0000	0.024	0.990	
-4640.783								
18	509	26.67	597	0.8534	0.0000	0.024	0.990	
-4640.782								
19	509	26.67	597	0.8534	0.0000	0.024	0.990	
-4640.742								
20	509	26.67	597	0.8534	0.0000	0.024	0.990	
-4640.472								
21	510	26.70	597	0.8538	0.0000	0.024	0.990	
-4638.337								
22	513	26.96	599	0.8571	0.0000	0.024	0.990	
-4622.303								
23	544	28.90	609	0.8934	0.0000	0.024	0.990	
-4533.242								
24	703	38.36	627	1.1210	0.0000	0.024	0.989	
-4274.488								

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Appendix C

Batch Results ChemCAD Report of the DMSO/Water Distillation

Simulation: DMSO distillation

BATCH DISTILLATION 1 CALCULATION RESULTS

Operation Step 1:

STREAM PROPERTIES

Name	Pot Charge	Accumulator	Pot Residue	Distillate
- - Overall - -				
Molar flow kmol	109.6953	14.7901	93.8819	0.1129
Mass flow kg	7669.8066	268.6261	7329.7456	3.5212
Temp C	25.0000	45.8518	118.5419	53.4486
Pres bar	0.1000	0.1000	0.1000	0.1000
Vapor mole fraction	0.000	7.811E-05	0.000	0.000
Enth MJ	-29012.	-4201.2	-23145.	-31.341
Tc C	449.5155	374.8857	452.8278	413.6053
Pc bar	84.6829	222.0417	56.6975	230.5752
Std. sp gr. wtr = 1	1.100	1.001	1.104	1.055
Std. sp gr. air = 1	2.414	0.627	2.696	1.077
Degree API	-2.8940	9.8589	-3.3614	2.6606
Average mol wt	69.9192	18.1625	78.0741	31.1975
Actual dens kg/m3	1091.6582	337.3329	1002.7520	1029.6150
Actual vol m3	7.0258	0.7963	7.3096	0.0034
Std liq m3	6.9709	0.2684	6.6376	0.0033
Std vap 0 C m3	2458.6716	331.5005	2104.2358	2.5298
- - Vapor only - -				
Molar flow kmol		0.0020		
Mass flow kg		0.0357		
Average mol wt		18.0163		
Actual dens kg/m3		0.0680		
Actual vol m3		0.5252		
Std liq m3		0.0000		
Std vap 0 C m3		0.0444		
Cp kJ/kg-K		1.8693		
Z factor		0.9987		
Visc N-s/m2		1.038e-05		
Th cond W/m-K		0.0199		
- - Liquid only - -				
Molar flow kmol	109.6953	14.7881	93.8819	0.1129
Mass flow kg	7669.8066	268.5904	7329.7461	3.5212
Average mol wt	69.9192	18.1626	78.0741	31.1975
Actual dens kg/m3	1091.6582	990.6572	1002.7521	1029.6150
Actual vol m3	7.0258	0.2711	7.3096	0.0034
Std liq m3	6.9709	0.2683	6.6376	0.0033
Std vap 0 C m3	2458.6716	331.4561	2104.2358	2.5298
Cp kJ/kg-K	2.0360	4.1532	2.0817	2.9660
Z factor	0.0004	0.0001	0.0003	0.0002
Visc N-s/m2	0.001861	0.0005881	0.0005887	0.0006244
Th cond W/m-K	0.1644	0.6281	0.1242	0.3031
Surf. tens. N/m	0.0447	0.0681	0.0308	0.0513
Component mole %				
Dimethyl Sulfoxi	86.334318	0.245411	99.898696	21.927060
Water	13.665682	99.754590	0.101302	78.072941

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Appendix D

Tray Properties ChemCAD Report of the DMSO/Water Distillation

Simulation: DMSO distillation

Tray Properties

Unit type : BATC Unit name: Eqp # 1

* End of Operation Step 1 *

Liquid		Average	Actual	Actual	viscosity	Thermal	Surface	Liq H
Stg	kg/h	mol wt	vol rate	density		conduct.	tension	
			m3/h	kg/m3	N-s/m2	W/m-K	N/m	MJ/h
1	431	21.81	0.43	1007.50	0.0006	0.483	0.062	
-5577.950								
2	937	51.44	0.91	1035.13	0.0006	0.179	0.040	
-4847.790								
3	1320	72.17	1.30	1015.23	0.0006	0.134	0.033	
-4605.299								
4	1423	76.54	1.41	1006.54	0.0006	0.127	0.031	
-4611.430								
5	1441	77.21	1.43	1004.96	0.0006	0.126	0.031	
-4616.174								
6	1443	77.30	1.44	1004.72	0.0006	0.126	0.031	
-4616.958								
7	1444	77.31	1.44	1004.69	0.0006	0.126	0.031	
-4617.073								
8	1444	77.32	1.44	1004.69	0.0006	0.126	0.031	
-4617.091								
9	1444	77.32	1.44	1004.69	0.0006	0.126	0.031	
-4617.089								
10	1444	77.32	1.44	1004.69	0.0006	0.126	0.031	
-4617.090								
11	1444	77.32	1.44	1004.68	0.0006	0.126	0.031	
-4617.089								
12	1444	77.32	1.44	1004.68	0.0006	0.126	0.031	
-4617.084								
13	1444	77.32	1.44	1004.68	0.0006	0.126	0.031	
-4617.081								
14	1444	77.32	1.44	1004.68	0.0006	0.126	0.031	
-4617.085								
15	1444	77.32	1.44	1004.68	0.0006	0.126	0.031	
-4617.086								
16	1444	77.32	1.44	1004.68	0.0006	0.126	0.031	
-4617.083								
17	1444	77.32	1.44	1004.68	0.0006	0.126	0.031	
-4617.083								
18	1444	77.32	1.44	1004.68	0.0006	0.126	0.031	
-4617.082								
19	1444	77.32	1.44	1004.68	0.0006	0.126	0.031	
-4617.083								
20	1444	77.32	1.44	1004.68	0.0006	0.126	0.031	
-4617.083								
21	1444	77.32	1.44	1004.68	0.0006	0.126	0.031	
-4617.083								
22	1444	77.32	1.44	1004.68	0.0006	0.126	0.031	
-4617.078								
23	1444	77.32	1.44	1004.68	0.0006	0.126	0.031	
-4617.076								

24	0	0.00	0.00	0.00	0.0000	0.000	0.000
0.000							

Vapor		Average	Actual	Actual		Thermal		Vap H
Stg	kg/h	mol wt	vol rate	density	viscosity	conduct.	Compr.	
			m3/h	kg/m3	N-s/m2	W/m-K	factor	MJ/h
1	0	0.00	0	0.0000	0.0000	0.000	0.000	
0.000								
2	479	22.10	6265	0.0765	0.0000	0.020	0.999	
-5151.557								
3	986	48.92	6318	0.1560	0.0000	0.017	0.998	
-4421.397								
4	1368	67.67	6498	0.2106	0.0000	0.016	0.997	
-4178.906								
5	1472	71.70	6623	0.2222	0.0000	0.016	0.997	
-4185.037								
6	1489	72.32	6649	0.2240	0.0000	0.016	0.997	
-4189.782								
7	1492	72.41	6652	0.2242	0.0000	0.016	0.997	
-4190.566								
8	1492	72.42	6653	0.2243	0.0000	0.016	0.997	
-4190.681								
9	1492	72.42	6653	0.2243	0.0000	0.016	0.997	
-4190.698								
10	1492	72.42	6653	0.2243	0.0000	0.016	0.997	
-4190.697								
11	1492	72.42	6653	0.2243	0.0000	0.016	0.997	
-4190.698								
12	1492	72.42	6653	0.2243	0.0000	0.016	0.997	
-4190.697								
13	1492	72.42	6653	0.2243	0.0000	0.016	0.997	
-4190.693								
14	1492	72.42	6653	0.2243	0.0000	0.016	0.997	
-4190.690								
15	1492	72.42	6653	0.2243	0.0000	0.016	0.997	
-4190.694								
16	1492	72.42	6653	0.2243	0.0000	0.016	0.997	
-4190.695								
17	1492	72.42	6653	0.2243	0.0000	0.016	0.997	
-4190.693								
18	1492	72.42	6653	0.2243	0.0000	0.016	0.997	
-4190.692								
19	1492	72.42	6653	0.2243	0.0000	0.016	0.997	
-4190.691								
20	1492	72.42	6653	0.2243	0.0000	0.016	0.997	
-4190.692								
21	1492	72.42	6653	0.2243	0.0000	0.016	0.997	
-4190.692								
22	1492	72.42	6653	0.2243	0.0000	0.016	0.997	
-4190.693								
23	1492	72.42	6653	0.2243	0.0000	0.016	0.997	
-4190.688								
24	1492	72.42	6653	0.2243	0.0000	0.016	0.997	
-4190.685								

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Appendix E

Heat Pump's Inputs and Outputs Table from ChemCAD

		Isopropanol	Water	DiMth Sulfoxide	Bottom Tempera	Vapour IPA	Vapour WT	Vapour DMSO
Time (h)	Reflux Ratio	kmol/h	kmol/h	kmol/h	C	kmol/h	kmol/h	kmol/h
0	0	0.000835523	0.000412389	0	80.24	0.000835523	0.000412389	0.00E+00
0.05	4	4.74825	0.00288502	5.25E-05	80.3893	23.74125	0.0144251	2.63E-04
0.1	4	4.74609	0.00284451	5.26E-05	80.5409	23.73045	0.01422255	2.63E-04
0.15	4	4.74349	0.00280624	5.26E-05	80.6946	23.71745	0.0140312	2.63E-04
0.2	4	4.74086	0.00276982	5.26E-05	80.8503	23.7043	0.0138491	2.63E-04
0.25	4	4.73856	0.00273532	5.26E-05	81.008	23.6928	0.0136766	2.63E-04
0.3	4	4.73628	0.0027014	5.26E-05	81.1676	23.6814	0.013507	2.63E-04
0.35	4	4.73417	0.00266993	5.26E-05	81.329	23.67085	0.01334965	2.63E-04
0.4	4	4.73199	0.0026383	5.26E-05	81.4922	23.65995	0.0131915	2.63E-04
0.45	4	4.73005	0.00260865	5.26E-05	81.657	23.65025	0.01304325	2.63E-04
0.5	4	4.72789	0.00257872	5.26E-05	81.8234	23.63945	0.0128936	2.63E-04
0.55	4	4.72593	0.00255025	5.26E-05	81.9915	23.62965	0.01275125	2.63E-04
0.6	4	4.724	0.00252264	5.26E-05	82.161	23.62	0.0126132	2.63E-04
0.65	4	4.72199	0.00249535	5.26E-05	82.332	23.60995	0.01247675	2.63E-04
0.7	4	4.72026	0.00246996	5.26E-05	82.5045	23.6013	0.0123498	2.63E-04
0.75	4	4.71817	0.00244314	5.26E-05	82.6784	23.59085	0.0122157	2.63E-04
0.8	4	4.71621	0.00241803	5.26E-05	82.8536	23.58105	0.01209015	2.63E-04
0.85	4	4.71428	0.00239336	5.26E-05	83.0302	23.5714	0.0119668	2.63E-04
0.9	4	4.71296	0.00237209	5.26E-05	83.2081	23.5648	0.01186045	2.63E-04
0.95	4	4.71093	0.00234792	5.26E-05	83.3873	23.55465	0.0117396	2.63E-04
1	4	4.70911	0.00232527	5.26E-05	83.5678	23.54555	0.01162635	2.63E-04
1.05	4	4.70705	0.00230216	5.26E-05	83.7496	23.53525	0.0115108	2.63E-04
1.1	4	4.70549	0.0022817	5.26E-05	83.9326	23.52745	0.0114085	2.63E-04
1.15	4	4.70379	0.00226099	5.25E-05	84.1168	23.51895	0.01130495	2.63E-04
1.2	4	4.70223	0.0022414	5.25E-05	84.3023	23.51115	0.011207	2.63E-04
1.25	4	4.7002	0.00222019	5.25E-05	84.489	23.501	0.01110095	2.63E-04
1.3	4	4.69865	0.00220157	5.25E-05	84.6768	23.49325	0.01100785	2.63E-04
1.35	4	4.69705	0.00218309	5.25E-05	84.8659	23.48525	0.01091545	2.63E-04

1.4	4	4.69511	0.0021636	5.25E-05	85.0562	23.47555	0.010818	2.63E-04
1.45	4	4.6934	0.00214547	5.25E-05	85.2477	23.467	0.01072735	2.63E-04
1.5	4	4.69175	0.00212796	5.25E-05	85.4404	23.45875	0.0106398	2.63E-04
1.55	4	4.69004	0.00211064	5.25E-05	85.6342	23.4502	0.0105532	2.63E-04
1.6	4	4.68822	0.00209314	5.25E-05	85.8293	23.4411	0.0104657	2.63E-04
1.65	4	4.68614	0.00207499	5.25E-05	86.0256	23.4307	0.01037495	2.63E-04
1.7	4	4.68465	0.00205973	5.25E-05	86.2231	23.42325	0.01029865	2.63E-04
1.75	4	4.68272	0.00204299	5.25E-05	86.4218	23.4136	0.01021495	2.63E-04
1.8	4	4.68112	0.00202784	5.25E-05	86.6218	23.4056	0.0101392	2.63E-04
1.85	4	4.67904	0.00201107	5.25E-05	86.8229	23.3952	0.01005535	2.63E-04
1.9	4	4.67717	0.0019956	5.25E-05	87.0254	23.38585	0.009978	2.63E-04
1.95	4	4.67549	0.00198116	5.25E-05	87.229	23.37745	0.0099058	2.63E-04
2	4	4.67357	0.00196614	5.25E-05	87.4339	23.36785	0.0098307	2.63E-04
2.05	4	4.67158	0.00195111	5.25E-05	87.6401	23.3579	0.00975555	2.63E-04
2.1	4	4.66943	0.0019358	5.25E-05	87.8476	23.34715	0.009679	2.63E-04
2.15	4	4.66773	0.00192248	5.25E-05	88.0564	23.33865	0.0096124	2.63E-04
2.2	4	4.6658	0.00190869	5.25E-05	88.2665	23.329	0.00954345	2.63E-04
2.25	4	4.66385	0.00189507	5.25E-05	88.478	23.31925	0.00947535	2.63E-04
2.3	4	4.6617	0.00188093	5.25E-05	88.6908	23.3085	0.00940465	2.63E-04
2.35	4	4.65965	0.00186754	5.25E-05	88.9049	23.29825	0.0093377	2.63E-04
2.4	4	4.65784	0.00185535	5.25E-05	89.1204	23.2892	0.00927675	2.63E-04
2.45	4	4.65556	0.00184159	5.25E-05	89.3373	23.2778	0.00920795	2.63E-04
2.5	4	4.65378	0.00183001	5.25E-05	89.5557	23.2689	0.00915005	2.63E-04
2.55	4	4.65113	0.00181548	5.25E-05	89.7755	23.25565	0.0090774	2.63E-04
2.6	4	4.64922	0.00180395	5.25E-05	89.9968	23.2461	0.00901975	2.63E-04
2.65	4	4.6469	0.00179112	5.25E-05	90.2195	23.2345	0.0089556	2.63E-04
2.7	4	4.64473	0.00177924	5.25E-05	90.4437	23.22365	0.0088962	2.63E-04
2.75	4	4.64254	0.00176748	5.25E-05	90.6695	23.2127	0.0088374	2.63E-04
2.8	4	4.64024	0.00175558	5.25E-05	90.8968	23.2012	0.0087779	2.63E-04
2.85	4	4.63805	0.00174424	5.25E-05	91.1257	23.19025	0.0087212	2.63E-04
2.9	4	4.63557	0.00173218	5.25E-05	91.3562	23.17785	0.0086609	2.63E-04
2.95	4	4.63314	0.00172065	5.25E-05	91.5882	23.1657	0.00860325	2.63E-04
3	4	4.63074	0.00170934	5.25E-05	91.822	23.1537	0.0085467	2.63E-04
3.05	4	4.62833	0.00169829	5.25E-05	92.0574	23.14165	0.00849145	2.63E-04

3.1	4	4.62548	0.00168603	5.25E-05	92.2945	23.1274	0.00843015	2.63E-04
3.15	4	4.62324	0.00167595	5.25E-05	92.5333	23.1162	0.00837975	2.63E-04
3.2	4	4.62039	0.00166416	5.25E-05	92.7739	23.10195	0.0083208	2.63E-04
3.25	4	4.61793	0.00165392	5.25E-05	93.0162	23.08965	0.0082696	2.62E-04
3.3	4	4.6152	0.00164289	5.25E-05	93.2604	23.076	0.00821445	2.62E-04
3.35	4	4.61251	0.0016323	5.25E-05	93.5064	23.06255	0.0081615	2.62E-04
3.4	4	4.60976	0.00162173	5.25E-05	93.7543	23.0488	0.00810865	2.62E-04
3.45	4	4.60729	0.00161231	5.25E-05	94.0041	23.03645	0.00806155	2.62E-04
3.5	4	4.60432	0.00160146	5.25E-05	94.2558	23.0216	0.0080073	2.62E-04
3.55	4	4.60138	0.00159093	5.25E-05	94.5094	23.0069	0.00795465	2.62E-04
3.6	4	4.59837	0.00158041	5.25E-05	94.7651	22.99185	0.00790205	2.62E-04
3.65	4	4.5955	0.00157053	5.25E-05	95.0228	22.9775	0.00785265	2.62E-04
3.7	4	4.59246	0.00156035	5.25E-05	95.2825	22.9623	0.00780175	2.62E-04
3.75	4	4.58941	0.00155037	5.25E-05	95.5444	22.94705	0.00775185	2.62E-04
3.8	4	4.58637	0.0015406	5.25E-05	95.8084	22.93185	0.007703	2.62E-04
3.85	4	4.5831	0.00153037	5.25E-05	96.0745	22.9155	0.00765185	2.62E-04
3.9	4	4.57989	0.00152043	5.25E-05	96.3429	22.89945	0.00760215	2.62E-04
3.95	4	4.5765	0.00151026	5.25E-05	96.6135	22.8825	0.0075513	2.62E-04
4	4	4.57323	0.00150054	5.25E-05	96.8864	22.86615	0.0075027	2.62E-04
4.05	4	4.56992	0.00149105	5.25E-05	97.1616	22.8496	0.00745525	2.62E-04
4.1	4	4.5664	0.00148111	5.25E-05	97.4392	22.832	0.00740555	2.62E-04
4.15	4	4.56307	0.00147178	5.24E-05	97.7191	22.81535	0.0073589	2.62E-04
4.2	4	4.55951	0.00146215	5.24E-05	98.0016	22.79755	0.00731075	2.62E-04
4.25	4	4.55559	0.00145155	5.24E-05	98.2865	22.77795	0.00725775	2.62E-04
4.3	4	4.55195	0.00144201	5.24E-05	98.5739	22.75975	0.00721005	2.62E-04
4.35	4	4.54813	0.00143219	5.24E-05	98.8639	22.74065	0.00716095	2.62E-04
4.4	4	4.54448	0.00142301	5.24E-05	99.1566	22.7224	0.00711505	2.62E-04
4.45	4	4.54032	0.00141255	5.24E-05	99.4518	22.7016	0.00706275	2.62E-04
4.5	4	4.53641	0.00140312	5.24E-05	99.7498	22.68205	0.0070156	2.62E-04
4.55	4	4.53232	0.00139328	5.24E-05	100.051	22.6616	0.0069664	2.62E-04
4.6	4	4.52857	0.00138461	5.24E-05	100.354	22.64285	0.00692305	2.62E-04
4.65	4	4.52415	0.00137429	5.24E-05	100.661	22.62075	0.00687145	2.62E-04
4.7	4	4.52	0.00136485	5.24E-05	100.97	22.6	0.00682425	2.62E-04
4.75	4	4.51578	0.00135549	5.24E-05	101.282	22.5789	0.00677745	2.62E-04

4.8	4	4.51148	0.0013461	5.24E-05	101.598	22.5574	0.0067305	2.62E-04
4.85	4	4.5067	0.00133553	5.24E-05	101.916	22.5335	0.00667765	2.62E-04
4.9	4	4.50223	0.00132605	5.24E-05	102.237	22.51115	0.00663025	2.62E-04
4.95	4	4.49752	0.00131608	5.24E-05	102.562	22.4876	0.0065804	2.62E-04
5	4	4.49281	0.00130638	5.24E-05	102.89	22.46405	0.0065319	2.62E-04
5.05	4	4.4878	0.00129607	5.24E-05	103.221	22.439	0.00648035	2.62E-04
5.1	4	4.48312	0.00128682	5.24E-05	103.556	22.4156	0.0064341	2.62E-04
5.15	4	4.4782	0.00127709	5.24E-05	103.894	22.391	0.00638545	2.62E-04
5.2	4	4.47315	0.00126726	5.24E-05	104.235	22.36575	0.0063363	2.62E-04
5.25	4	4.46758	0.00125627	5.23E-05	104.58	22.3379	0.00628135	2.62E-04
5.3	4	4.46232	0.00124632	5.23E-05	104.929	22.3116	0.0062316	2.62E-04
5.35	4	4.4569	0.00123614	5.23E-05	105.281	22.2845	0.0061807	2.62E-04
5.4	4	4.45142	0.00122604	5.23E-05	105.637	22.2571	0.0061302	2.62E-04
5.45	4	4.44576	0.00121562	5.23E-05	105.997	22.2288	0.0060781	2.62E-04
5.5	4	4.43994	0.00120509	5.23E-05	106.361	22.1997	0.00602545	2.62E-04
5.55	4	4.43417	0.00119491	5.23E-05	106.728	22.17085	0.00597455	2.62E-04
5.6	4	4.42853	0.00118519	5.23E-05	107.1	22.14265	0.00592595	2.62E-04
5.65	4	4.42221	0.00117398	5.23E-05	107.476	22.11105	0.0058699	2.61E-04
5.7	4	4.41623	0.00116395	5.23E-05	107.856	22.08115	0.00581975	2.61E-04
5.75	4	4.40982	0.00115304	5.23E-05	108.24	22.0491	0.0057652	2.61E-04
5.8	4	4.40335	0.0011422	5.23E-05	108.629	22.01675	0.005711	2.61E-04
5.85	4	4.39659	0.00113084	5.23E-05	109.022	21.98295	0.0056542	2.61E-04
5.9	4	4.39007	0.0011204	5.23E-05	109.419	21.95035	0.005602	2.61E-04
5.95	4	4.38297	0.00110872	5.23E-05	109.821	21.91485	0.0055436	2.61E-04
6	4	4.37611	0.00109786	5.22E-05	110.228	21.88055	0.0054893	2.61E-04
6.05	4	4.36914	0.00108701	5.22E-05	110.64	21.8457	0.00543505	2.61E-04
6.1	4	4.36169	0.00107519	5.22E-05	111.056	21.80845	0.00537595	2.61E-04
6.15	4	4.35422	0.00106365	5.22E-05	111.477	21.7711	0.00531825	2.61E-04
6.2	4	4.34668	0.00105224	5.22E-05	111.904	21.7334	0.0052612	2.61E-04
6.25	4	4.33895	0.00104061	5.22E-05	112.335	21.69475	0.00520305	2.61E-04
6.3	4	4.33119	0.00102924	5.22E-05	112.772	21.65595	0.0051462	2.61E-04
6.35	4	4.32343	0.00101799	5.22E-05	113.214	21.61715	0.00508995	2.61E-04
6.4	4	4.315	0.00100563	5.22E-05	113.661	21.575	0.00502815	2.61E-04
6.45	4	4.30664	0.000993692	5.22E-05	114.115	21.5332	0.00496846	2.61E-04

6.5	4	4.29806	0.000981575	5.22E-05	114.573	21.4903	0.004907875	2.61E-04
6.55	4	4.28916	0.000968996	5.21E-05	115.038	21.4458	0.00484498	2.61E-04
6.6	4	4.2804	0.000957082	5.21E-05	115.508	21.402	0.00478541	2.61E-04
6.65	4	4.2711	0.000944237	5.21E-05	115.985	21.3555	0.004721185	2.61E-04
6.7	4	4.26198	0.000932146	5.21E-05	116.467	21.3099	0.00466073	2.61E-04
6.75	4	4.25245	0.000919521	5.21E-05	116.956	21.26225	0.004597605	2.61E-04
6.8	4	4.24232	0.000905972	5.21E-05	117.451	21.2116	0.00452986	2.60E-04
6.85	4	4.23272	0.000893803	5.21E-05	117.952	21.1636	0.004469015	2.60E-04
6.9	4	4.22228	0.000880313	5.21E-05	118.46	21.1114	0.004401565	2.60E-04
6.95	4	4.21201	0.000867467	5.21E-05	118.975	21.06005	0.004337335	2.60E-04
7	4	4.20137	0.000854331	5.20E-05	119.497	21.00685	0.004271655	2.60E-04
7.05	4	4.19058	0.00084131	5.20E-05	120.026	20.9529	0.00420655	2.60E-04
7.1	4	4.17958	0.000828147	5.20E-05	120.561	20.8979	0.004140735	2.60E-04
7.15	4	4.16822	0.000814788	5.20E-05	121.104	20.8411	0.00407394	2.60E-04
7.2	4	4.15673	0.000801459	5.20E-05	121.655	20.78365	0.004007295	2.60E-04
7.25	4	4.14485	0.00078789	5.20E-05	122.213	20.72425	0.00393945	2.60E-04
7.3	4	4.13301	0.000774939	5.20E-05	122.778	20.66505	0.003874695	2.60E-04
7.35	4	4.12065	0.000761357	5.19E-05	123.352	20.60325	0.003806785	2.60E-04
7.4	4	4.10794	0.000747647	5.19E-05	123.933	20.5397	0.003738235	2.60E-04
7.45	4	4.09522	0.00073438	5.19E-05	124.523	20.4761	0.0036719	2.60E-04
7.5	4	4.08194	0.000720657	5.19E-05	125.12	20.4097	0.003603285	2.59E-04
7.55	4	4.06858	0.000707225	5.19E-05	125.727	20.3429	0.003536125	2.59E-04
7.6	4	4.05493	0.000693863	5.19E-05	126.341	20.27465	0.003469315	2.59E-04
7.65	4	4.04065	0.000680128	5.18E-05	126.965	20.20325	0.00340064	2.59E-04
7.7	4	4.02614	0.000666456	5.18E-05	127.597	20.1307	0.00333228	2.59E-04
7.75	4	4.01157	0.000653377	5.18E-05	128.238	20.05785	0.003266885	2.59E-04
7.8	4	3.99702	0.000640752	5.18E-05	128.889	19.9851	0.00320376	2.59E-04
7.85	4	3.98125	0.000627097	5.18E-05	129.548	19.90625	0.003135485	2.59E-04
7.9	4	3.96602	0.000614797	5.17E-05	130.218	19.8301	0.003073985	2.59E-04
7.95	4	3.95022	0.000602191	5.17E-05	130.897	19.7511	0.003010955	2.59E-04
8	4	3.93393	0.00058987	5.17E-05	131.585	19.66965	0.00294935	2.58E-04
8.05	4	3.91765	0.000578075	5.17E-05	132.284	19.58825	0.002890375	2.58E-04
8.1	4	3.90052	0.000565998	5.16E-05	132.993	19.5026	0.00282999	2.58E-04
8.15	4	3.88355	0.00055493	5.16E-05	133.712	19.41775	0.00277465	2.58E-04

8.2	4	3.86559	0.000543577	5.16E-05	134.441	19.32795	0.002717885	2.58E-04
8.25	4	3.84773	0.000533154	5.16E-05	135.181	19.23865	0.00266577	2.58E-04
8.3	4	3.82936	0.000523233	5.15E-05	135.932	19.1468	0.002616165	2.58E-04
8.35	4	3.81082	0.000514212	5.15E-05	136.693	19.0541	0.00257106	2.58E-04
8.4	4	3.792	0.000506069	5.15E-05	137.466	18.96	0.002530345	2.57E-04
8.45	4	3.77217	0.000498134	5.14E-05	138.249	18.86085	0.00249067	2.57E-04
8.5	4	3.75316	0.000492822	5.14E-05	139.044	18.7658	0.00246411	2.57E-04
8.55	4	3.73312	0.000488285	5.14E-05	139.85	18.6656	0.002441425	2.57E-04
8.6	4	3.71261	0.000485572	5.14E-05	140.668	18.56305	0.00242786	2.57E-04
8.65	4	3.69179	0.000485592	5.13E-05	141.497	18.45895	0.00242796	2.57E-04
8.7	4	3.67112	0.000490071	5.13E-05	142.338	18.3556	0.002450355	2.56E-04
8.75	4	3.64946	0.000499763	5.12E-05	143.191	18.2473	0.002498815	2.56E-04
8.8	4	3.62776	0.000520207	5.12E-05	144.055	18.1388	0.002601035	2.56E-04
8.85	4	3.60578	0.000562569	5.12E-05	144.932	18.0289	0.002812845	2.56E-04
8.9	4	3.58342	0.000673605	5.11E-05	145.82	17.9171	0.003368025	2.56E-04
8.95	4	3.51617	0.0498227	4.62E-05	146.713	17.58085	0.2491135	2.31E-04
9	4	3.26039	0.310652	2.57E-05	147.575	16.30195	1.55326	1.28E-04
9.05	4	3.01136	0.571136	1.28E-05	148.407	15.0568	2.85568	6.42E-05
9.1	5	2.42163	1.19538	2.90E-14	149.014	14.52978	7.17228	1.74E-13
9.15	5	2.41565	1.19244	0	149.631	14.4939	7.15464	0.00E+00
9.2	5	2.40936	1.18937	0	150.259	14.45616	7.13622	0.00E+00
9.25	5	2.28667	1.31811	0	150.875	13.72002	7.90866	0.00E+00
9.3	5	2.05798	1.56567	0	151.456	12.34788	9.39402	0.00E+00
9.35	5	1.84321	1.79125	0	152.003	11.05926	10.7475	0.00E+00
9.4	5	1.64351	2.00183	1.00E-15	152.517	9.86106	12.01098	6.00E-15
9.45	5	1.45888	2.19885	1.00E-15	153	8.75328	13.1931	6.00E-15
9.5	5	1.28953	2.38184	1.00E-15	153.453	7.73718	14.29104	6.00E-15
9.55	5	1.13503	2.54986	2.00E-15	153.879	6.81018	15.29916	1.20E-14
9.6	5	0.995232	2.70335	2.00E-15	154.279	5.971392	16.2201	1.20E-14
9.65	5	0.869561	2.84279	3.00E-15	154.656	5.217366	17.05674	1.80E-14
9.7	5	0.75709	2.96817	3.00E-15	155.012	4.54254	17.80902	1.80E-14
9.75	5	0.657082	3.08063	4.00E-15	155.35	3.942492	18.48378	2.40E-14
9.8	5	0.568617	3.18106	4.00E-15	155.671	3.411702	19.08636	2.40E-14
9.85	5	0.490776	3.27082	5.00E-15	155.977	2.944656	19.62492	3.00E-14

9.9	5	0.422482	3.35007	5.00E-15	156.27	2.534892	20.10042	3.00E-14
9.95	5	0.362908	3.42076	5.00E-15	156.552	2.177448	20.52456	3.00E-14
10	5	0.311072	3.48331	6.00E-15	156.825	1.866432	20.89986	3.60E-14
10.05	5	0.266129	3.53872	6.00E-15	157.089	1.596774	21.23232	3.60E-14
10.1	5	0.227259	3.58737	6.00E-15	157.347	1.363554	21.52422	3.60E-14
10.15	5	0.193761	3.63051	6.00E-15	157.599	1.162566	21.78306	3.60E-14
10.2	5	0.164942	3.6683	7.00E-15	157.847	0.989652	22.0098	4.20E-14
10.25	5	0.140217	3.70159	7.00E-15	158.091	0.841302	22.20954	4.20E-14
10.3	5	0.119049	3.73102	7.00E-15	158.331	0.714294	22.38612	4.20E-14
10.35	5	0.100947	3.75655	7.00E-15	158.57	0.605682	22.5393	4.20E-14
10.4	5	0.0854974	3.77874	7.00E-15	158.807	0.5129844	22.67244	4.20E-14
10.45	5	0.0723335	3.79807	7.00E-15	159.043	0.434001	22.78842	4.20E-14
10.5	5	0.0611276	3.8144	7.00E-15	159.278	0.3667656	22.8864	4.20E-14
10.55	5	0.0516095	3.82869	7.00E-15	159.512	0.309657	22.97214	4.20E-14
10.6	5	0.0435304	3.84079	8.00E-15	159.747	0.2611824	23.04474	4.80E-14
10.65	5	0.0366803	3.85099	8.00E-15	159.982	0.2200818	23.10594	4.80E-14
10.7	5	0.03088	3.85957	8.00E-15	160.217	0.18528	23.15742	4.80E-14
10.75	5	0.025974	3.8668	8.00E-15	160.453	0.155844	23.2008	4.80E-14
10.8	5	0.0218284	3.87282	8.00E-15	160.69	0.1309704	23.23692	4.80E-14
10.85	4	0.0182999	4.61991	1.10E-14	160.972	0.0914995	23.09955	5.50E-14
10.9	4	0.0153682	4.62298	1.10E-14	161.256	0.076841	23.1149	5.50E-14
10.95	4	0.0128925	4.62487	1.10E-14	161.543	0.0644625	23.12435	5.50E-14
11	4	0.0108061	4.62666	1.10E-14	161.831	0.0540305	23.1333	5.50E-14
11.05	4	0.00904742	4.62735	1.10E-14	162.122	0.0452371	23.13675	5.50E-14
11.1	4	0.00756724	4.62739	1.10E-14	162.414	0.0378362	23.13695	5.50E-14
11.15	4	0.00632327	4.62727	1.10E-14	162.71	0.03161635	23.13635	5.50E-14
11.2	4	0.00527839	4.62675	1.10E-14	163.007	0.02639195	23.13375	5.50E-14
11.25	4	0.00440161	4.62574	1.10E-14	163.308	0.02200805	23.1287	5.50E-14
11.3	4	0.00366674	4.62455	1.10E-14	163.61	0.0183337	23.12275	5.50E-14
11.35	4	0.00305118	4.6227	1.10E-14	163.916	0.0152559	23.1135	5.50E-14
11.4	4	0.00253648	4.62098	1.10E-14	164.224	0.0126824	23.1049	5.50E-14
11.45	4	0.00210643	4.61924	1.10E-14	164.535	0.01053215	23.0962	5.50E-14
11.5	4	0.00174728	4.61692	1.10E-14	164.849	0.0087364	23.0846	5.50E-14
11.55	4	0.0014479	4.61482	1.10E-14	165.166	0.0072395	23.0741	5.50E-14

11.6	4	0.0011984	4.61219	1.10E-14	165.485	0.005992	23.06095	5.50E-14
11.65	4	0.000990835	4.60968	1.10E-14	165.808	0.004954175	23.0484	5.50E-14
11.7	4	0.000818307	4.60714	1.10E-14	166.134	0.004091535	23.0357	5.50E-14
11.75	4	0.000675026	4.60435	1.10E-14	166.462	0.00337513	23.02175	5.50E-14
11.8	4	0.00055617	4.60137	1.10E-14	166.794	0.00278085	23.00685	5.50E-14
11.85	4	0.000457717	4.59846	1.10E-14	167.129	0.002288585	22.9923	5.50E-14
11.9	4	0.000376243	4.59553	1.10E-14	167.468	0.001881215	22.97765	5.50E-14
11.95	4	0.000308865	4.59202	1.10E-14	167.81	0.001544325	22.9601	5.50E-14
12	4	0.000253272	4.58908	1.10E-14	168.155	0.00126636	22.9454	5.50E-14
12.05	4	0.000207409	4.58577	1.10E-14	168.503	0.001037045	22.92885	5.50E-14
12.1	4	0.000169634	4.58236	1.10E-14	168.855	0.00084817	22.9118	5.50E-14
12.15	4	0.000138559	4.57899	1.10E-14	169.211	0.000692795	22.89495	5.50E-14
12.2	4	0.000113025	4.5755	1.10E-14	169.57	0.000565125	22.8775	5.50E-14
12.25	4	9.21E-05	4.57195	1.10E-14	169.932	0.000460361	22.85975	5.50E-14
12.3	4	7.49E-05	4.56805	1.10E-14	170.299	0.000374481	22.84025	5.50E-14
12.35	4	6.08E-05	4.56459	1.10E-14	170.669	0.000304225	22.82295	5.50E-14
12.4	4	4.94E-05	4.56072	1.10E-14	171.043	0.000246774	22.8036	5.50E-14
12.45	4	4.00E-05	4.55683	1.10E-14	171.421	0.000199885	22.78415	5.50E-14
12.5	4	3.23E-05	4.55291	1.10E-14	171.803	0.000161663	22.76455	5.50E-14
12.55	4	2.61E-05	4.54914	1.10E-14	172.188	0.000130556	22.7457	5.50E-14
12.6	4	2.11E-05	4.54519	1.10E-14	172.578	0.000105266	22.72595	5.50E-14
12.65	4	1.69E-05	4.54073	1.10E-14	172.972	8.47335E-05	22.70365	5.50E-14
12.7	4	1.36E-05	4.5366	1.10E-14	173.37	6.80995E-05	22.683	5.50E-14
12.75	4	1.09E-05	4.53217	1.10E-14	173.773	0.000054638	22.66085	5.50E-14
12.8	4	8.75E-06	4.52781	1.10E-14	174.179	4.37634E-05	22.63905	5.50E-14
12.85	4	7.00E-06	4.52375	1.10E-14	174.591	3.49938E-05	22.61875	5.50E-14
12.9	4	5.59E-06	4.51899	1.10E-14	175.006	2.79275E-05	22.59495	5.50E-14
12.95	4	4.45E-06	4.51464	1.10E-14	175.426	2.22492E-05	22.5732	5.50E-14
13	4	3.54E-06	4.50984	1.10E-14	175.851	1.76903E-05	22.5492	5.50E-14
13.05	4	2.81E-06	4.50525	1.10E-14	176.28	1.40388E-05	22.52625	5.50E-14
13.1	4	2.22E-06	4.5003	1.10E-14	176.715	1.1118E-05	22.5015	5.50E-14
13.15	4	1.76E-06	4.49526	1.10E-14	177.153	8.7865E-06	22.4763	5.50E-14
13.2	4	1.39E-06	4.49028	1.10E-14	177.597	6.9293E-06	22.4514	5.50E-14
13.25	4	1.09E-06	4.48517	1.10E-14	178.046	5.45245E-06	22.42585	5.50E-14

13.3	4	8.56E-07	4.47999	1.10E-14	178.5	4.28054E-06	22.39995	5.50E-14
13.35	4	6.71E-07	4.47476	1.10E-14	178.959	3.35253E-06	22.3738	5.50E-14
13.4	4	5.24E-07	4.46941	1.10E-14	179.423	2.61919E-06	22.34705	5.50E-14
13.45	4	4.08E-07	4.46391	1.10E-14	179.893	2.04099E-06	22.31955	5.50E-14
13.5	4	3.17E-07	4.4584	1.10E-14	180.368	1.58625E-06	22.292	5.50E-14
13.55	4	2.46E-07	4.38525	0.0345403	180.841	1.2285E-06	21.92625	1.73E-01
13.6	2	1.90E-07	4.28948	2.58564	181.296	5.71392E-07	12.86844	7.76E+00
13.65	2	1.46E-07	4.08307	2.79765	181.735	4.38843E-07	12.24921	8.39E+00
13.7	2	1.12E-07	3.88055	3.00814	182.157	3.35766E-07	11.64165	9.02E+00
13.75	2	8.53E-08	3.68229	3.21656	182.562	2.55925E-07	11.04687	9.65E+00
13.8	2	6.48E-08	3.4888	3.42273	182.95	1.94334E-07	10.4664	1.03E+01
13.85	2	4.90E-08	3.30022	3.62599	183.321	1.47005E-07	9.90066	1.09E+01
13.9	2	3.69E-08	3.11706	3.82619	183.675	1.10786E-07	9.35118	1.15E+01
13.95	2	2.77E-08	2.93926	4.02256	184.013	8.31714E-08	8.81778	1.21E+01
14	2	2.07E-08	2.76726	4.21493	184.333	6.22068E-08	8.30178	1.26E+01
14.05	2	1.55E-08	2.60128	4.40305	184.638	4.63536E-08	7.80384	1.32E+01
14.1	2	1.15E-08	2.44143	4.58648	184.926	3.44124E-08	7.32429	1.38E+01
14.15	2	8.48E-09	2.28766	4.76447	185.199	2.54522E-08	6.86298	1.43E+01
14.2	2	6.25E-09	2.14045	4.93761	185.457	1.87581E-08	6.42135	1.48E+01
14.25	2	4.59E-09	1.99965	5.10519	185.7	1.37749E-08	5.99895	1.53E+01
14.3	2	3.36E-09	1.8654	5.26731	185.928	1.00801E-08	5.5962	1.58E+01
14.35	2	2.45E-09	1.73756	5.42348	186.143	7.35042E-09	5.21268	1.63E+01
14.4	2	1.78E-09	1.61612	5.57358	186.345	5.34144E-09	4.84836	1.67E+01
14.45	2	1.29E-09	1.50106	5.71768	186.533	3.8685E-09	4.50318	1.72E+01
14.5	2	9.31E-10	1.39219	5.8553	186.71	2.79228E-09	4.17657	1.76E+01
14.55	2	6.70E-10	1.28955	5.9872	186.874	2.009E-09	3.86865	1.80E+01
14.6	2	4.80E-10	1.19283	6.11256	187.028	1.44072E-09	3.57849	1.83E+01
14.65	2	3.43E-10	1.10191	6.23158	187.171	1.0299E-09	3.30573	1.87E+01
14.7	2	2.45E-10	1.01668	6.34476	187.304	7.33971E-10	3.05004	1.90E+01
14.75	2	1.74E-10	0.936861	6.45181	187.427	5.21463E-10	2.810583	1.94E+01
14.8	2	1.23E-10	0.862229	6.55254	187.542	3.69354E-10	2.586687	1.97E+01
14.85	2	8.70E-11	0.792691	6.64815	187.648	2.60862E-10	2.378073	1.99E+01
14.9	2	6.12E-11	0.727856	6.73746	187.746	1.83687E-10	2.183568	2.02E+01
14.95	2	4.30E-11	0.667617	6.82169	187.836	1.28979E-10	2.002851	2.05E+01

15	2	3.01E-11	0.611621	6.89975	187.92	9.0297E-11	1.834863	2.07E+01
15.05	2	2.10E-11	0.55978	6.97335	187.997	6.3045E-11	1.67934	2.09E+01
15.1	2	1.46E-11	0.511794	7.04199	188.068	4.3899E-11	1.535382	2.11E+01
15.15	2	1.02E-11	0.467408	7.10533	188.133	3.048E-11	1.402224	2.13E+01
15.2	2	7.04E-12	0.426461	7.16444	188.193	2.1105E-11	1.279383	2.15E+01
15.25	2	4.86E-12	0.388726	7.21937	188.248	1.4577E-11	1.166178	2.17E+01
15.3	2	3.35E-12	0.353995	7.27033	188.299	1.0041E-11	1.061985	2.18E+01
15.35	2	2.30E-12	0.322052	7.31729	188.345	6.897E-12	0.966156	2.20E+01
15.4	2	1.58E-12	0.292717	7.36048	188.387	4.725E-12	0.878151	2.21E+01
15.45	2	1.08E-12	0.265808	7.40025	188.426	3.231E-12	0.797424	2.22E+01
15.5	2	7.34E-13	0.241158	7.43687	188.461	2.202E-12	0.723474	2.23E+01
15.55	2	4.99E-13	0.218607	7.47082	188.494	1.497E-12	0.655821	2.24E+01
15.6	2	3.38E-13	0.197978	7.50154	188.523	1.014E-12	0.593934	2.25E+01
15.65	2	2.29E-13	0.179149	7.53001	188.55	6.87E-13	0.537447	2.26E+01
15.7	2	1.54E-13	0.161965	7.55581	188.574	4.62E-13	0.485895	2.27E+01
15.75	2	1.04E-13	0.146306	7.57955	188.597	3.12E-13	0.438918	2.27E+01
15.8	2	7.00E-14	0.132047	7.60114	188.617	2.1E-13	0.396141	2.28E+01
15.85	2	4.70E-14	0.119076	7.62081	188.635	1.41E-13	0.357228	2.29E+01
15.9	2	3.10E-14	0.107289	7.63873	188.652	9.3E-14	0.321867	2.29E+01
15.95	2	2.10E-14	0.0965867	7.65502	188.667	6.3E-14	0.2897601	2.30E+01
16	2	1.40E-14	0.0868799	7.66999	188.681	4.2E-14	0.2606397	2.30E+01
16.05	2	9.00E-15	0.0780818	7.6835	188.693	2.7E-14	0.2342454	2.31E+01
16.1	2	6.00E-15	0.0701136	7.69558	188.704	1.8E-14	0.2103408	2.31E+01
16.15	2	4.00E-15	0.0629064	7.70666	188.715	1.2E-14	0.1887192	2.31E+01

time	Temp. after compress °C	Vap frac. after compr °C	Temp. after first heat ex °C	Vap frac. Af °C	Heat Duty of second [MJ/hr]	Heat D. Intercooler [MJ/hr]
0	193.6109009	1	-270.3722229	0	999.9386597	0.002782357
0.05	177.2104797	1	115.1567459	0	-169.3969574	24.07682419
0.1	177.2103424	1	115.072319	0	-168.855545	24.06489372
0.15	177.2101135	1	114.9708939	0	-168.2058411	24.05089188
0.2	177.2098694	1	114.8682556	0	-167.5495605	24.03661919
0.25	177.2096252	1	114.7782288	0	-166.9747772	24.0241642
0.3	177.2094269	1	114.6889114	0	-166.4049072	24.01169205
0.35	177.2092285	1	114.6061478	0	-165.8776398	24.00061417
0.4	177.2089844	1	114.5205612	0	-165.3331757	23.98855209
0.45	177.208786	1	114.4442673	0	-164.8483124	23.97802925
0.5	177.2086487	1	114.3593216	0	-164.3091583	23.96640968
0.55	177.2084503	1	114.2821121	0	-163.8194885	23.95578384
0.6	177.2082367	1	114.2060242	0	-163.3374634	23.94542503
0.65	177.2080383	1	114.1267471	0	-162.8359528	23.93466187
0.7	177.207901	1	114.0583878	0	-162.4036865	23.92526054
0.75	177.2077637	1	113.975853	0	-161.8827515	23.91400146
0.8	177.2076263	1	113.8983765	0	-161.3939819	23.90351295
0.85	177.207428	1	113.8219757	0	-160.9126892	23.89308167
0.9	177.2072601	1	113.7695236	0	-160.5824432	23.88596153
0.95	177.2071228	1	113.6890945	0	-160.0766602	23.8751297
1	177.2069855	1	113.6168976	0	-159.6230469	23.86525726
1.05	177.2068481	1	113.5351791	0	-159.1102753	23.85438156
1.1	177.2067108	1	113.4730911	0	-158.7211761	23.84594727
1.15	177.206543	1	113.4054489	0	-158.2975159	23.83691788
1.2	177.2064514	1	113.3433304	0	-157.9085846	23.82859421
1.25	177.2063141	1	113.262558	0	-157.403656	23.81768227
1.3	177.2061462	1	113.200676	0	-157.0173187	23.80940628
1.35	177.2060699	1	113.1368256	0	-156.6190948	23.80078697

1.4	177.2059326	1	113.0594788	0	-156.1367645	23.79053116
1.45	177.2058411	1	112.9911194	0	-155.7112122	23.78158379
1.5	177.205658	1	112.9251328	0	-155.3005066	23.77280426
1.55	177.2055969	1	112.8567429	0	-154.8755188	23.76363754
1.6	177.2054596	1	112.7838364	0	-154.4230042	23.7541256
1.65	177.2053986	1	112.700592	0	-153.9065247	23.74316978
1.7	177.2052917	1	112.6408157	0	-153.536087	23.73522186
1.75	177.2051544	1	112.5633621	0	-153.056778	23.7250576
1.8	177.2050476	1	112.4990768	0	-152.6592102	23.71665382
1.85	177.2049561	1	112.4155273	0	-152.1429749	23.70571709
1.9	177.2048798	1	112.3403168	0	-151.6789856	23.69586372
1.95	177.2047882	1	112.2726669	0	-151.2617798	23.68695259
2	177.2046814	1	112.1953201	0	-150.7855377	23.67694092
2.05	177.2045746	1	112.115097	0	-150.2917938	23.66657066
2.1	177.2045135	1	112.0283585	0	-149.7589874	23.65520477
2.15	177.2044373	1	111.9596252	0	-149.337204	23.64630318
2.2	177.2043457	1	111.8815994	0	-148.8585815	23.63625526
2.25	177.2042389	1	111.8026657	0	-148.3752594	23.62600136
2.3	177.2041321	1	111.7155838	0	-147.8425598	23.61483192
2.35	177.2040405	1	111.6324768	0	-147.3345032	23.60415649
2.4	177.2039337	1	111.5589676	0	-146.8859406	23.59471703
2.45	177.2039032	1	111.466362	0	-146.3213806	23.58279037
2.5	177.2038269	1	111.3939667	0	-145.8805389	23.57350731
2.55	177.2037659	1	111.2861404	0	-145.224472	23.55980492
2.6	177.2036591	1	111.208252	0	-144.7514954	23.54985809
2.65	177.2036285	1	111.1136093	0	-144.1773987	23.5378418
2.7	177.2035217	1	111.024971	0	-143.6401978	23.52659607
2.75	177.2034302	1	110.9353867	0	-143.0982666	23.51522636
2.8	177.2033997	1	110.8412552	0	-142.5293121	23.50331497
2.85	177.2032928	1	110.7514725	0	-141.9873962	23.49194717
2.9	177.2032623	1	110.6497116	0	-141.3740234	23.47912979
2.95	177.2031555	1	110.549881	0	-140.7728424	23.46656799
3	177.2030487	1	110.4511414	0	-140.179245	23.45413399
3.05	177.2030182	1	110.35186	0	-139.5834198	23.44166565

3.1	177.2029572	1	110.2343292	0	-138.8789215	23.42708206
3.15	177.2029114	1	110.1417923	0	-138.3250885	23.41546822
3.2	177.2028198	1	110.0239334	0	-137.6206055	23.40076256
3.25	177.2027893	1	109.9220657	0	-137.0127258	23.38806534
3.3	177.2026825	1	109.8087845	0	-136.3378754	23.37397385
3.35	177.202652	1	109.6970901	0	-135.6732635	23.36008835
3.4	177.2026062	1	109.5827103	0	-134.9938812	23.34601402
3.45	177.2025146	1	109.4797668	0	-134.3834076	23.33314705
3.5	177.2024841	1	109.3559036	0	-133.6498108	23.31795311
3.55	177.2023773	1	109.233017	0	-132.9234467	23.30279922
3.6	177.2023468	1	109.107048	0	-132.1800995	23.28739929
3.65	177.2022705	1	108.986702	0	-131.4711609	23.27260971
3.7	177.202179	1	108.8590393	0	-130.7204895	23.25707054
3.75	177.2021332	1	108.7307281	0	-129.9672852	23.24137688
3.8	177.2021027	1	108.6026611	0	-129.2167053	23.22582436
3.85	177.2020416	1	108.4646149	0	-128.4092712	23.20892334
3.9	177.2019958	1	108.3288498	0	-127.6167221	23.19251251
3.95	177.2019043	1	108.1852112	0	-126.7797546	23.17508507
4	177.2018738	1	108.0464249	0	-125.9724808	23.15839005
4.05	177.201828	1	107.9056702	0	-125.1554794	23.14138985
4.1	177.2017975	1	107.7556915	0	-124.2866898	23.123312
4.15	177.2017365	1	107.6134796	0	-123.4646149	23.10630608
4.2	177.2016296	1	107.4611969	0	-122.5858994	23.08810616
4.25	177.2015991	1	107.2931519	0	-121.6185455	23.06801987
4.3	177.2015686	1	107.1367645	0	-120.7203827	23.04933167
4.35	177.2014923	1	106.9722748	0	-119.7776718	23.02984619
4.4	177.2014618	1	106.8147659	0	-118.8770676	23.01121521
4.45	177.201355	1	106.6348495	0	-117.8504868	22.98989296
4.5	177.2013245	1	106.4653397	0	-116.8857727	22.9698658
4.55	177.2012939	1	106.2876663	0	-115.8767624	22.94889069
4.6	177.2012634	1	106.1243286	0	-114.9515152	22.92977524
4.65	177.2011566	1	105.9313507	0	-113.8610001	22.90725517
4.7	177.2011261	1	105.7497406	0	-112.8374557	22.88607597
4.75	177.2011261	1	105.5645981	0	-111.7965012	22.86438751

4.8	177.2010193	1	105.3754272	0	-110.7356491	22.84238243
4.85	177.2009583	1	105.1646194	0	-109.5565796	22.8180027
4.9	177.2009125	1	104.9669266	0	-108.4540482	22.79523659
4.95	177.200882	1	104.7580566	0	-107.2923737	22.77114296
5	177.2008209	1	104.5485687	0	-106.1306381	22.74707031
5.05	177.2007751	1	104.3251419	0	-104.8950577	22.72144318
5.1	177.2007446	1	104.1157913	0	-103.7408676	22.69760704
5.15	177.2006836	1	103.89505	0	-102.5275192	22.67255783
5.2	177.2006073	1	103.6677933	0	-101.2821198	22.6467514
5.25	177.2005768	1	103.4163666	0	-99.90856934	22.6182785
5.3	177.2005157	1	103.1781235	0	-98.61140442	22.59151077
5.35	177.20047	1	102.9318466	0	-97.27487183	22.56394005
5.4	177.2004395	1	102.6820145	0	-95.92357635	22.53594398
5.45	177.2003479	1	102.4230881	0	-94.52783203	22.5070591
5.5	177.2003479	1	102.1559525	0	-93.09293365	22.47725868
5.55	177.2003021	1	101.8901138	0	-91.67028046	22.44790459
5.6	177.2002106	1	101.6293564	0	-90.27964783	22.41910744
5.65	177.2001648	1	101.3360825	0	-88.72136688	22.38696861
5.7	177.2001038	1	101.0574951	0	-87.24694824	22.35644341
5.75	177.2000732	1	100.7577744	0	-85.66661835	22.323843
5.8	177.1999969	1	100.4539871	0	-84.07152557	22.29084396
5.85	177.1999664	1	100.1352844	0	-82.40489197	22.25628853
5.9	177.1999359	1	99.82661438	0	-80.79763794	22.22313499
5.95	177.1998291	1	99.4890213	0	-79.04714203	22.18702698
6	177.1997986	1	99.16143799	0	-77.35603333	22.15215683
6.05	177.1996918	1	98.82709503	0	-75.63787079	22.11663055
6.1	177.1996918	1	98.46810913	0	-73.80140686	22.07858658
6.15	177.1996307	1	98.10644531	0	-71.95989227	22.04063034
6.2	177.1995544	1	97.73962402	0	-70.10124969	22.00222397
6.25	177.1995239	1	97.36170959	0	-68.19589996	21.96275711
6.3	177.1994629	1	96.98044586	0	-66.28316498	21.92334557
6.35	177.1994324	1	96.59728241	0	-64.37055206	21.88380051
6.4	177.1993561	1	96.1787796	0	-62.29258728	21.84089279
6.45	177.1993256	1	95.76150513	0	-60.23195267	21.79832268

6.5	177.1992493	1	95.33087158	0	-58.11724472	21.754673
6.55	177.1991577	1	94.88158417	0	-55.92351151	21.7094593
6.6	177.1991577	1	94.43683624	0	-53.76473999	21.66479301
6.65	177.1990814	1	93.96174622	0	-51.47247314	21.61746216
6.7	177.1990204	1	93.49306488	0	-49.22492218	21.57106209
6.75	177.1989136	1	93.00017548	0	-46.87609863	21.5225811
6.8	177.1989136	1	92.47286224	0	-44.37957001	21.47107506
6.85	177.1988525	1	91.96979523	0	-42.01380539	21.42225838
6.9	177.198822	1	91.41895294	0	-39.44094467	21.36908913
6.95	177.1987152	1	90.87323761	0	-36.90997696	21.3169384
7	177.1986389	1	90.30384064	0	-34.28788757	21.26276207
7.05	177.1986389	1	89.72214508	0	-31.62915802	21.20792007
7.1	177.1985474	1	89.1246109	0	-28.91847038	21.15200043
7.15	177.1985168	1	88.50273132	0	-26.11924744	21.09426498
7.2	177.1984406	1	87.86871338	0	-23.28790283	21.03595734
7.25	177.1983795	1	87.20773315	0	-20.36065674	20.97542572
7.3	177.1983337	1	86.54346466	0	-17.44334602	20.91526222
7.35	177.1982727	1	85.84400177	0	-14.39780712	20.85240555
7.4	177.1982117	1	85.11826324	0	-11.26625443	20.78783989
7.45	177.1981659	1	84.38526154	0	-8.132292747	20.7232399
7.5	177.1981049	1	83.61276245	0	-4.860282421	20.65579033
7.55	177.1980286	1	82.82801819	0	-1.568857074	20.58792305
7.6	177.1980286	1	82.01825714	0	1.793938875	20.51845741
7.65	177.1979675	1	81.16230011	0	5.312459469	20.445961
7.7	177.197937	1	80.28315735	0	8.886652946	20.37231064
7.75	177.1978607	1	79.39069366	0	12.47587967	20.29835129
7.8	177.1977997	1	78.48960876	0	16.06019974	20.22448349
7.85	177.1977692	1	77.50154877	0	19.94509125	20.14446831
7.9	177.1976624	1	76.53590393	0	23.69673347	20.06716728
7.95	177.1976929	1	75.5220108	0	27.5885849	19.98692322
8	177.1976624	1	74.4634552	0	31.60115242	19.90426826
8.05	177.1976318	1	73.39181519	0	35.61038971	19.82177925
8.1	177.1975555	1	72.24909973	0	39.82950974	19.73487854
8.15	177.1974945	1	71.10147095	0	44.00894165	19.6488018

8.2	177.1975555	1	69.86954498	0	48.43193817	19.55762672
8.25	177.1974945	1	68.62625885	0	52.8303299	19.46715927
8.3	177.1974945	1	67.32814026	0	57.35403824	19.37409973
8.35	177.197525	1	65.997612	0	61.91934586	19.28009987
8.4	177.197464	1	64.62537384	0	66.55371857	19.18477058
8.45	177.197525	1	63.15537262	0	71.43585205	19.08433914
8.5	177.1976013	1	61.72230148	0	76.11590576	18.98809433
8.55	177.1975555	1	60.18536758	0	81.04943848	18.8866787
8.6	177.1976318	1	58.58387375	0	86.09830475	18.78291512
8.65	177.1977234	1	56.92775345	0	91.22298431	18.67769623
8.7	177.1979065	1	55.25237274	0	96.3081131	18.57324982
8.75	177.1980286	1	53.46224213	0	101.6366425	18.46406746
8.8	177.1983337	1	51.63271332	0	106.9722366	18.35482788
8.85	177.1988525	1	49.74225235	0	112.3713684	18.24467278
8.9	177.1999664	1	47.78271866	0	117.8468475	18.13440132
8.95	177.655365	1	45.66524124	0	122.0250015	18.98668289
9	180.311615	1	43.45315552	0	120.1175995	23.96788788
9.05	183.4264069	1	41.81497192	0	117.6058884	28.88408279
9.1	193.6128693	1	98.71665955	0	-66.57486725	48.38911819
9.15	193.6130066	1	98.05029297	0	-63.94444275	48.26996231
9.2	193.6135101	1	97.34601593	0	-61.18433762	48.14525986
9.25	196.459198	1	98.24255371	0	-62.69842148	50.88578033
9.3	203.0704956	1	101.0635986	0	-68.8225708	56.15716934
9.35	211.4131927	1	104.1904602	0	-75.01062775	61.44734573
9.4	220.6092224	1	107.7329102	0	-81.7694397	66.2472229
9.45	229.9230499	1	111.4274979	0	-88.32724762	70.7897644
9.5	239.2223969	1	115.2201614	0	-94.62324524	75.13739014
9.55	248.4512177	1	118.988266	0	-100.3711395	79.26013947
9.6	257.5636292	1	122.7634964	0	-105.7596664	83.1533432
9.65	266.5106506	1	126.5382767	0	-110.828743	86.8024292
9.7	275.2362976	1	130.2068329	0	-115.334816	90.18628693
9.75	283.6824646	1	133.7971497	0	-119.4155731	93.3089447
9.8	291.7902832	1	137.2900238	0	-123.0691986	96.17306519
9.85	299.5057068	1	140.7204132	0	-126.4310379	98.79779053

9.9	306.778656	1	143.9635773	0	-129.2310486	101.1688919
9.95	313.5728149	1	147.1393127	0	-131.8182678	103.3283081
10	319.8612061	1	150.1683044	0	-134.0435791	105.2750168
10.05	325.6345215	1	153.0631104	0	-135.9870605	107.0280304
10.1	330.8782654	1	155.7550964	0	-137.5412598	108.5895233
10.15	335.6063843	1	158.3137512	0	-138.9160156	109.9901428
10.2	339.8360291	1	160.6689758	0	-139.9959106	111.2304916
10.25	343.5935669	1	162.857254	0	-140.9045105	112.3321152
10.3	346.9067078	1	164.8948364	0	-141.710144	113.3121109
10.35	349.8186951	1	166.7205353	0	-142.3008423	114.1677094
10.4	352.3551636	1	168.3546143	0	-142.7406006	114.9146042
10.45	354.5639648	1	169.8261108	0	-143.0990906	115.5679016
10.5	356.4709778	1	171.0738678	0	-143.2461548	116.1223679
10.55	358.1086731	1	172.2011871	0	-143.3994598	116.60849
10.6	359.5185242	1	173.1614685	0	-143.4467163	117.0214539
10.65	360.7255859	1	173.9729767	0	-143.4120636	117.3711624
10.7	361.7565613	1	174.6564026	0	-143.3226166	117.6660004
10.75	362.6346436	1	175.2335205	0	-143.2059326	117.9150696
10.8	363.3811035	1	175.7108765	0	-143.056427	118.1236038
10.85	364.5516052	1	171.9577942	0	-133.2799377	117.463707
10.9	365.0007019	1	172.1005554	0	-132.8692169	117.5565338
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11	365.7026062	1	172.1946716	0	-131.9446564	117.6734619
11.05	365.9742126	1	172.1453094	0	-131.4045715	117.7000885
11.1	366.2032166	1	172.0400543	0	-130.8157501	117.7084961
11.15	366.3963623	1	171.9333801	0	-130.2817078	117.7121735
11.2	366.5585938	1	171.7940216	0	-129.7294312	117.704155
11.25	366.6950378	1	171.6091156	0	-129.1253815	117.6836472
11.3	366.8095093	1	171.4131927	0	-128.534668	117.6569977
11.35	366.9054871	1	171.1476593	0	-127.8314209	117.6129303
11.4	366.9857788	1	170.9046021	0	-127.200798	117.5723495
11.45	367.0530396	1	170.6653137	0	-126.5999832	117.5296936
11.5	367.1093445	1	170.361969	0	-125.8867569	117.4730835
11.55	367.1561279	1	170.0888367	0	-125.2516174	117.4210968

11.6	367.1953125	1	169.7559204	0	-124.5079041	117.3552933
11.65	367.2278748	1	169.4399261	0	-123.8105469	117.2925262
11.7	367.2549438	1	169.1225433	0	-123.1203003	117.2287521
11.75	367.2774353	1	168.776947	0	-122.3812485	117.1584015
11.8	367.2962341	1	168.4098816	0	-121.6055145	117.0832443
11.85	367.311615	1	168.0521545	0	-120.8556747	117.0099106
11.9	367.3245239	1	167.6927643	0	-120.1079636	116.9354401
11.95	367.335022	1	167.2637787	0	-119.2246475	116.8472595
12	367.3437805	1	166.9039459	0	-118.4852676	116.7722549
12.05	367.3511047	1	166.499527	0	-117.6596985	116.6881866
12.1	367.3569336	1	166.0827484	0	-116.812912	116.6016846
12.15	367.3618774	1	165.6707306	0	-115.97892	116.5163193
12.2	367.3658752	1	165.2437286	0	-115.1177444	116.4278107
12.25	367.3693542	1	164.809082	0	-114.2440338	116.3376007
12.3	367.3719788	1	164.3310394	0	-113.2864914	116.2384033
12.35	367.3742371	1	163.9062042	0	-112.4374466	116.1500702
12.4	367.3760681	1	163.4304657	0	-111.4895477	116.0515671
12.45	367.3776855	1	162.9515839	0	-110.5378036	115.9531021
12.5	367.3787537	1	162.4679871	0	-109.5791931	115.8528061
12.55	367.3799133	1	162.0022125	0	-108.6580048	115.7571182
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12.65	367.3813477	1	160.9600677	0	-106.6046219	115.5430908
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12.75	367.3820801	1	159.8949127	0	-104.5158844	115.3256989
12.8	367.3826294	1	159.350708	0	-103.4525528	115.2144699
12.85	367.382843	1	158.8427582	0	-102.4623032	115.1109772
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13	367.3835144	1	157.0947571	0	-99.07102203	114.757576
13.05	367.3835754	1	156.5151825	0	-97.95220184	114.6404343
13.1	367.3836365	1	155.8885651	0	-96.74549103	114.5150223
13.15	367.383728	1	155.2489777	0	-95.5171051	114.3864975
13.2	367.3837891	1	154.6154022	0	-94.30329895	114.2600708
13.25	367.383728	1	153.9634705	0	-93.05773163	114.1293793

13.3	367.3837891	1	153.3008728	0	-91.79528046	113.998085
13.35	367.3838501	1	152.630127	0	-90.52064514	113.8643494
13.4	367.3838501	1	151.9420624	0	-89.21673584	113.7286606
13.45	367.3839111	1	151.2327576	0	-87.87639618	113.588974
13.5	367.3839111	1	150.5200806	0	-86.53353119	113.4485855
13.55	373.5588379	1	151.2781982	0	-87.54368591	116.9095154
13.6	321.4978333	1	181.0343933	0	-143.6678314	76.22077179
13.65	318.5427856	1	184.6828156	0	-150.9433441	73.01770782
13.7	315.8691711	1	188.2410278	0	-158.1621552	69.86045074
13.75	313.4510193	1	191.6974182	0	-165.288269	66.75937653
13.8	311.2644043	1	195.072113	0	-172.3768005	63.72338104
13.85	309.2869263	1	198.3374329	0	-179.3345795	60.75603867
13.9	307.4988708	1	201.5149841	0	-186.2223663	57.86685944
13.95	305.8817139	1	204.5668488	0	-192.8991547	55.05499268
14	304.4192505	1	207.5118561	0	-199.414505	52.32880402
14.05	303.0961914	1	210.3540955	0	-205.7702942	49.69319916
14.1	301.8995667	1	213.0847778	0	-211.9204865	47.14942551
14.15	300.8170166	1	215.6760559	0	-217.7398682	44.69771194
14.2	299.8372498	1	218.1826019	0	-223.4305573	42.34759521
14.25	298.9505615	1	220.5750732	0	-228.864624	40.09652328
14.3	298.1478882	1	222.8711853	0	-234.1068726	37.94799423
14.35	297.4211426	1	225.0540466	0	-239.0749512	35.89956665
14.4	296.7627869	1	227.1270294	0	-243.7740021	33.95241547
14.45	296.1672974	1	229.1012115	0	-248.2456818	32.10494232
14.5	295.6275024	1	230.9593964	0	-252.4030151	30.35580444
14.55	295.1384583	1	232.739212	0	-256.417511	28.70628929
14.6	294.6954041	1	234.4078522	0	-260.1264038	27.15096474
14.65	294.6546326	0.998775065	235.9760132	0	-263.5743103	25.68877792
14.7	295.1001282	0.995985746	237.4648895	0	-266.861908	24.31720352
14.75	295.5115662	0.993374705	238.8616486	0	-269.9186401	23.03254128
14.8	295.8909302	0.990947008	240.159668	0	-272.7023621	21.83037186
14.85	296.2406006	0.988680661	241.40065	0	-275.4276123	20.71118164
14.9	296.5623779	0.986581147	242.5400391	0	-277.8482971	19.6668663
14.95	296.8583374	0.984633803	243.6184998	0	-280.1765137	18.69710159

15	297.1302185	0.982828438	244.5962372	0	-282.1832886	17.79479408
15.05	297.3797913	0.981158614	245.5276337	0	-284.1619873	16.95993233
15.1	297.6086731	0.979616225	246.3922424	0	-285.9890137	16.19366646
15.15	297.8183594	0.978194594	247.1762695	0	-287.5744324	15.47935581
15.2	298.0103455	0.976881504	247.9102173	0	-289.0818481	14.82279015
15.25	298.1860657	0.975682855	248.5918427	0	-290.4883118	14.21297264
15.3	298.3465881	0.974577248	249.2237854	0	-291.7988281	13.65301704
15.35	298.4933472	0.973564327	249.8016663	0	-292.9758301	13.13643837
15.4	298.6271973	0.972637653	250.3291321	0	-294.029541	12.66506577
15.45	298.7493591	0.971782625	250.8130493	0	-294.9897766	12.22649574
15.5	298.8606873	0.971006513	251.2580566	0	-295.8735046	11.82717037
15.55	298.9620972	0.97029382	251.6747284	0	-296.7344971	11.46115208
15.6	299.0543823	0.969651759	252.0439606	0	-297.4462891	11.12675095
15.65	299.1383362	0.96905607	252.3907013	0	-298.1516418	10.82505894
15.7	299.2146301	0.968520463	252.7003937	0	-298.7512817	10.54525661
15.75	299.2839355	0.968025506	252.9873962	0	-299.324646	10.2831049
15.8	299.3468933	0.967582524	253.2467499	0	-299.8314209	10.05102253
15.85	299.4039917	0.967176199	253.4825134	0	-300.2910767	9.839715958
15.9	299.455719	0.966808736	253.6970978	0	-300.7095947	9.647149086
15.95	299.502594	0.966475129	253.8917999	0	-301.0874939	9.467474937
16	299.5449829	0.966179609	254.0730286	0	-301.4584045	9.313630104
16.05	299.5833435	0.96589905	254.2352142	0	-301.7808533	9.17149353
16.1	299.6180725	0.965651453	254.3776398	0	-302.0441284	9.039224625
16.15	299.6180725	0.965651453	254.3776398	0	-302.0441284	9.039224625

[MJ/hr]	[MJ/hr]	°C	°C	°C	°C	
Heat D. Comp.1	Heat D. Comp.2	Temp. of inter	LMTD	T1 after first cor	T2 after interco	F safety factor
0.005245263	0.005519986	25.0029068	107.0683823	145.5064392	119.6709747	1
97.6080246	99.72086334	50.05176926	89.29512024	131.6281433	122.4860535	1
97.56270599	99.67485809	50.03936005	89.30162811	131.6278992	122.4861526	1
97.50853729	99.61934662	50.02477646	89.30931854	131.6277008	122.4862595	1
97.45367432	99.56330109	50.00992584	89.31712341	131.6274567	122.4863586	1
97.40570831	99.51416016	49.99697113	89.32395935	131.6272888	122.4864578	1
97.35806274	99.46547699	49.9839859	89.3307724	131.6270447	122.4865646	1
97.31420898	99.4204483	49.97245789	89.33691406	131.6270142	122.4866333	1
97.26863098	99.37387085	49.95987701	89.34349823	131.6267395	122.4867325	1
97.22808838	99.33244324	49.94892502	89.34925842	131.6265717	122.4868011	1
97.18312073	99.2864151	49.9368515	89.35561371	131.6264038	122.4869003	1
97.1421814	99.24459076	49.92580032	89.36142731	131.626236	122.486969	1
97.10195923	99.20339203	49.91501617	89.36711121	131.6260986	122.4870377	1
97.06009674	99.16052246	49.90379333	89.37302399	131.6259613	122.4871063	1
97.02400208	99.12365723	49.89402771	89.37817383	131.625824	122.487175	1
96.98049927	99.07913208	49.8822937	89.38437653	131.6257019	122.4872742	1
96.93968201	99.0374527	49.87137604	89.39012146	131.6255646	122.4873428	1
96.8994751	98.99624634	49.86052704	89.39582825	131.6253967	122.4874115	1
96.87192535	98.96804047	49.85309982	89.39971161	131.6252594	122.487442	1
96.8296814	98.92488861	49.84184265	89.40566254	131.6251526	122.4875107	1
96.79174805	98.88613892	49.83156586	89.41105652	131.6249847	122.4875793	1
96.74897766	98.84237671	49.82024384	89.41704559	131.6248779	122.487648	1
96.71648407	98.80912781	49.81145859	89.42168427	131.6247864	122.4877167	1
96.68113708	98.77293396	49.8020668	89.42660522	131.624649	122.4877472	1
96.64868164	98.73970795	49.79338837	89.43121338	131.6245728	122.4878159	1
96.60645294	98.69660187	49.7820282	89.43717957	131.6244049	122.4878845	1
96.57418823	98.66355896	49.77341461	89.4417038	131.6242981	122.4879227	1
96.54090118	98.62960052	49.76446152	89.44641876	131.624176	122.4879837	1

96.50062561	98.58839417	49.75378036	89.45207977	131.6240997	122.4880524	1
96.46517944	98.55210876	49.74445724	89.45700836	131.6240387	122.4880905	1
96.43087769	98.51698303	49.7353363	89.46179199	131.6239014	122.488121	1
96.39533997	98.48073578	49.72577286	89.46683502	131.6237946	122.4881897	1
96.35764313	98.44203186	49.71587372	89.47206116	131.6237335	122.4882278	1
96.3144989	98.39801025	49.70447922	89.47810364	131.6236572	122.4882889	1
96.2835083	98.36639404	49.69620514	89.48243713	131.6235199	122.488327	1
96.24354553	98.32540894	49.68562698	89.48804474	131.6234589	122.4883957	1
96.21037292	98.29147339	49.67687607	89.49266052	131.6233826	122.4884262	1
96.16724396	98.24743652	49.66548538	89.49866486	131.623291	122.4884644	1
96.12851715	98.20782471	49.65524292	89.50408936	131.6232147	122.488533	1
96.09362793	98.17222595	49.64595413	89.5089798	131.6231232	122.4885635	1
96.05389404	98.13157654	49.63554382	89.51447296	131.6230469	122.4885941	1
96.01270294	98.08945465	49.62475967	89.52017212	131.6229858	122.4886322	1
95.96815491	98.04399872	49.61292648	89.52642822	131.622879	122.4887009	1
95.93293762	98.00798035	49.60363388	89.53131104	131.6227722	122.4887314	1
95.893013	97.96711731	49.59319305	89.53684235	131.6227417	122.4887695	1
95.8526535	97.9258728	49.5825119	89.54247284	131.6226501	122.4888	1
95.80820465	97.88037872	49.57088089	89.5486145	131.6225739	122.4888382	1
95.76579285	97.83700562	49.55979156	89.55446625	131.6225128	122.4888687	1
95.72834015	97.79869843	49.54995728	89.55966187	131.6224365	122.4888992	1
95.68119812	97.75051117	49.53754807	89.5662384	131.6223755	122.4889679	1
95.64437103	97.71294403	49.52788544	89.57133484	131.6222992	122.489006	1
95.58963013	97.65698242	49.51360703	89.57888031	131.6222382	122.4890366	1
95.55012512	97.61657715	49.50326538	89.58434296	131.6221619	122.4890747	1
95.50218964	97.5676651	49.49075317	89.59095764	131.6221008	122.4891052	1
95.45741272	97.52177429	49.47905731	89.59713745	131.6220398	122.4891434	1
95.41214752	97.47548676	49.46722031	89.6033783	131.6219635	122.4891739	1
95.36463165	97.42700195	49.4548111	89.60995483	131.621933	122.4892044	1
95.31937408	97.38071442	49.44297791	89.61619568	131.6218567	122.4892426	1
95.26815033	97.32844543	49.42964935	89.62324524	131.6217957	122.4892731	1
95.21803284	97.27713013	49.41656113	89.63014984	131.6217346	122.4893112	1
95.16845703	97.22642517	49.40361023	89.63699341	131.6216583	122.4893417	1
95.11869049	97.17564392	49.39065552	89.64383698	131.6215973	122.4893799	1

95.05995941	97.11551666	49.37546539	89.65189362	131.6215973	122.4894104	1
95.01367188	97.0683136	49.36336136	89.65828705	131.621521	122.4894485	1
94.95484924	97.00817108	49.34806824	89.66636658	131.62146	122.4894791	1
94.90413666	96.9563446	49.33484268	89.6733551	131.6213837	122.4895096	1
94.84778595	96.89870453	49.32019424	89.68109894	131.6213531	122.4895477	1
94.79225922	96.84204865	49.30571747	89.68874359	131.6212921	122.4895782	1
94.73560333	96.78413391	49.29106903	89.69650269	131.6212463	122.4896164	1
94.68457794	96.731987	49.27767181	89.70355225	131.6211548	122.4896469	1
94.62339783	96.66944885	49.26187134	89.71192932	131.6211548	122.4896851	1
94.56272888	96.60741425	49.24610519	89.72026062	131.6210785	122.4897156	1
94.50071716	96.54405212	49.23006439	89.72875214	131.621048	122.4897537	1
94.4414978	96.48361206	49.21467209	89.73684692	131.6209412	122.4897537	1
94.37887573	96.419487	49.19849777	89.74541473	131.6209412	122.4897842	1
94.31595612	96.35527802	49.18215179	89.75404358	131.6208801	122.4898148	1
94.25334167	96.2912674	49.16597748	89.76262665	131.6208801	122.4898529	1
94.18589783	96.22234344	49.14838028	89.77189636	131.6207733	122.4898834	1
94.11977386	96.15477753	49.13129044	89.78093719	131.6207428	122.4899216	1
94.04987335	96.08332825	49.11315155	89.79051208	131.6206818	122.4899521	1
93.98252869	96.01447296	49.09578705	89.79971313	131.6206818	122.4899902	1
93.91435242	95.94481659	49.07808685	89.80905151	131.6206055	122.4900208	1
93.84178162	95.87073517	49.05926895	89.81898499	131.6205444	122.4900589	1
93.77320862	95.80056	49.04156876	89.82836151	131.6205444	122.4900894	1
93.69981384	95.7256012	49.02261353	89.83834839	131.6204681	122.4900894	1
93.61910248	95.64312744	49.00169373	89.84939575	131.6204071	122.4901199	1
93.54405975	95.56653595	48.98226166	89.85967255	131.6203766	122.4901581	1
93.46541595	95.48606873	48.96198654	89.87038422	131.6203308	122.4901886	1
93.39026642	95.40927887	48.94258881	89.88064575	131.6203003	122.4902267	1
93.30454254	95.32164001	48.92037964	89.89237213	131.6202698	122.4902573	1
93.22403717	95.23937225	48.8995285	89.90338135	131.6202087	122.4902954	1
93.13975525	95.15335083	48.87768936	89.91490173	131.6201324	122.4903259	1
93.06255341	95.07445526	48.8578186	89.92541504	131.6201324	122.4903641	1
92.97158051	94.98139191	48.8343544	89.93781281	131.6201019	122.4903946	1
92.88614655	94.89408875	48.81231308	89.94946289	131.6200714	122.4904251	1
92.79920197	94.80535126	48.78973007	89.96137238	131.6199951	122.4904633	1

92.71061707	94.71479034	48.76684189	89.97344208	131.6199341	122.4904938	1
92.61222839	94.61423492	48.74144745	89.98686218	131.6199036	122.4905319	1
92.52023315	94.52022552	48.71774292	89.99938965	131.6199036	122.4905624	1
92.42321777	94.42118835	48.69268417	90.01260376	131.6198273	122.4906006	1
92.32627869	94.32204437	48.66762543	90.02584076	131.6197968	122.4906311	1
92.22309113	94.21669769	48.64094162	90.03990936	131.6197205	122.4906693	1
92.12677765	94.1182785	48.61611938	90.05300903	131.6196899	122.4906998	1
92.02552795	94.0147171	48.59004593	90.06678772	131.6196899	122.4907303	1
91.92153168	93.9085083	48.56319046	90.08094788	131.6196289	122.4907684	1
91.80683136	93.79139709	48.53355408	90.09657288	131.6195526	122.490799	1
91.69860077	93.6807251	48.50568008	90.11126709	131.6195221	122.4908371	1
91.58708954	93.5667572	48.47699356	90.1264267	131.6195221	122.4908676	1
91.47424316	93.45153809	48.44786835	90.14176941	131.6194611	122.4909058	1
91.35778809	93.33242798	48.41778946	90.15763855	131.6194153	122.4909363	1
91.23795319	93.21009064	48.38676453	90.1740036	131.6193542	122.4910049	1
91.11923981	93.08880615	48.35621262	90.19011688	131.6193237	122.4910355	1
91.00312805	92.97012329	48.32623672	90.2059021	131.6192474	122.4910736	1
90.87310028	92.83727264	48.29277039	90.22355652	131.6192474	122.4911041	1
90.74998474	92.71147156	48.26099777	90.24029541	131.6191864	122.4911423	1
90.61812592	92.57672119	48.2270546	90.25817871	131.6191559	122.4911728	1
90.48493958	92.44071198	48.19270706	90.27626801	131.6190796	122.4912109	1
90.34579468	92.29853821	48.15673065	90.29524231	131.6190491	122.4912796	1
90.21166992	92.16149139	48.12224579	90.3134079	131.6190186	122.4913101	1
90.06554413	92.01215363	48.0846405	90.33322144	131.618988	122.4913406	1
89.92443085	91.86797333	48.04835892	90.3523407	131.618988	122.4913788	1
89.78098297	91.72136688	48.01136398	90.37180328	131.6189117	122.4914093	1
89.62766266	91.56481171	47.9717598	90.39266205	131.6188507	122.491478	1
89.47400665	91.40771484	47.93225479	90.41345215	131.6188049	122.4915161	1
89.31884766	91.24918365	47.89227676	90.43449402	131.6187744	122.4915466	1
89.15976715	91.08670807	47.85118103	90.45613098	131.6187134	122.4916153	1
89.00016785	90.92355347	47.81015015	90.47770691	131.6186829	122.4916458	1
88.84047699	90.76047516	47.76898575	90.49938965	131.6186371	122.4917145	1
88.66703033	90.58322906	47.72432709	90.52285767	131.618576	122.4917526	1
88.49500275	90.40755463	47.68001175	90.54614258	131.6184998	122.4917831	1

88.31855011	90.22718048	47.63457489	90.57007599	131.6184998	122.4918518	1
88.13544464	90.04005432	47.58750916	90.59481812	131.6184692	122.49189	1
87.95520782	89.85601807	47.54101944	90.61925507	131.6184082	122.491951	1
87.76386261	89.66049194	47.49175262	90.64512634	131.6183319	122.4919891	1
87.57624817	89.46876526	47.44343185	90.67053223	131.6183014	122.4920578	1
87.38018799	89.26844025	47.39297485	90.69702148	131.6182404	122.4920883	1
87.17186737	89.05561066	47.33936691	90.72520447	131.6182404	122.492157	1
86.97438812	88.85381317	47.28853607	90.75190735	131.6181946	122.4922256	1
86.7596283	88.63438416	47.23320007	90.78096008	131.6181335	122.4922943	1
86.54837036	88.41850281	47.17891312	90.8094635	131.618103	122.4923248	1
86.32949829	88.19487762	47.12252045	90.83905029	131.6180267	122.4923935	1
86.10756683	87.96821594	47.06542206	90.86901855	131.6179962	122.4924622	1
85.88130951	87.7370224	47.00719833	90.89953613	131.6179352	122.4925003	1
85.64765167	87.49828339	46.94711304	90.93106842	131.6178894	122.4925613	1
85.41139984	87.25682831	46.88641739	90.96292114	131.6178894	122.49263	1
85.16699219	87.00717926	46.82338333	90.99594116	131.6177979	122.4926987	1
84.92347717	86.75836945	46.76075363	91.02877045	131.6177673	122.4927673	1
84.66928101	86.4986496	46.69534302	91.06305695	131.6177216	122.492836	1
84.40789032	86.23159027	46.62810516	91.09828186	131.617691	122.4929047	1
84.1463089	85.96430206	46.56086349	91.13350677	131.6176605	122.4929733	1
83.87320709	85.68527985	46.49067307	91.1702652	131.61763	122.493042	1
83.5983963	85.4046402	46.4200058	91.20722961	131.6175537	122.4930725	1
83.31769562	85.11787415	46.34767914	91.24508667	131.6174927	122.4931717	1
83.02405548	84.81785583	46.27223587	91.28455353	131.6174622	122.4932404	1
82.72570038	84.51302338	46.19556808	91.32466125	131.6174164	122.493309	1
82.42610931	84.20692444	46.11856079	91.36492157	131.6173859	122.4933777	1
82.12692261	83.90124512	46.04165649	91.40512085	131.6173553	122.4934464	1
81.80268097	83.56996155	45.95837784	91.44864655	131.6173248	122.493515	1
81.4895401	83.24997711	45.87791443	91.49067688	131.6172791	122.4935837	1
81.16468811	82.91814423	45.79436493	91.53432465	131.6172485	122.4936829	1
80.82976532	82.57604218	45.70830536	91.57923889	131.617218	122.4937515	1
80.49514008	82.23406982	45.62244797	91.62408447	131.6172485	122.4938202	1
80.14296722	81.87425995	45.53197861	91.67127991	131.617218	122.4938889	1
79.7940979	81.51780701	45.44235992	91.71800995	131.6171875	122.4939575	1

79.42486572	81.14070892	45.34745026	91.76750183	131.617157	122.4940567	1
79.05780029	80.76559448	45.25328827	91.81661224	131.6171875	122.4941254	1
78.68024445	80.37985229	45.15641022	91.86708832	131.6171875	122.494194	1
78.2991333	79.99059296	45.05855179	91.91805267	131.6171875	122.4942627	1
77.91234589	79.5953598	44.95930099	91.96973419	131.6171875	122.4943314	1
77.50482178	79.179039	44.85475922	92.02418518	131.6172485	122.4944305	1
77.11422729	78.78000641	44.75456238	92.07633972	131.6172791	122.4944992	1
76.70240784	78.35927582	44.64900589	92.13121033	131.6172791	122.4945297	1
76.28101349	77.92878723	44.54097366	92.18740082	131.6173553	122.4945984	1
75.85334778	77.49184418	44.43144989	92.24436951	131.6174927	122.4946671	1
75.42879486	77.0582428	44.32270432	92.30088806	131.6175842	122.4946976	1
74.98407745	76.60388184	44.2090416	92.3599472	131.6177673	122.4946976	1
74.53874969	76.14897156	44.09531403	92.41905212	131.6179962	122.4946976	1
74.08810425	75.68878174	43.9806366	92.47865295	131.6184082	122.4945984	1
73.63122559	75.22225189	43.86585617	92.53849792	131.6194153	122.4942627	1
73.31827545	75.00643158	44.75310516	92.19654846	132.0110931	122.3180008	1
73.74188232	75.97717285	49.93837738	90.25382996	134.2877045	121.4276352	1
74.31585693	77.08005524	55.05475235	88.58869171	136.9406433	120.647644	1
91.21907043	95.99689484	75.32249451	81.818573	145.5080719	119.6710129	1
90.994133	95.76013184	75.19889832	81.88710022	145.5082397	119.6710129	1
90.75814819	95.51186371	75.0695343	81.95890808	145.5086517	119.6710129	1
91.22792053	96.28478241	77.91139984	81.74156952	147.8746796	119.7729416	1
92.47431183	98.14853668	83.37252045	82.13748169	153.3570251	120.6220093	1
93.89387512	100.1341705	88.84554291	84.11064148	160.589447	122.8228683	1
95.38558197	102.2046051	93.80420685	86.88551331	168.3753967	125.4861679	1
96.85870361	104.2056198	98.4903183	89.72654724	176.1836243	127.9423904	1
98.30282593	106.1175995	99.96736908	94.15620422	183.9524841	130.1170807	1
99.68465424	107.9105072	99.96736908	99.15026855	191.6452332	132.0180817	1
101.01548	109.6039047	99.96736908	103.8932724	199.2257233	133.6702728	1
102.294632	111.2048798	99.96736908	108.3832855	206.649765	135.1021576	1
103.4982071	112.6926956	99.96736908	112.6184998	213.8742981	136.3398132	1
104.6368484	114.0843811	99.96736908	116.5927963	220.8477325	137.4078217	1
105.709137	115.3831635	99.96736908	120.300766	227.5213928	138.3278351	1
106.7275848	116.6052704	99.96736908	123.7394638	233.852829	139.1194611	1

107.6659851	117.7247314	99.96736908	126.9062653	239.8033142	139.7992554	1
108.5565262	118.7797623	99.96736908	129.8028107	245.3449097	140.3822784	1
109.3843536	119.7555313	99.96736908	132.4337616	250.4589691	140.8816833	1
110.155426	120.6647415	99.96736908	134.806778	255.1366119	141.3087463	1
110.8586502	121.4877319	99.96736908	136.9320679	259.377594	141.6733398	1
111.5124741	122.2498779	99.96736908	138.8234253	263.1920471	141.9844055	1
112.1046829	122.9382858	99.96736908	140.4962006	266.5973816	142.2491302	1
112.6461334	123.5657196	99.96736908	141.9675903	269.6164856	142.4746552	1
113.1429749	124.1373291	99.96736908	143.2547607	272.2760925	142.6661682	1
113.5828476	124.6471558	99.96736908	144.3751373	274.6052551	142.8283997	1
113.9735413	125.0945969	99.96736908	145.3468475	276.6350403	142.966568	1
114.3216705	125.4975967	99.96736908	146.1860809	278.3960571	143.0834503	1
114.6156921	125.836319	99.96736908	146.9079285	279.9164429	143.1823273	1
114.8796158	126.1392365	99.96736908	147.5275269	281.225647	143.2660217	1
115.1035767	126.3965149	99.96736908	148.0575867	282.3486938	143.3367462	1
115.2926788	126.6134949	99.96736908	148.5101929	283.309967	143.3964539	1
115.451416	126.7961655	99.96736908	148.8955841	284.1299744	143.4469147	1
115.5852051	126.9501114	99.96736908	149.223114	284.828064	143.4894409	1
115.6961899	127.0775757	99.96736908	149.5012817	285.4218445	143.5253143	1
114.9060898	126.2189026	99.96736908	149.9363098	286.3515015	143.5813293	1
114.9420853	126.2621002	99.96736908	150.1027527	286.7078552	143.6025238	1
114.9549789	126.2794495	99.96736908	150.2435608	287.0093994	143.6204529	1
114.9712143	126.2995911	99.96736908	150.3627625	287.2650146	143.6355133	1
114.9644165	126.2941818	99.96736908	150.4633026	287.4805908	143.6482239	1
114.9450912	126.2749557	99.96736908	150.5478516	287.6619568	143.658905	1
114.9254456	126.2547302	99.96736908	150.6193085	287.8153381	143.6678925	1
114.8981857	126.226181	99.96736908	150.6791992	287.9438782	143.6754608	1
114.8617249	126.1866913	99.96736908	150.7298737	288.0527954	143.6817627	1
114.8220749	126.1441956	99.96736908	150.7720642	288.1433411	143.687088	1
114.7676544	126.0853348	99.96736908	150.807373	288.2191772	143.6915283	1
114.718399	126.0313339	99.96736908	150.8372955	288.2835083	143.6952667	1
114.6688614	125.9780273	99.96736908	150.8617706	288.3360596	143.698349	1
114.6068726	125.9099274	99.96736908	150.882782	288.3812561	143.7009583	1
114.5505981	125.8485184	99.96736908	150.9000244	288.4183044	143.703125	1

114.4817963	125.7734985	99.96736908	150.9143524	288.4490967	143.7049255	1
114.416687	125.7022171	99.96736908	150.9263611	288.474884	143.7064209	1
114.3512955	125.6305695	99.96736908	150.9363098	288.4962769	143.7076721	1
114.2800217	125.5525742	99.96736908	150.9445648	288.5140076	143.708725	1
114.204483	125.46978	99.96736908	150.9514771	288.5288696	143.7095642	1
114.131073	125.3890152	99.96736908	150.9572906	288.5413818	143.7102814	1
114.0570145	125.3080139	99.96736908	150.961853	288.5511475	143.7108917	1
113.9695816	125.2113342	99.96736908	150.9661713	288.5605164	143.7113647	1
113.8954926	125.1304016	99.96736908	150.9691315	288.566864	143.7117767	1
113.8126602	125.0395813	99.96736908	150.9717865	288.5725098	143.7121124	1
113.7275848	124.9460144	99.96736908	150.9740143	288.5773621	143.7123871	1
113.643692	124.8537292	99.96736908	150.9759521	288.5815125	143.712616	1
113.5568542	124.7582169	99.96736908	150.9775085	288.5848999	143.7127991	1
113.4684906	124.6612015	99.96736908	150.9787598	288.5875549	143.7129669	1
113.3714066	124.5545807	99.96736908	150.9797211	288.5896301	143.7130585	1
113.2850647	124.4600677	99.96736908	150.9803467	288.5909729	143.7131653	1
113.188797	124.3543854	99.96736908	150.9809875	288.5923157	143.7132721	1
113.0925064	124.2482758	99.96736908	150.9817657	288.5940552	143.7133331	1
112.9946594	124.1411972	99.96736908	150.9819336	288.5943298	143.7134094	1
112.9011688	124.0384445	99.96736908	150.9823914	288.5953369	143.7134399	1
112.8032532	123.9305801	99.96736908	150.9828186	288.596283	143.7134705	1
112.692276	123.8089752	99.96736908	150.9829102	288.596405	143.713501	1
112.5900421	123.696312	99.96736908	150.9833069	288.5973206	143.7135315	1
112.4800491	123.575386	99.96736908	150.9834442	288.5976257	143.7135315	1
112.3715973	123.4565659	99.96736908	150.983429	288.5975647	143.7135773	1
112.2706604	123.3458176	99.96736908	150.983429	288.5975647	143.7135773	1
112.1530457	123.2160263	99.96736908	150.9838562	288.5985413	143.7135773	1
112.0446472	123.0973282	99.96736908	150.9836731	288.5980835	143.7136078	1
111.9258041	122.9665451	99.96736908	150.9839172	288.5986633	143.7136078	1
111.8116074	122.8413849	99.96736908	150.9837646	288.5983276	143.7136078	1
111.6891861	122.7063522	99.96736908	150.9840851	288.5990601	143.7136078	1
111.5638657	122.5689468	99.96736908	150.9839783	288.5987549	143.7136078	1
111.4404984	122.4331894	99.96736908	150.9841461	288.5991516	143.7136078	1
111.3131409	122.2937698	99.96736908	150.9837952	288.5983887	143.7136078	1

111.1849899	122.1525116	99.96736908	150.9840851	288.5990295	143.7136078	1
111.0546646	122.0099487	99.96736908	150.9837494	288.5982361	143.7136078	1
110.9222412	121.8640594	99.96736908	150.9839935	288.5988159	143.7136078	1
110.7859573	121.7141418	99.96736908	150.9841461	288.5991516	143.7136078	1
110.6490631	121.5638962	99.96736908	150.9840546	288.5989685	143.7136078	1
112.4883728	122.7746582	99.96736908	160.018631	298.8968506	151.5462494	1
110.0062332	117.9115677	99.96736908	175.8639221	265.3945618	211.7306366	1
110.090683	117.7195129	99.96736908	175.7397461	263.2410583	213.8247223	1
110.2076492	117.5625534	97.53217316	176.924881	261.2940674	215.7749634	1
110.3573151	117.4393539	94.33287811	178.5485229	259.5371094	217.5927277	1
110.5453873	117.3575974	91.19776917	180.1712036	257.9519653	219.2893066	1
110.7640228	117.3095474	88.13082123	181.7861786	256.522644	220.8726501	1
111.0192795	117.3028183	85.14216614	183.3838196	255.2341614	222.3506927	1
111.2980194	117.3243408	82.23130035	184.9602051	254.0727692	223.7302246	1
111.6047897	117.3797836	79.40706635	186.5069733	253.0261841	225.0173492	1
111.9392776	117.4696655	76.67493439	188.0176239	252.0829163	226.2175751	1
112.2962952	117.5891647	74.0364151	189.4886169	251.2328339	227.3365173	1
112.6633072	117.7258148	71.49202728	190.9167633	250.4669037	228.3783875	1
113.0573959	117.8979645	69.0519104	192.2944489	249.7764282	229.3479309	1
113.4650497	118.0920563	66.71362305	193.6210175	249.1540375	230.2493134	1
113.8901138	118.3124237	64.48098755	194.8927917	248.5929565	231.0864868	1
114.3227081	118.549118	62.35167694	196.1096802	248.0869751	231.8633575	1
114.7608948	118.8001175	60.32703781	197.2697296	247.6306763	232.5832062	1
115.2055435	119.0670624	58.40558243	198.3732758	247.2190857	233.2503815	1
115.6459503	119.3383942	56.58599091	199.4198151	246.8476563	233.867218	1
116.0949631	119.6277695	54.86974716	200.4083405	246.5123749	234.4373474	1
116.5347595	119.9165726	53.25126648	201.3414001	246.2097778	234.9636383	1
116.9664383	120.205658	51.72955704	202.2191315	245.9364777	235.4487305	1
117.3964767	120.5020599	50.30197906	203.0431671	245.689621	235.8960876	1
117.8158264	120.7957382	48.96476746	203.8152771	245.4665985	236.3079376	1
118.2175903	121.0801392	47.71337509	204.5377197	245.2649231	236.6866913	1
118.6197968	121.3727036	46.54831696	205.2105865	245.0826874	237.0348053	1
118.9972153	121.6476212	45.46117783	205.8381958	244.9178314	237.3544159	1
119.3676147	121.923027	44.45162201	206.4210052	244.7687073	237.6476593	1

119.7080612	122.1746902	43.51232529	206.9629822	244.6338196	237.9164886	1
120.0441589	122.4288559	42.64321899	207.464386	244.511673	238.1627655	1
120.3684769	122.6721573	41.84552383	207.9260712	244.4034729	238.3882141	1
120.6620178	122.8944778	41.10194778	208.3549194	244.3037109	238.5944519	1
120.9451523	123.1102448	40.41848755	208.7495422	244.2142029	238.78302	1
121.2093582	123.3164749	39.78372192	209.114624	244.1312866	238.9552307	1
121.4598312	123.5124512	39.20087051	209.4500732	244.0566864	239.1125031	1
121.6904526	123.6943054	38.66318512	209.7590332	243.9886017	239.256012	1
121.9056854	123.8620529	38.17256546	210.0416107	243.9280396	239.3868256	1
122.1006088	124.0183945	37.71608734	210.3028564	243.8710022	239.5060425	1
122.2838745	124.164444	37.30047226	210.5412292	243.8202209	239.6145935	1
122.4579315	124.3054657	36.91954803	210.7594299	243.7738342	239.7133942	1
122.6125107	124.4288635	36.57154465	210.9588928	243.732193	239.8032837	1
122.762886	124.5483093	36.2575531	211.1396179	243.6955719	239.8849792	1
122.8937912	124.6545181	35.96638107	211.3060608	243.6609802	239.9591675	1
123.0109558	124.7547913	35.69358826	211.4602203	243.6271057	240.026535	1
123.122406	124.8457336	35.45205688	211.5982361	243.5988312	240.0876465	1
123.2242355	124.9288864	35.23216248	211.7238464	243.5731964	240.1430817	1
123.3169403	125.0052567	35.0317955	211.8381195	243.5498352	240.1932678	1
123.3976059	125.0747604	34.84482574	211.943573	243.5270386	240.238739	1
123.4810486	125.1410751	34.68474579	212.0359802	243.5095825	240.2798767	1
123.5535889	125.2000809	34.53686905	212.1206512	243.4928894	240.3170624	1
123.6142273	125.2511368	34.39923477	212.1986389	243.4767151	240.350708	1
123.6142273	125.2511368	34.39923477	212.1986389	243.4767151	240.350708	1

Appendix F

Excel Code used for the Heat Pump

Option Explicit

Sub DynamicSim()

Dim IPA As Double, WT As Double, DMSO As Double

Dim i As Long

Dim MolRateIPA As Double, MolRateWT As Double, MolRateDMSO As Double

Dim retFlag As Variant

Sheets("Compressor").Select

Dim Temp2 As Double, Temp3 As Double, Vapf2 As Double, Vapf3 As Double, HeatD As Double

Dim HeatInt As Double, Duty1 As Double, Duty2 As Double, DelT As Double, F As Double, T1 As Double, T2 As Double, LMTD As Double

i = 7

Do While i <= 330

IPA = Cells(i, 7).Value

WT = Cells(i, 8).Value

DMSO = Cells(i, 9).Value

Cells(3, 12).Value = IPA

Cells(3, 13).Value = WT

Cells(3, 14).Value = DMSO

retFlag = objCHEMCAD.DynamicRunStep

Temp2 = Cells(3, 16).Value

Vapf2 = Cells(3, 17).Value

Temp3 = Cells(3, 18).Value

Vapf3 = Cells(3, 19).Value

HeatD = Cells(3, 20).Value

HeatInt = Cells(3, 21).Value

Duty1 = Cells(3, 22).Value

Duty2 = Cells(3, 23).Value

DelT = Cells(3, 24).Value

LMTD = Cells(3, 25).Value

T1 = Cells(3, 26).Value

T2 = Cells(3, 27).Value

F = Cells(3, 28).Value

Cells(i, 12).Value = Temp2

Cells(i, 13).Value = Vapf2

Cells(i, 14).Value = Temp3

```
Cells(i, 15).Value = Vapf3  
Cells(i, 16).Value = HeatD  
Cells(i, 17).Value = HeatInt  
Cells(i, 18).Value = Duty1  
Cells(i, 19).Value = Duty2  
Cells(i, 20).Value = DelT  
Cells(i, 21).Value = LMTD  
Cells(i, 22).Value = T1  
Cells(i, 23).Value = T2  
Cells(i, 24).Value = F
```

```
i = i + 1
```

Loop

Call RestoreToInitialState

End Sub

Appendix G

Stream Properties ChemCAD Report of IPA at 75°C

Simulation: Batch distillation

STREAM PROPERTIES

Stream No. 10

Name

- - Overall - -

Molar flow kmol/h 0.0166

Mass flow kg/h 1.0000

Temp C **75.2000**Pres bar **1.0130**

Vapor mole fraction 0.000

Enth MJ/h -5.1366

Tc C 235.1600

Pc bar 47.6430

Std. sp gr. wtr = 1 0.794

Std. sp gr. air = 1 2.075

Degree API 46.7638

Average mol wt 60.0960

Actual dens kg/m3 730.6159

Actual vol m3/h 0.0014

Std liq m3/h 0.0013

Std vap 0 C m3/h 0.3730

- - Liquid only - -

Molar flow kmol/h 0.0166

Mass flow kg/h 1.0000

Average mol wt 60.0960

Actual dens kg/m3 730.6158

Actual vol m3/h 0.0014

Std liq m3/h 0.0013

Std vap 0 C m3/h 0.3730

Cp kJ/kg-K 3.2962

Z factor 0.0034

Visc N-s/m2 0.0005851**Th cond W/m-K 0.1235**

Surf. tens. N/m 0.0170

Component mole fractions

Isopropanol 1.000000

Water 0.000000

DiMth Sulfoxide 0.000000

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